Government of India
Ministry of Jal Shakti
Depart of Water Resources, River Development and Ganga Rejuvenation
CENTRAL GROUND WATER BOARD

ANNUAL REPORT 2021 - 22
Faridabad

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GOVERNMENT OF INDIA
MINISTRY OF JAL SHAKTI
DEPARTMENTOF WATER RESOURCES, RIVER DEVELOPMENT & GANGA REJUVENATION
CENTRAL GROUND WATER BOARD

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2022

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1. ABOUT CENTRAL GROUND WATER BOARD

Central Ground Water Board (CGWB) is a scientific department under Department of Water Resources, RD & GR, Ministry of Jal Shakti, Government of India and national apex organization for all aspects of ground water surveys, exploration, development and management. Though it has been designated as 'Board', it is a regular department in Government of India, which is governed by the General Central Services Rules and Regulations.

1.1 VISION

Sustainable Development and Managemement of Ground Water Resources of the Country.

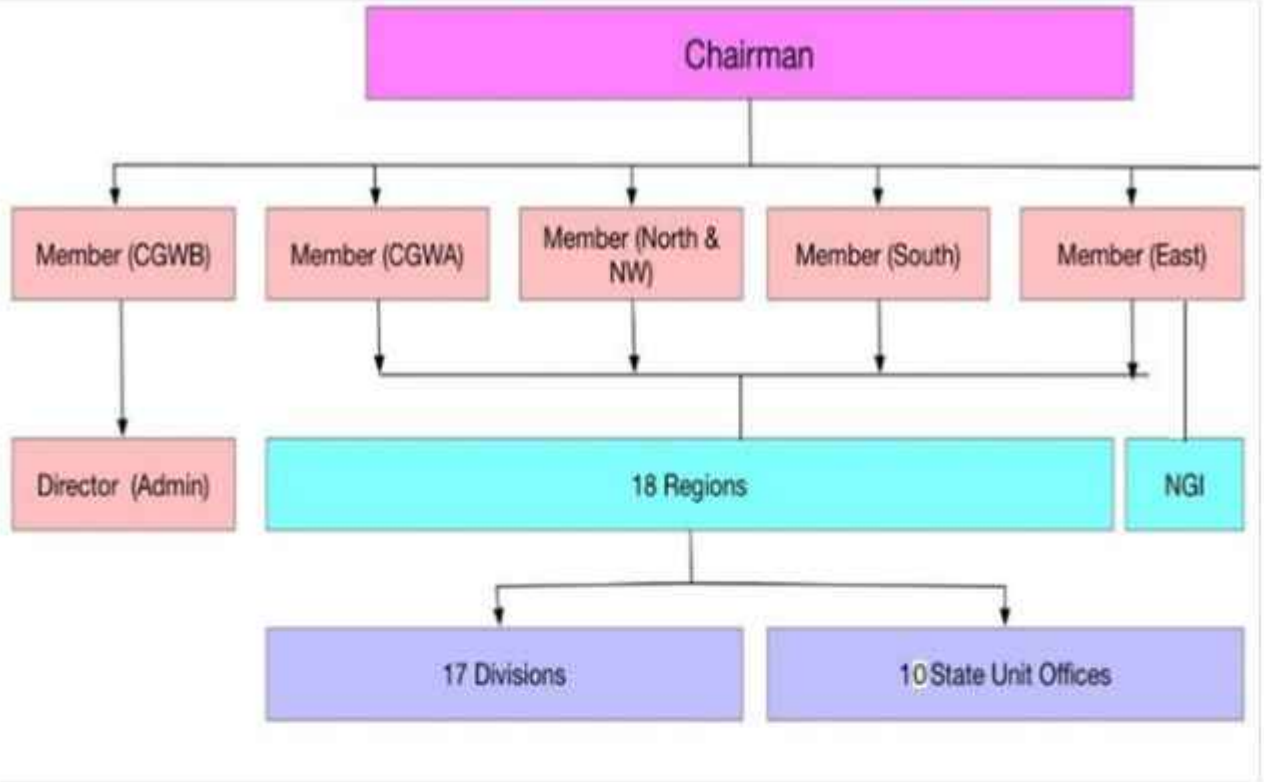
1.2 MANDATE

CGWB is a multidisciplinary Scientific Organization with a mandate to "Develop and disseminate technologies for Scientific and Sustainable development and management of India's Ground Water Resources, including monitoring exploration, assessment and augmentation."
The Central Ground Water Board was formed in 1970 by renaming erstwhile Exploratory Tube well Organization. Subsequently, Ground Water Division of Geological Survey of India got merged with it in 1972. The Board was placed under various ministries in the past and since 1982, it is an organization under Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, Government of India.
Central Ground Water Board is also discharging the functions as Central Ground Water Authority (CGWA) to regulate and control the development and management of ground water in the country since 1997.
All the policy decision related to the functioning of the CGWB and CGWA is taken by a governing body, "Board", constituted by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shkati.

1.3 ORGANIZATIONAL SET UP

The Central Ground Water Board is headed by the Chairman and has six full time Members namely, Member (Head Quarter- HQ), Member (North & West), Member (South), Member (East & NGI), Member (CGWA) and Member (Finance). The other Members of the Board are all ex-officio being thenominees of institutions in related fields of expertise. The ex-officio members are:

- The Joint Secretary (A), Ministry of Water Resources, River Development and Ganga Rejuvenation.
- The Joint Secretary (GW), Ministry of Water Resources, River Development and Ganga Rejuvenation.
- The JointSecretary& Financial Adviser, Ministry of Water Resources, River Development and Ganga Rejuvenation.
- The Joint Secretary, Ministry of Environment & Forests.
- The Chief Engineer, IMO (WP &P), CWC.
- The General Manager, ONGC, Ministry of Petroleum & Natural Gas.



1.4 OFFICES OF CENTRAL GROUND WATER BOARD

CGWB executes its function through 18 Regions, 10 State Unit Offices and 17 divisions located throughout India. All the functional units have their defined area of jurisdiction for performing the functions of the Board. The Regional Director is the over all in-charge of Region, Division and the State Unit Offices under its jurisdiction. The Regional Offices are located mostly in the State Capitals. The location of the CGWB offices is shown in Plate –I and listed in the table -1:-

1.4.1 REGIONS& REGIONAL OFFICES

Sr. No.	REGIONS & REGIONAL OFFICES		STATES/ UT's
1	NWR, Chandigarh	North Western Region, Chandigarh	Punjab
			Haryana
			Chandigarh
2	NWHR, Jammu	North Western Himalayan Region, Jammu	Jammu, Kashmir, Leh&Ladakkh
3	NHR, Dharamshala	North Himalayan Region, Dharamshala	Himachal Pradesh
4	WCR, Ahmedabad	West Central Region, Ahmedabad	Gujarat
			Daman & Diu
5	NCR, Bhopal	North Central Region, Bhopal	Madhya Pradesh
6	WR, Jaipur	Western Region, Jaipur	Rajasthan
7	NR, Lucknow	Northern Region, Lucknow	Uttar Pradesh
8	UR, Dehradun	Uttaranchal, Dehradun	Uttarakhand
9	ER, Kolkata	Eastern Region, West Bengal	West Bengal
			Sikkim
			Andaman & Nicobar Islands

Sr. No.	REGIONS & REGIONAL OFFICES		STATES/ UT's
10	NER, Guwahati	North Eastern Region, Guwahati	Assam
			Arunachal Pradesh
			Manipur
			Meghalaya
			Mizoram
			Nagaland
			Tripura
11	MER, Patna	Mid-Eastern Region, Patna	Bihar
			Jharkhand
12	SER, Bhubaneswar	South Eastern Region, Bhubaneswar	Odisha
13	NCCR, Raipur	North Central Chhattisgarh Region, Raipur	Chhattisgarh
14	CR, Nagpur	Central Region, Nagpur	Maharashtra
			Pune
			Dadra & Nagar Haveli
15	SWR, Bengaluru	South Western Region, Bengaluru	Karnataka
			Goa
16	SECR, Chennai	South East Central Region, Chennai	Tamil Nadu
			Puducherry
17	SR, Hyderabad	Southern Region, Hyderabad	Andhra Pradesh
			Telangana
18	KR, Thiruvananthapuram	Kerala Region, Thiruvananthapuram	Kerala

1.4.2ENGINEERING DIVISION OFFICES OF CGWB

DIVISION		STATE
I	Ahmedabad	Gujarat
		Daman & Div
II	Ambala	Punjab
		Haryana
		Chandigarh
		New Delhi
III	Varanasi	Uttar Pradesh
IV	Chennai	Tamil Nadu
		Puducherry (UTP)
		Kerala
V	Ranchi	Bihar
		Jharkhand

DIVISION		STATE
VI	Nagpur	Maharashtra
		Dadra & Nagar Haveli
VII	Guwahati	Assam
		Arunachal Pradesh
		Meghalaya
		Manipur
		Nagaland
		Tripura
VIII	Jammu	Jammu & Kashmir
IX	Hyderabad	Andhra Pradesh
		Telangana
X	Bhubneshwar	Odisha
XI	Jodhpur	Rajasthan
XII	Bhopal	Madhya Pradesh
XIII	Raipur	Chhattisgarh
XIV	Bangalore	Karnataka
		Goa
XV	Kolkata	West Bengal
		Sikkim
		Andaman & Nicobar
XVI	Bareilly	Uttar Pradesh
		Uttarakhand
XVII	Dharamshala	Himachal Pradesh

1.4.3 STATE UNIT OFFICE (SUO)

	SUO	STATE	RESPECTIVE REGIONAL OFFICE
1	Agartala	Tripura	NER, Guwahati
2	Allahabad	Uttar Pradesh	NR, Lucknow
3	Belagavi	Karnataka	SWR, Bengaluru
4	Itanagar	Arunachal Pradesh	NER, Guwahati
5	Jodhpur	Rajashtan	WR, Jaipur
6	R. K. Puram	New Delhi	
7	Pune	Maharashtra	CR, Maharashtra
8	Ranchi	Jharkhand	MER, Patna
9	Shillong	Meghalaya	NER, Guwahati
10	Vishakhapatnam	Andhra Pradesh	SR, Hyderabad

Table -1 Location and Jurisdiction of the CGWB offices

1.5 MAJOR SCHEMES AND ACTIVITIES OF CENTRAL GROUND WATER BOARD

1.5.1 Ground Water Management & Regulation Scheme (GWMR)

Government of India is implementing the Ground Water management and Regulation scheme. It is a Central Sector Scheme, which deals exclusively with ground water and is being implemented by CGWB, DoWR, RD and GR, Ministry of Jal Shakti. The GWM&R Scheme is being continued since 2007-08 and has been approved for continuation during the period 2021-2026.

Activities like preparation of aquifer maps and management plans, monitoring of ground water levels; assessment of ground water quality; estimation of dynamic ground water resources; innovative projects on aquifer rejuvenation and springshed mapping; regulation and control of ground water extraction, organization of workshops, seminars, providing technical assistance to State and Central Government organsiation on groundwater issues and other related activities are being taken up under this scheme in various parts of the country.

A) **National Aquifer Mapping and Management Program (NAQUIM) :** NAQUIM which is being implemented by CGWB, envisages mapping of aquifers (water bearing formations), their characterization and development of Aquifer Management Plans to facilitate sustainable management of Ground Water Resources. NAQUIM was initiated in 2012 as a part of the Ground Water Management and Regulation plan scheme. Out of ~33 lakh km2 of the entire country, a mappable area of ~25 lakh km2 has been identified to be covered under this programme by March 2023.

B) **Ground water level monitoring:** Ground water level monitoring is one of the key activities of Central Ground Water Board. The primary objective of ground water monitoring is to record the response of various natural and anthropogenic stresses on the groundwater regime which impacts the recharge and discharge parameters. At present, CGWB has a network of nearly 23000 ground water observation wells throughout the country. Ground water levels are measured four times a year during the months of January, March/April/May, August and November.

C) **Groundwater quality studies:** Central Ground Water Board has 16 Regional Chemical Laboratories to carry out chemical analysis of water samples. The Chemical laboratories are well equipped with sophisticated instruments like ICP-MS, Atomic Absorption Spectrophotometer (AAS) etc. Ground water sampling for quality monitoring is done once a year during the premonsoon period. In addition to it ground water samples are also collected and analysed as a part of other scientific studies. CGWB has also taken up sampling and analysis of ground water samples for URANIUM content across the country. So far nearly 20000 groundwater samples have been collected and analysed by CGWB. As a part of the NAQUIM studies, CGWB has constructed more than 500 wells in ARSENIC affected areas with innovative designs to tap deeper arsenic safe

aquifers avoiding contamination from shallow aquifers. The technique has been replicated by the states and nearly 1500 such wells have been constructed by the state governments of Uttar Pradesh and West Bengal catering to safe drinking water requirements of nearly 70 lakh persons.

D) **Geophysical Studies:** Geophysical studies are non-invasive techniques and provide indirect information about disposition and characteristics of aquifers. CGWB has inhouse facility for various geophysical studies. Outputs of geophysical studies are used mostly in aquifer mapping studies and other case specific investigations.

E) **High Resolution mapping of Aquifer through Heli-borne Geophysical Studies**: Heli-borne Transient Elecromagnetic (Heli-TEM) studies in conjunction with the classical ground water surveys and exploration have the advantage of faster spatial coverage. An area of nearly 4 lakh square km has been identified for coverage through heliborne surveys in the arid regions of North Western India. Under Phase-I of the project, Central Ground Water Board, Ministry of Jal Shakti and CSIR-NGRI, Hyderabad have signed an agreement for use of advanced heli-borne geophysical survey and other scientific studies in parts of the States of Rajasthan, Gujarat and Haryana under the Aquifer Mapping Programme.

F) **Exploratory Drilling:** Ground Water Exploration is one of the core activities of Central Ground Water Board. Drilling aided ground water exploration provides direct information about disposition and characteristics of the aquifers. Every year around 650 wells are drilled by CGWB for ground water exploration through in-house resources. CGWB has a fleet of 78 operational drilling rigs and has capability to construct wells in various types of terrain in the country. In view of enhanced requirements of ground water exploration under the NAQUIM programme, CGWB has also taken up exploratory drilling through outsourcing. After conducting necessary tests and chemical quality assessment, successful wells are handed over to the state user agencies.

G) **Aquifer Rejuvenation and Water Conservation:** CGWB implemented innovative Aquifer Rejuvenation interventions in three aspirational districts of Maharashtra, Telangana and Andhra Pradesh. Construction of innovative Bridge-cum-Bandhara was undertaken at five sites in the State of Maharashtra to Conserve surface water & base flows available in the stream channel. Existing bridges were improvised as water retaining structure.

H) **Assessment of Dynamic Ground Water Resources of India:** Periodic Assessment of dynamic ground water resources is done jointly by CGWB and the respective state governments. The latest assessment for the year 2020 has been carried out jointly by the Central Ground Water Board and State/UT Nodal/Ground Water Departments based on the methodology recommended by Ground Water Estimation Committee -2015 (GEC-2015). For the first time, the entire assessment has been

done using the GIS based web portal 'India- Groundwater Resource Estimation System (IN-GRES) that has been developed by CGWB in association with IIT-Hyderabad. IN-GRES provides a common and standardized platform for GW Resource Assessment for the entire country (Central and State Governments). High Resolution mapping of Aquifer through Heli-borne Geophysical Studies: Helib-borne Transient Elecromagnetic (Heli-TEM) studies in conjunction with the classical ground water surveys and exploration have the advantage of faster spatial coverage. An area of nearly 4 lakh square km has been identified for coverage through heliborne surveys in the arid regions of North Western India. Under Phase-I of the project, Central Ground Water Board, Ministry of Jal Shakti and CSIR-NGRI, Hyderabad have signed an agreement for use of advanced heli-borne geophysical survey and other scientific studies in parts of the States of Rajasthan, Gujarat and Haryana under the Aquifer Mapping Programme.

I) **Outreach Programs:** Public Interaction programs (PIPs) were initiated in the year 2018 as a medium to disseminate findings of NAQUIM studies at grassroots level. The participants are sensitized on various aspects of water conservation, ground water management and findings of the NAQUIM study.

J) **CGWA- Regulation and Control of Ground Water Extraction:** Central Ground Water Authority (CGWA) has been constituted under Section 3(3) of the Environment Protection Act, 1986. CGWA has been entrusted with the responsibility of regulating and controlling ground water development and management in the country and issuing necessary directives for the purpose. To promote ease of doing business, CGWA has put in place a web-based system NOCAP for processing of applications and issuance of NOCs. New Guidelines for regulation and control of ground water extraction have been issued in September 2020 by Central Ground Water Authority.

1.5.2 Rajiv Gandhi National Ground Water Training & Research Institute (RGNGWTRI)

The institute caters to training and research in the field of ground water in the country. Since 2012 (XII Plan) RGNGWTRI has been implementing a three tiered (Tier I- National Level, Tier II- State/ District Tier -III: Block Level) training programme. It also occasionally carries out training programmes related to ground water management for foreign nationals and institutes. Other important activities of RGI include conducting exams for accreditation of ground water consultants and conducting research on various aspects of ground water. RGI is in the process of conducting a course for accreditation of water auditors. The course is being designed to comprehensively cover the aspects of industry water auditing and create the capacity of water auditor to cater to the demand of industries for water audit as per CGWA notification

1.5.3 PMKSY- Har Khet Ko Pani- Ground Water (PMKSY-HKKP-GW)

It is a centrally sponsored scheme launched by Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti, which envisages to provide irrigation facility for Small and Marginal Farmers in areas having sufficient potential for future development of ground water. The Scheme is being implemented with an objective to support goal of Hon'ble Prime Minister of India, for doubling the farmer incomes. The scheme was approved for 2015-20, but keeping in view various requirements to implement the scheme, guidelines were revised during May 2019 and scheme has been re-launched in July 2019. Thirteen projects amounting Rs. 1020.54 crore are at various stage of implementation in the States of Assam, Arunachal Pradesh, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, and Uttarakhand.

1.5.4 National Hydrology Project (NHP)

CGWB is one of the implementing agencies of the NHP. Major activity traken up under NHP in CGWB include installation of Digital Water Level Recorders with telemetry systems. A Centre of Excellence for CGWB has been established with its headquarters in Faridabad and four regional Hubs each at NR, Lucknow/ SECR, Chennai/ SER, Bhubaneswar/ WCR, Ahmedabad. Major role of CoE is to coordinate the activities related to Ground Water Modelling and organize regular presentations for sharing of knowledge through Ground Water Modeling Talk Series.

2.0 GROUND WATER SCENARIO OF INDIA

Introduction

India is one of the largest users of groundwater in various sectors like agriculture, domestic and industrial. Potential use of Groundwater irrigation has been increased at a very rapid pace since 1970s. Groundwater sustainability and its availability play a vital role in ensuring livelihood as well as food security across the country, especially in (UP, Panjab, Haryana, WB, etc.) where economies depend on agriculture. Rural agriculture and drinking water sector are dependent partly or wholly on groundwater through pumping and/or spring or river due to facile access. However, urban population highly rely on groundwater due to unreliable and deficient quality of water supplies from other sources. Unplanned Intensive irrigation, domestic and industrial use of Groundwater created serious problems for scientific community in planning management of groundwater. Saltwater encroachment near coast, depletion in water tables, water logging, drying of aquifers, pollution of groundwater, and salinity, etc. are major consequences of overexploitation. Deterioration in groundwater quality by various causes is another serious issue. Despite the cumulative effort of different stakeholders, 20% of groundwater blocks are semi-critical, critical, or overexploited. Moreover, aquifers depletion in the most populated and economically productive areas of the country encourages us to understand and create an overall scenario of Indian aquifer system.

CGWB, a largest government organisation in the world working under Ground water, keep an India-focus ground water management and regulation for the reason that it is one of the pioneer organisations to Develop and disseminate technologies, and monitor and implement national policies for the scientific and sustainable development and management of India's ground water resources, including their exploration, assessment, conservation, augmentation, protection from pollution and distribution, based on principles of economic and ecological efficiency and equity.

Combined together, it can be understood that various issues of ground water are going to reduce the fresh water availability for irrigation, domestic and industrial uses. Realizing this, central Ground water board instigate several protective and legislative measures on ground water quality, aquifer mapping and ground water regime monitoring to overcome the groundwater management-related problems.In this chapter, CGWB attempt to unpack myriad mystery of Indian aquifer system using water table, ground water resource, and aquifer disposition

Ground water regime monitoring is one of the key activities of CGWB to generate information on ground water level/ quality through representative sampling. Ground water levels are being monitored throughout the Country four times in a year by Central Ground Water Board through a network of 23196 monitoring wells during the months of January, Premonsoon (March/April/ May), August and Postmonsoon. The ground water regime monitoring was started in the year 1969 by Central Ground Water Board. At present CGWB has a network of 22339 ground water observation

wells, out of which 16190 observation wells are dugwells and 6149 are piezometers. The water level / piezometric head data collected from these observations were entered into the National database and are analysed for obtaining background information of ground water regime and changes on regional scale.

Aquifer system of India

Aquifer System itisan outcome of the extensive hydrogeological studies carried out by Central Ground Water Board (CGWB) include Hydrogeological information collected by CGWB through its survey, investigations and exploration were collated, analyzed and integrated to define the extent and characteristics of aquifer systems.

Based on the hydrogeological characteristics, the entire country has been classified into 14 Principal Aquifer Systems and 42 major aquifers. Alluvium is the major aquifer system which covers maximum area of around 31% of the entire country and available in Uttar Pradesh, Bihar, West Bengal, Assam, Odisha and Rajasthan. Thes and stone aquifer covers around 8% area in the country and available in Chhattisgarh, Andhra Pradesh, Madhya Pradesh, Gujarat, Karnataka and Rajasthan. The rest of the country is covered with the other formations that cover around 60% of the area. Among these, Basalt aquifer covers maximum of around 17% area of the country and mainly spread over Maharashtra, Madhya Pradesh, Gujarat, Rajasthan and Karnataka.Shale aquifer accounts for around 7% of area in the country and is available mostly in Chhattisgarh, Andhra Pradesh, Madhya Pradesh, Rajasthan and in the north-eastern states as well as in the Himalayan terrain. Limestoneaquifer covers averysmallareaof around2% inthe country and mainly available inthe states of Chhattisgarh, Andhra Pradesh, Karnataka, and Gujarat and in the Himalayan states.Around20%ofthearea ofthecountryiscoveredby Banded Gneissic Complex (BGC) and Gneiss aquifers which are available almost in all the peninsular statesas well as the Himalayan states. Therest 15% of the entire area is covered by aquifers namely; Schist, Granite, Quartzite, Charnockite, Khondalite, Lat- erites and Intrusive.

Ground Water level scenario

Ground Water Year Book is prepared annually by CGWB depicting changes in ground water regime of the country through different seasons. It is an effort to obtain information on ground water levels through representative monitoring wells. The important attributes of ground water regime monitoring are ground water level.

Ground water levels are being measured by Central Ground Water Board four times a year during January, March/April/ May, August and November to obtain information on ground water levels through representative monitoring wells. The factor affecting the ground water regime can be either natural or anthropogenic such as rainfall, evapotranspiration, pumpage from the aquifer, recharge due to irrigation systems and other practices like waste disposal etc.

A network of 23209 observation wells, as on 31.03.2022, located all over the country is being monitored. During August 2021, almost 68 % of the wells analysed shows water level in the range of 0- 5 m bgl. However, major major parts of north-western and western states have generally deeper depth to water level i.e., ~10- 40 m bgl.

DECADAL WATER LEVEL FLUCTUATION

Water Level Fluctuation with Decadal Mean (Augest-2011 to Augest-2020) to Augest-2021

CGWB has made a comparison of depth to water level from August 2021 with decadal mean of August (2011-2020). The output indicates that out of total 14060 wells analyzed 8124 nos. (57.8%) of well showing rise in the water level while 5928 nos. (42.2%) wells showing decline in water level with following trend-

Sr. No.	Trend	No of wells analyzed	Range (m)
1	Rise	4738	0 - 2
2		1534	2 - 4
3		1004	>4
4	Fall	4308	0 - 2
5		955	2 - 4
6		665	>4

The rise in water level mostly observed in southern peninsula covering Maharashtra, Telangana, Andhra Pradesh, Karnataka, Tamil Nadu, Kerala, southern Gujarat, eastern Uttar Pradesh, Jharkhand, Bihar and northern West Bengal. Decline in water level observed in Odisha, Jharkhand, Madhya Pradesh, Konkan area of Maharashtra, Punjab, Haryana, Rajasthan and eastern Gujarat. More than 4 m fall in water level observed in Western India comprising of Delhi, Gujarat, Haryana, Punjab, Rajasthan.

DECADAL WATER LEVEL FLUCTUATION IN DEEPER AQUIFER

Water Level Fluctuation with Decadal Mean (August-2011 to Augest-2020) to Augest-2021 IN DEEPER AQUIFER

CGWB has made a comparison of depth to water level from August 2021 with decadal mean of August (2011-2020). The output indicates that out of total 956 wells analyzed 555 nos. (58.1%) of well showing rise in the water level while 401 nos. (41.9%) wells showing decline in water level with following trend-

Sr. No.	Trend	No of wells analyzed	Range (m)
1	Rise	226	0 - 2
2		102	2 - 4
3		227	>4
4	Fall	160	0 - 2
5		100	2 - 4
6		141	>4

Rise in water level in majority of wells observed in Andhra Pradesh, Bihar, Karnataka, Maharashtra, Telangana and Tamil Nadu. Similarly fall in significant number of wells is observed in Chandigarh, Chhatisgarh, Delhi, Gujarat, Haryana, Madhya Pradesh, Odisha, Pondichery, Punjab and Uttaranchal.

Ground Water Resources scenario

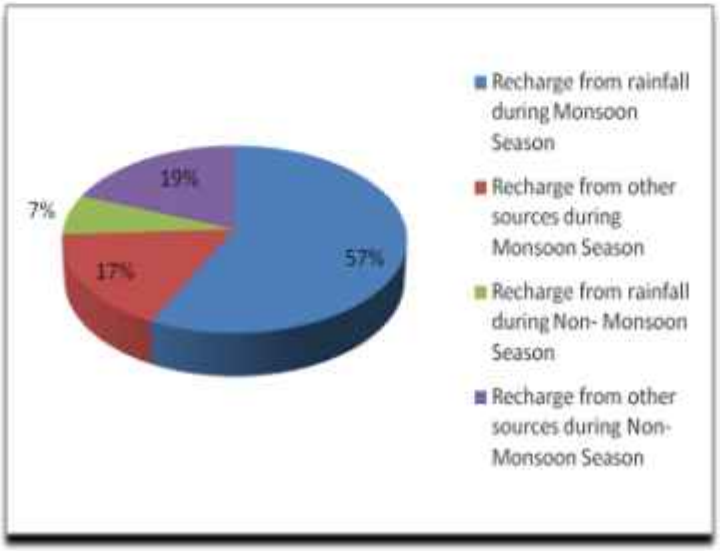
Ground water samples collected from observation wells once a year during the month of March/April/ May are utilized to obtain background information of ground water quality changes on regional scale. And database thus generated is used for assessment of ground water resources and changes in the regime consequent to various development and management activities.

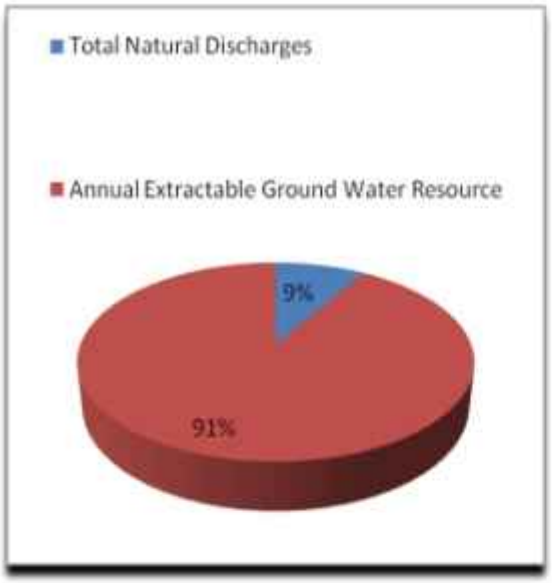
As per the 2020 assessment of Dynamic Ground Water Resources, the Total Annual Ground Water Recharge for the entire country has been assessed as 436.15 billion cubic meter (bcm) and Total natural discharges works out to be 38.51 bcm. Hence, Annual Extractable Ground Water Resources for the entire country is 397.62 bcm.

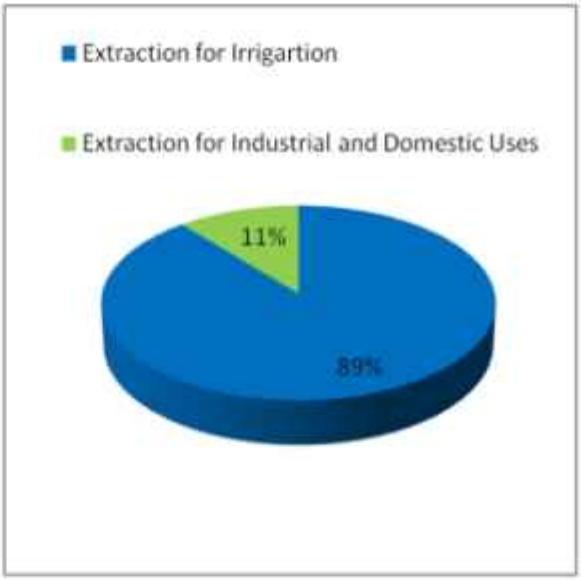
Major source of ground water recharge is the monsoon rainfall. Therefore, Annual Ground Water Recharge is significantly high in the Indus-Ganga-Brahmaputra alluvial belt in the North, East and North East India covering the states of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal and valley areas of North Eastern States. In western India, particularly Rajasthan and parts of northern Gujarat that have arid climate, the annual ground water recharge is scanty. Similarly, in major parts of the southern peninsula covered with hard-rock terrains. This is primarily because of comparatively low infiltration and storage capacity of the rock formations prevailing in the region. The remaining part of Central India is mostly characterized by moderate recharge.

The estimate of Annual Ground Water Recharge for the entire country reflects an increase of 4 bcm in the present assessment as compared to the last assessment i.e., 2017 and the Annual Extractable Ground Water Resources shows an increase of 5 bcm for the country.

Ground Water Resources and Extraction Scenario in India, 2020







Categorization of Assessment Unit

Out of the total 6965 assessment units (Blocks/ Districts/ Mandals/ Talukas/Firkas) in the country,

- 1114 units in various States (16 %) have been categorized as 'Over-Exploited'
- A total of 270 (4 %) assessment units have been categorized as 'Critical'.
- There are 1057 'Semi-Critical' units (15 %).
- And 4427 (64 %) assessment units have been categorized as 'Safe'.

Apart from this, there are 97 assessment units (1 %), which have been categorized as 'Saline' as major part of the ground water in phreatic aquifers is brackish or saline

3.0 GROUND WATER EXPLORATION

Ground Water Exploration aided by drilling is one of the major activities of the Board. It is aimed at delineation of aquifers in different hydrogeological setups and determination of their hydraulic parameters. The exploratory drilling operations have enabled demarcation of aquifers both in lateral and vertical extensions and evaluation of various aquifer parameters, designing of suitable structures and assessment of their yield potential in various hydrogeological settings. These studies have helped in identifying areas worthy for future ground water development. Ground Water exploration contributes to a large extent in guiding the States to implement ground water development schemes.

Groundwater exploration is being carried out by the Board through its fleet of 78 drilling rigs (24 Direct Rotary, 52 Down the Hole and 02 Dual Rotary). During the year 2021-22, Central Ground Water Board under its in-house Ground Water Exploration programme constructed 749 wells (EW-452, OW- 162, PZ- 135). The region/division/state wise deployment of rigs during AAP 2021-22 is mentioned in table 3.1 and the division-wise details of pumping test units is mentioned in table 3.2. Priority was accorded for exploration in Over Exploited/ Critical/ Semi-Critical/ Drought Prone and areas affected with ground water pollution etc. The statement showing State, Division & Region wise distribution of boreholes drilled/ completed during 2021-22 is presented in Table 3.3& 3.4(a) & (b). Out of 452 exploratory wells, 399 wells were constructed in hard rock, 48 wells in alluvium and 5 well in bouldary formations (Table 3.6) including 50 wells constructed in Tribal areas and 25 wells in Drought prone areas (Table 3.7) of the country.

The Board since inception has drilled 44925 bore holes (including 9023 bore holes through outsourcing) as on 31.03.2022 to identify worthy areas for ground water development in the country. The statement showing State-wise distribution of boreholes drilled/completed till March, 2022 in the country is presented in Table 3.5.

Table 3.1 Region/Division/State wise Deployment of Rigs during AAP 2021-22

Region	Division	State	Number Of Rigs			
			DR	DTH	DUR	T
NWHR,Jammu	VIII,Jammu	Jammu & Kashmir		3	1	**4**
NWR, Chandigarh	II,Ambala	Punjab,Haryana and Chandigarh	2			**2**
SUO,Delhi		Delhi	1			**1**
WR,Jaipur	XI,Jodhpur	Rajasthan	2	3		**5**
WCR,Ahmedabad	I,Ahmedabad	Gujarat	4	2		**6**
NCR,Bhopal	XII,Bhopal	Madhya Pradesh		4		**4**
NCCR,Raipur	XIII,Raipur	Chhattisgarh		4		**4**
CR,Nagpur	VI,Nagpur	Maharashtra		5		**5**
NR,Lucknow	III,Varanasi	Eastern U.P	3	2		**5**
	XVI,Bareilly	Western U.P	3			**3**
MER,Patna	V,Ranchi	Bihar and Jharkhand	1	4		**5**
ER,Kolkatta	XV,Kolkata	West Bengal, Sikkim and Andaman & Nicobar	3	2		**5**

Region	Division	State	Number Of Rigs			
			DR	DTH	DUR	T
SER,Bhubaneswar	X,Bhubaneswar	Orissa	1	4		**5**
SR,Hyderabad	IX, Hyderabad	A.P, Telangana		5		**5**
SWR,Bangalore	XIV,Bangalore	Karnataka		4		**4**
SECR,Chennai	IV,Chennai	Tamil Nadu	1	4		**5**
KR,Trivendrum	IV,Chennai	Kerla		2		**2**
UR,Dehradun	XVI,Bareilly	Uttarakhand			1	**1**
NHR,Dharamshala	XVII,Dharamshala	Himachal Pradesh		2		**2**
Total			**24**	**52**	**2**	**78**

Abbreviation used
DR: Direct Rotary DUR: Dual Rotary DTH: Down the Hole

Table 3.2 State wise Wells constructed by CGWB during the Year 2021-22(till 31th Mar., 2022)

STATE	TARGET 2021-22				CUMULATIVE ACHIEVEMENT 2021-22 (SINCE APRIL 2021)			
	EW	OW	PZ	T	EW	OW	PZ	T
Gujarat	19	16	5	**40**	20	17	3	**40**
Daman &Div				**0**				**0**
Punjab			10	**10**			5	**5**
Haryana			10	**10**			10	**10**
Chandigarh				**0**				**0**
New Delhi			10	**10**	0	0	9	**9**
Uttar Pradesh	32	22		**54**	35	19	2	**56**
Tamil Nadu	17	12	50	**79**	26	2	53	**81**
Puducherry (UTP)				**0**				**0**
Kerala	15	8		**23**	17	5		**22**
Bihar	5	4		**9**	4	2	1	**7**
Jharkhand	21	9		**30**	22	10	1	**33**
Maharashtra	56	12		**68**	59	13		**72**
Dadra & Nagar Haveli				**0**				**0**
Assam	7	5	8	**20**	8	3	6	**17**
Arunachal Pradesh				**0**				**0**
Meghalaya	5	3		**8**	3	2		**5**
Manipur				**0**				**0**
Nagaland				**0**				**0**
Tripura				**0**				**0**
Jammu & Kashmir	14	10	6	**30**	13	3		**16**
Andhra Pradesh	34	16		**50**	46	13		**59**
Telangana			20	**20**			25	**25**

STATE	TARGET 2021-22				CUMULATIVE ACHIEVEMENT 2021-22 (SINCE APRIL 2021)			
	EW	OW	PZ	T	EW	OW	PZ	T
Odisha	30	18		**48**	38	11		**49**
Rajasthan	30	15	5	**50**	37	20		**57**
Madhya Pradesh	28	20		**48**	42	7		**49**
Chhatisgarh	30	15		**45**	35	10		**45**
Karnataka	30	9	15	**54**	20	11	15	**46**
Goa				**0**				**0**
West Bengal	17	11		**28**	18	11		**29**
Sikkim				**0**				**0**
Andaman & Nicobar Islands				**0**				**0**
Uttarakhand	2	1		**3**	1	1		**2**
Himachal Pradesh	8	2	6	**16**	8	2	5	**15**
TOTAL	**400**	**208**	**145**	**753**	**452**	**162**	**135**	**749**

Abbreviation used
EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table3.3(a) Division- wise wells constructed by Central Ground Water Board during 2021-22

DIVISION	TARGET 2021-22				CUMULATIVE ACHIEVEMENT 2021-22 (SINCE APRIL 2021)				Achievement (%)
	EW	OW	PZ	T	EW	OW	PZ	T	
I.Ahmedabad	19	16	5	**40**	20	17	3	**40**	**100.00%**
II.Ambala	0	0	30	**30**	0	0	24	**24**	**80.00%**
III.Varanasi	23	16	0	**39**	25	12	2	**39**	**100.00%**
IV.Chennai	24	17	50	**91**	35	3	53	**91**	**100.00%**
V.Ranchi	26	13	0	**39**	26	12	2	**40**	**102.56%**
VI.Nagpur	56	12	0	**68**	59	13	0	**72**	**105.88%**
VII.Guwahati	12	8	8	**28**	11	5	6	**22**	**78.57%**
VIII.Jammu	14	10	6	**30**	13	3	0	**16**	**53.33%**
IX.Hyderabad	34	16	20	**70**	46	13	25	**84**	**120.00%**
X. Bhubneshwar	30	18	0	**48**	38	11	0	**49**	**102.08%**
XI.Jodhpur	30	15	5	**50**	37	20	0	**57**	**114.00%**
XII.Bhopal	28	20	0	**48**	42	7	0	**49**	**102.08%**
XIII.Raipur	30	15	0	**45**	35	10	0	**45**	**100.00%**
XIV. Bangalore	38	12	15	**65**	28	15	15	**58**	**89.23%**
XV.Kolkata	17	11	0	**28**	18	11	0	**29**	**103.57%**
XVI.Bareilly	11	7	0	**18**	11	8	0	**19**	**105.56%**
XVII. Dharamshala	8	2	6	**16**	8	2	5	**15**	**93.75%**
TOTAL	**400**	**208**	**145**	**753**	**452**	**162**	**135**	**749**	**99.47%**

Abbreviation used
EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table3.3(b). Region - wise wells constructed by CGWB during 2021-22

REGION	TENTATIVE TARGET 2021-22				ACHIEVEMENT 2021-22 (01.04.21 TO 31.03.2022)				% ACHIEVEMENT
	EW	OW	PZ	T	EW	OW	PZ	T	
NWHR. Jammu	14	10	6	**30**	13	3	0	**16**	53.33%
NWR. Chandigarh	0	0	30	**30**	0	0	24	**24**	80.00%
WR. Jaipur	30	15	5	**50**	37	20	0	**57**	114.00%
WCR. Ahmedabad	19	16	5	**40**	20	17	3	**40**	100.00%
NCR. Bhopal	28	20	0	**48**	42	7	0	**49**	102.00%
NCCR. Raipur	30	15	0	**45**	35	10	0	**45**	100.00%
CR. Nagpur	56	12	0	**68**	59	13	0	**72**	105.88%
NR. Lucknow	32	22	0	**54**	35	19	2	**56**	103.70%
MER. Patna	26	13	0	**39**	26	12	2	**40**	102.56%
ER. Kolkata	17	11	0	**28**	18	11	0	**29**	103.57%
NER. Guwahati	12	8	8	**28**	11	5	6	**22**	78.57%
SER. Bhubaneshwar	30	18	0	**48**	38	11	0	**49**	102.08%
SR. Hyderabad	34	16	20	**70**	46	13	25	**84**	120.00%
SWR. Bangalore	30	9	15	**54**	20	11	15	**46**	85.19%
SECR. Chennai	17	12	50	**79**	46	2	53	**81**	102.53%
KR. Trivandrum	15	8	0	**23**	17	5	0	**22**	95.65%
UR. Dehradun	2	1	0	**3**	1	1	0	**2**	66.67%
NHR. Dharamsala	8	2	6	**16**	8	2	5	**15**	93.75%
TOTAL	**400**	**208**	**145**	**753**	**452**	**162**	**135**	**749**	**99.47%**

Abbreviation used

EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table 3.4: Status of Cumulative Boreholes Drilled by CGWB till 2021-22 in State's & UT's

S No.	State /UT	EW	OW	PZ	T	EW	OW	PZ	SH	DW	Total	TOTAL (I+II)
		(I) Through Outsourcing (Contractual)				(II) Through Departmental Rigs						
A.	**STATES**											
1	Andhra Pradesh	456	122	0	**578**	924	425	307	9	4	**1669**	**2247**
2	Arunachal Pradesh	0	0	0	**0**	46	10	0	1	1	**58**	**58**
3	Assam	17	20	0	**37**	455	214	65	16	42	**792**	**829**
4	Bihar	151	39	0	**190**	322	195	75	10	514	**1116**	**1306**
5	Chhattisgarh	414	15	105	**534**	886	285	161	0	28	**1360**	**1894**
6	Goa	13	1	0	**14**	58	18	14	0	31	**121**	**135**
7	Gujarat	392	116	0	**508**	1176	546	501	27	255	**2505**	**3013**
8	Haryana	75	55	80	**210**	410	273	239	23	170	**1115**	**1325**
9	Himachal Pradesh	0	0	0	**0**	272	49	10	1	0	**332**	**332**
10	Jammu & Kashmir	21	0	0	**21**	501	124	37	8	114	**784**	**805**
11	Jharkhand	275	28	0	**303**	496	249	49	4	71	**869**	**1172**
12	Karnataka	804	146	0	**950**	1631	738	369	7	5	**2750**	**3700**
13	Kerala	10	0	0	**10**	613	213	231	16	13	**1086**	**1096**
14	Madhya Pradesh	725	57	80	**862**	1433	747	176	8	149	**2513**	**3375**
15	Maharashtra	92	2	88	**182**	1792	571	167	2	166	**2698**	**2880**
16	Manipur	0	0	0	**0**	29	14	1	0	2	**46**	**46**
17	Meghalaya	0	0	0	**0**	120	40	2	2	8	**172**	**172**
18	Mizoram	0	0	0	**0**	3	3	0	0	0	**6**	**6**
19	Nagaland	0	0	0	**0**	15	6	1	0	3	**25**	**25**
20	Orissa	575	40	67	**682**	1751	441	151	21	191	**2555**	**3237**
21	Punjab	121	105	0	**226**	232	215	113	20	14	**594**	**820**
22	Rajasthan	809	195	0	**1004**	1488	558	573	93	591	**3303**	**4307**
23	Sikkim	0	0	0	**0**	31	9	0	0	0	**40**	**40**
24	Tamil Nadu	425	206	179	**810**	1427	451	331	13	93	**2315**	**3125**
25	Tripura	0	0	0	**0**	64	31	1	5	22	**123**	**123**
26	Telangana	373	78	0	**451**	796	534	534	5	27	**1896**	**2347**
27	Uttarakhand	28	12	0	**40**	75	7	3	1	129	**215**	**255**
28	Uttar Pradesh	694	274	0	**968**	1132	768	210	40	501	**2651**	**3619**
29	West Bengal	263	80	100	**443**	602	306	177	12	82	**1179**	**1622**
TOTAL(A)		**6733**	**1591**	**699**	**9023**	**18780**	**8040**	**4498**	**344**	**3226**	**34888**	**43911**

Table 3.4: Status of Cumulative Boreholes Drilled by CGWB till 2021-22 in State's & UT's

S No.	State /UT	EW	OW	PZ	T	EW	OW	PZ	SH	DW	Total	TOTAL (I+II)
		(I) Through Outsourcing (Contractual)				(II) Through Departmental Rigs						
B.	**UNION TERRITORIES**											
1	Andaman & Nicobar	0	0	0	**0**	46	13	0	1	0	**60**	**60**
2	Chandigarh	0	0	0	**0**	9	17	14	2	15	**57**	**57**
3	Dadra &NagarHaveli	0	0	0	**0**	14	1	0	0	0	**15**	**15**
4	Delhi	0	0	0	**0**	149	64	192	13	380	**798**	**798**
5	Daman & Diu	0	0	0	**0**	0	0	7	0	0	**7**	**7**
6	Pondicherry	0	0	0	**0**	30	20	8	5	14	**77**	**77**
TOTAL(B)		**0**	**0**	**0**	**0**	**248**	**115**	**221**	**21**	**409**	**1014**	**1014**
GRAND TOTAL(A+B)		**6733**	**1591**	**699**	**9023**	**19028**	**8155**	**4719**	**365**	**3635**	**35902**	**44925**

Abbreviation used

EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table 3.5 Division / State / Formation wise Achievement during 2021-22 (as on 31.03.2022)

DIVISION	STATE/ UT	HARD ROCK				ALLUVIUM				BOULDARY				TOTAL			
		EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T
I.AHMEDABAD	Gujarat	15	9	2	26	5	8	1	14				0	**20**	**17**	**3**	**40**
	Haryana				0			10	10				0	0	0	10	10
	Punjab				0			5	5				0	0	0	5	5
II.AMBALA	Delhi				0			9	9				0	0	0	9	9
III.VARANASI	Uttar Pradesh	19	6		25	6	6	2	14				0	25	12	2	39
	Tamil Nadu	14	1	53	68	4			4				0	18	1	53	72
IV.CHENNAI	Kerala	17	2		19				0				0	17	2	0	19
	Bihar	3	1	1	5	1	1		2				0	4	2	1	7
V.RANCHI	Jharkhand	22	10	1	33				0				0	22	10	1	33
VI.NAGPUR	Maharashtra	59	13		72				0				0	59	13	0	72
	Assam	4	2	6	12	7	3		10				0	11	5	6	22
	Arunachal Pradesh				0				0				0	0	0	0	0
	Meghalaya				0				0				0	0	0	0	0
VII.GUWAHATI	Tripura				0				0				0	0	0	0	0
VIII.JAMMU	Jammu & Kashmir	9	2		11				0	4	1		5	13	3	0	16
DIVISION	STATE/ UT	HARD ROCK				ALLUVIUM				BOULDARY				TOTAL			
		EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T
	Andhra Pradesh	46	13		59				0				0	46	13	0	59
IX.HYDERABAD	Telangana			25	25				0				0	0	0	25	25
X.BHUBANESWAR	Orissa	37	9		46	1	2		3				0	38	11	0	49
XI.JODHPUR	Rajasthan	31	14		45	6	6		12				0	37	20	0	57
XII.BHOPAL	Madhya Pradesh	42	7		49				0				0	42	7	0	49
XIII.RAIPUR	Chattisgarh	35	10		45				0				0	35	10	0	45
	Karnataka	20	11	15	46				0				0	20	11	15	46
XIV.BANGALORE	Kerala	8	4		12				0				0	8	4	0	12
XV.KOLKATTA	West Bengal	10	7		17	8	4		12				0	18	11	0	29
	Uttarkhand				0				0	1	1		2	1	1	0	2
XVI.BAREILLY	Uttar Pradesh				0	10	7		17				0	10	7	0	17
XVII.DHARAMSHALA	Himachal Pradesh	8	2	5	15				0				0	8	2	5	15
	TOTAL	**399**	**123**	**108**	**630**	**48**	**37**	**27**	**112**	**5**	**2**	**0**	**7**	**452**	**162**	**135**	**749**

Abbreviation used EW-Exploratory Well OW-Observation Well PZ-Piezometers

Table 3.6: Division/ State /Head wise Achievement in Normal/ Tribal/ Drought areas during 2021-22 (as on 31.03.2022)

DIVISION	STATE/ UT	NORMAL				TRIBAL				DROUGHT				TOTAL			
		EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T	EW	OW	PZ	T
I.AHMEDABAD	Gujarat	20	17	3	**40**				**0**				**0**	20	17	3	40
	Haryana			10	**10**				**0**				**0**	0	0	10	10
	Punjab			5	**5**				**0**				**0**	0	0	5	5
II.AMBALA	Delhi			9	**9**				**0**				**0**	0	0	9	9
III.VARANASI	Uttar Pradesh	6	6	2	**14**				**0**	19	6		**25**	25	12	2	39
	Tamil Nadu	26	2	53	**81**				**0**				**0**	26	2	53	81
IV.CHENNAI	Kerala	9	1		**10**				**0**				**0**	9	1	0	10
	Bihar	4	2	1	**7**				**0**				**0**	4	2	1	7
V.RANCHI	Jharkhand	22	10	1	**33**				**0**				**0**	22	10	1	33
VI.NAGPUR	Maharashtra	59	13		**72**				**0**				**0**	59	13	0	72
	Assam	11	5	6	**22**				**0**				**0**	11	5	6	22
	Arunachal Pradesh				**0**				**0**				**0**	0	0	0	0
	Meghalaya				**0**				**0**				**0**	0	0	0	0
VII.GUWAHATI	Tripura				**0**				**0**				**0**	0	0	0	0
VIII.JAMMU	Jammu & Kashmir	13	3		**16**				**0**				**0**	13	3	0	16
	Andhra Pradesh	46	13		**59**				**0**				**0**	46	13	0	59
IX.HYDERABAD	Telangana			25	**25**				**0**				**0**	0	0	25	25
X.BHUBANESWAR	Orissa	19	6		**25**	19	5		**24**				**0**	38	11	0	49
XI.JODHPUR	Rajasthan				**0**	31	14		**45**	6	6		**12**	37	20	0	57
XII.BHOPAL	Madhya Pradesh	42	7		**49**				**0**				**0**	42	7	0	49
XIII.RAIPUR	Chattisgarh	35	10		**45**				**0**				**0**	35	10	0	45
	Karnataka	20	11	15	**46**				**0**				**0**	20	11	15	46
XIV.BANGALORE	Kerala	8	4		**12**				**0**				**0**	8	4	0	12
XV.KOLKATTA	West Bengal	18	11		**29**				**0**				**0**	18	11	0	29
	Uttarkhand	1	1		**2**				**0**				**0**	1	1	0	2
XVI.BAREILLY	Uttar Pradesh	10	7		**17**				**0**				**0**	10	7	0	17
XVII.DHARAMSHALA	Himachal Pradesh	8	2	5	**15**				**0**				**0**	8	2	5	15
	TOTAL	**377**	**131**	**135**	**643**	**50**	**19**	**0**	**69**	**25**	**12**	**0**	**37**	**452**	**162**	**135**	**749**

Abbreviation used EW-Exploratory Well OW-Observation Well PZ-Piezometers

3.1 Well Development/Pumping Test

Once constructed, a tube-well is developed to increase its specific capacity to prevent sand rushing into the well and to obtain maximum well life. Thereafter, pumping tests are conducted for evaluating aquifer parameters i.e. Transmissivity, storage co-efficient and well parameters *viz.* specific capacity and well efficiency, with a view to evolve efficient design for tube wells, assessment of yield capabilities and spacing criteria for tube wells. Total of 52 tubewells were developed and tested during the year 2021-22 (Table 3.8).

Table 3.7 Status of Development & Pumping Test of Wells by CGWB

DIVISION	STATE	No. of Wells developed and tested during the year 2021-22	Balance No. of wells to be tested (Backlog)
I.Ahmedabad	Gujarat	0	43
II.Ambala	Haryana	0	6
	Punjab	1	18
	Delhi	0	0
	Chandigarh	0	2
III.Varanasi	Uttar Pradesh	0	118
IV.Chennai	Tamilnadu	2	13
	Kerala	0	0
V.Ranchi	Bihar	0	0
	Jharkhand	0	0
VI.Nagpur	Maharashtra	6	34
VII.Guwahati	Assam	0	53
	Meghalya	0	20
	Tripura	0	4
	Arunachal Pradesh	0	5
VIII.Jammu	Jammu & Kashmir	4	16
IX.Hyderabad	Andhra Pradesh	8	6
	Telangana	0	0

DIVISION	STATE	No. of Wells developed and tested during the year 2021-22	Balance No. of wells to be tested (Backlog)
X.Bhubneshwar	Orissa	6	23
XI.Jodhpur	Rajasthan	6	26
XII.Bhopal	Madhya Pradesh	0	19
XIII.Raipur	Chhattisgarh	0	34
XIV. Bangalore	Karnatka	4	23
	Kerala	2	4
XV.Kolkata	West Bangal	7	54
XVI.Bareilly	Uttar Pradesh	2	43
	Uttranchal	0	0
XVII. Dharamshala	Himachal Pradesh	4	1
	TOTAL	**52**	**565**

3.2 Handing Over/Taking Over Of Exploratory Wells

The exploratory drilling sites are selected in consultation with the State Government Departments considering that successful exploratory wells on completion of scientific studies can be converted into production wells once taken over by States for water supply.

Till March, 2022, a total of 19028 exploration wells have been drilled by the Board. Out of a total of 15092 successful wells having adequate ground water discharge, 14532 exploratory wells have been handed over to the State agencies while 560wells are yet to be handed over after the completion of scientific studies by the Board. The status of handing over of exploratory wells drilled by Central Ground Water Board to the State Government as on 31st March, 2022 is indicated in Table 3.9.

Table 3.8 Status of Handing over of Wells drilled by CGWB in 2021-22 (as on March 2022)

S.No.	States	Total Wells drilled (EW)	No. of Successful Wells (EW)	No. of Wells Handed Over to the user agencies
1	Andhra Pradesh	924	625	618
2	Arunachal Pradesh	46	41	41
3	Assam	455	428	421
4	Bihar	322	280	257
5	Chhattisgarh	886	810	783
6	Goa	58	49	49
7	Gujrat	1176	823	808
8	Haryana	410	236	221
9	Himachal Pradesh	272	231	224
10	Jammu & Kashmir	501	377	345
11	Jharkhand	496	370	298
12	Karnataka	1631	1398	1341
13	Kerala	613	453	444
14	Madhya Pradesh	1433	1058	1032
15	Maharashtra	1792	1535	1494
16	Manipur	29	21	21
17	Meghalaya	120	116	116
18	Mizoram	3	3	3
19	Nagaland	15	9	9
20	Orissa	1751	1650	1639
21	Punjab	232	205	189
22	Rajasthan	1488	1102	1068
23	Sikkim	31	10	10
24	Tamil Nadu	1427	1117	1093
25	Tripura	64	62	62
26	Telangana	796	613	607
27	Uttrakhand	75	66	37
28	Uttar Pradesh	1132	926	864
29	West Bengal	602	305	292
TOTAL(A)		**18780**	**14919**	**14386**
1	Andaman & Nicobar	46	12	10
2	Chandigarh	9	9	7
3	Dadra &Nagar Haveli	14	8	8
4	Delhi	149	131	108
	Daman & Diu	0	0	0
5	Pondicherry	30	13	13
Total(B)		**248**	**173**	**146**
Total(A+B)		**19028**	**15092**	**14532**

Abbreviation used EW-Exploratory Well

3.3. High Yielding Wells

Ground Water Exploration is carried to study the sub-surface hydrogeological set up of the aquifer to evaluate aquifer parameters. During 2021-22, CGWB under its scientific exploratory drilling programme has explored high yielding aquifers in various parts of the Country based on hydrogeological studies coupled with remote sensing and geophysical techniques. Such studies will help in identifying similar ground water sources in other parts of the state having similar hydrogeological conditions and in guiding the State agencies to adopt the follow up action with regard to ground water development for drinking water supply and meeting other demands. High Yielding Wells explored during 2019-20are indicatedinTable 3.10 below.During the year 2021-22, the Board has constructed 749 wells (452 Exploratory Wells (EW), 162 Observation Wells (OW) and 135Piezometers (PZ)to assess the ground water potential in different hydrogeological set-up of the country.

Table 3.9 State wise details of high yielding wells constructed during the year 2021-22

S No.	Name of State	Location	Yield in (LPS)	Remark
1	MP	Police Line II	35.3	
		ShahpuraRundi	22.9	
		Raghunathpur	18	
		Basai	44.5	
2	Kerala	Balussery	14	
		Peruvannamoozhi	10	
		Munderi- Kalpetta	16.44	
3	Karnataka	Hunsur	5.5	
		Gulagnjanahalli	4.55	
		Bangalore University	6.44	
4	Himachal Pradesh	Basantpur	15	
		Minjgram	9	
5	Odisa	Palasole	20	
		Palasole	10	
		Sabideipur	8	
		Lakhanpur	9.28	
6	J&K	Daraba	10.23	
		Panasa	3.70	
		Hasplote	07	
7	Gujrat	Kamlej	13.97	
		Kamlej	13.84	
		Valasan	20.56	
		Valasan	21	

S No.	Name of State	Location	Yield in (LPS)	Remark
		Segva	22	
		Segva	32.66	
		DhoranPardi	20.29	
		DhoranPardi	14.24	
		Fulsar	10	
		Fulsar	12.5	
		Chhaya	6.9	
		Chhaya	5.5	
		Kharakadi	6.13	
		Panchmahal	15	
		Panchmahal	18.35	
8	Andhra Pradesh	Yerraguntla	9.86	
		Mallela	11.72	
		Gattupalle	9.86	
9	Maharashtra	Vihirgaon	8.79	
		Vihirgaon	9.22	
		Suleran	22.1	
		Tamnakwada	14.34	
		Tamnakwada	12.5	
		Hingalwadi Bk	8.76	
		Nakoda	9.05	
		Dhanewadi	10.32	
		Khirwade	26.04	
		Nelkaranji	24.98	
		Nelkaranji	25.01	
		Dahegaon	16.28	
		Dahegaon	16.28	
10	Himachal	Basantpur	15	
		Minjgram	9	
11	Bihar	Tankupa,	6	
	Jharkhand	Sarpatia, Japla, Golwadhab, Koderma, Rahe, Khagradih, Ranchi, Nil, Kailatarn, Giridih	6-21	
12	Assam	Barpeta, Johrat and Darrang, Majuli,	3.4-6.3	
13	Meghalaya	Mendipathar,	4.73	
14	UP	Pipri Saidpur	18.5	
		Rehera	25.68	
		Adampur	17.9	
		Bansa	18.5	
		Husepur Luckman	25.68	
		KrishiVigyan Kendra	22.30	
		Dalelnagar	22	

S No.	Name of State	Location	Yield in (LPS)	Remark
14	UP	Bhadeicha	34.50	
		PahariKolBasti	3	
		Khariatkhurd	4	
		Bhatwari	3.1	
		Hateda	22.5	
		Khimpur	25.8	
		Dumardiha	6.2	
15	Rajasthan	Police Line Dungarpur	3.33	
		33LLW	13.33	
		Ummewala	14.16	
		Rakho	13.33	
		Shivpur	18.33	
		Ummewala	8	
		33 SSW	25	
		Dudhia	11.16	
		SALIWAL	20	
		3 SSW	20	
		Sinala	4.16	
		Shahjipura	13.33	
		Modawas	4	
		BHAGATPURA	21	
		Shahjipura	13.33	

High yielding well at Anjora, Amgaon taluka, Gondia district:
Total Depth-200mbgl
Discharge- 12.18 lps (43, 848 liters per hour)
Aquifer- fractured Granite Gneiss
Potential aquifer Zones: 40-44, 105-108 and 179-180 mbgl.
EC:719

High yielding well at Kanheriram, SadakArjuni taluka, Gondia district:
Total Depth-123.30mbgl
Discharge- 12.18 lps (43, 848 liters per hour)
Aquifer- fractured Granite Gneiss
Potential aquifer Zones: 31-34 and 121-123 mbgl.
EC: 526 µS/cm.

High yielding well at Itkheda, ArjuniMorgaon taluka, Gondia district:
Total Depth-200mbgl
Discharge- 7.24 lps (26,064 liters per hour)
Aquifer- fractured Granite Gneiss
Potential aquifer Zones: 87-89 and 157-159 mbgl.
EC: 701 µS/cm.

High yielding well at Hinigalwadi Bk, Taluka kalwan, district-Nashik
Total Depth-200mbgl
Discharge- 4.43lps (15,948 liters per hour)
Aquifer- fractured Massive Basalt
Potential aquifer Zones: 35-38.10, 53.30-56.40 mbgl
EC: 933 µS/cm.

High yielding well at Jamb, Taluka and District Yavatmal
Total Depth-135.80mbgl
Discharge- 4.43 lps (21,384 liters per hour)
Aquifer- fractured Granite Gneiss
Potential aquifer Zones: 20.00 to 22.50 and 45.00 to 45.50 mbgl
EC: 701 µS/cm.

Fig.3.4: High Yielding Well from Mahrashtra State

4.0 GEOPHYSICAL STUDIES

Geophysical investigations are used for exploration of groundwater and in delineating the underground structures which control the occurrence, distribution and movement of ground water. Application of geophysical techniques for ground water investigations on regular basis commenced in CGWB during the seventies. The Board has made extensive use of both the surface and the subsurface (well logging) geophysical techniques in the search of groundwater and proper construction of water wells. The findings of the geophysical studies, as a practice, are combined with the hydrogeological and geomorphologic investigations to place them on firm footing. The techniques have become an integral part of the ground water exploration programme.

The Borehole geophysics is used in groundwater to obtain information pertaining to lithology, fractures, permeability, porosity and water quality so as to delineate subsurface disposition of aquifers. Borehole-geophysical logging determines the character and thickness of the different lithological/Hydro-geologic units in drilled pilot boreholes. Saline / brackish water bearing aquifers are present in different parts of India. Fresh water bearing aquifers are often intervened by the saline water aquifers. Such information is essential for proper placement of casing and screens in water-supply wells and for characterizing and remediation of problems related to ground-water salinity. The proper positioning and condition of casing and screen pipes in a well can be rapidly evaluated with geophysical borehole logging.

Surface geophysical surveys specially the traditional Electrical Resistivity survey in soft and hard rock formations are commonly employed in CGWB to delineate the ground water bearing zones/structures, pin-pointing sites for construction of boreholes and providing inputs for formulating proposals for constructing artificial recharge structures. Geophysical survey has also been conducted for delineating the bedrock topography and sandy horizon of non – perennial channel. In recent times, Transient-Electromagnetic techniques (TEM) are also being used for identifying the sub-surface layers parameters in term of Resistivity and thickness as is done through Electricalresistivity surveys. TEM surveys, however, takes less time in comparison tothe conventional Electrical resitivitysurvey. Imaging Resistivity 2-D survey at present is conducting by CGWB by Multi Electrodes Resistivity Meter. Various other techniques like Self Potential, Induced Polarization, Mise-a-la-masse of electrical method, refraction seismic, electromagnetic – the Horizontal Loop, Very Low Frequency (VLF) *and* magnetic were also conducted by CGWB in past. The Heliborne Survey has alsobeenincorporated through pilot project of Aquifer mapping and identification of palaeo-channel in parts of Prayagraj and Kaushambi district in collaboration with CSIR-NGRI. Apart from these, resistivity survey (VES) were also carried out during the year for short-term water supply investigation on request of other Government organization and Public Sector Undertakings. Regionwise details of availability of Geophysical instruments in CGWB are presented below in table 4.1

Sl. No	Region	Geophysical Logger	Resistivity Meter	EM
1	ER-Kolkata	Uptron (18Y) Logger u-1	1.CRM Auto –C (7 y) 2. Res Meter(2 y)	TEM – 1 (New)
2	KR- Trivandrum	Nil	1. Terrameter (21 Y) 2. CRM-500 (15Y)	TEM – 1 (New)
3	NCCR- Raipur	Nil	SSR-MP-ATS (adv) (7Y)	TEM – 1 (New)
4	NCR - Bhopal	Nil	1. DDR-4 MP (20Y) 2. SSR-MP1 (18Y)	
5	WCR- Ahmadabad	Uptron (25 y) Logger	1.SSR-1 (7Y) 2.DDR-4 (25Y)	
6	NWHR- Jammu	Uptron (22 y) Logger	1.DDR-4 (18 Y) 2. Terrameter (18Y)	
7	NHR- Dharmshala	Nil		
8	UR - Dehradun	Nil	1.Terrameter SAS 300 (21Y)	
9	NER-Guwahati	Uptron (>27Y) Logger (+Gamma)	1. SSR-MP1 (13Y) 2. Terrameter (18 Y)	
10	CR- Nagpur	1.Uptron (27Y) Logger (+ Gamma), 2. Digital (25Y)Portable Logger (+ Gamma, Flow meter &Temp.) 3.Terrameter spot Logger (part of Terrameter)	1.Terrameter (19Y) 2.CRM – 20 (15Y)	
11	MER - Patna	Uptron (> 20 Y) Logger	1. CRM 500 – 40 W (Ranchi) (15Y) 2. CRM-Auto C – 40 W (5Y)	
12	SER - Bhubaneswar	Uptron (> 15 y) Logger	1. CRM 20 (5Y)	
13	NWR - Chandigarh	Uptron (27Y) Logger (+Gamma)	1. CRM – 500 (20Y)	
14	SR - Hyderabad	1.Upron (3no and 25 y) Logger (+Gamma)	1. DDR- 4–80 W (21Y) 2. SSR- MP1–80W (7Y) 3. DDR- 3 – 80 W (7Y) 4. Terrameter with manual logger (22Y) 5. Terrameter LS (6Y) 6. ERT System Terrameter LS–2D (6)	

Sl. No	Region	Geophysical Logger	Resistivity Meter	EM
15	SWR- Bangalore	Uptron (20Y) Logger(+Gamma)	1. SSR-MP-ATS - 100W – 2 no. (2Y) 2. Res Meter (5Y)	
16	SECR-Chennai	Uptron (19 Y) Logger	1. DDR – 3= (7Y) 2. CRM – 500- (15Y) 3. Adv WTS – 2 No (5Y)	
17	WR - Jaipur	Uptron (27Y) Logger (+Gamma), u-13	1. DDR- 4 – 80 W (19 Y) 2. CRM-Auto-C – 40 W (7 Y)	
18	NR - Lucknow	Uptron (21Y) Logger(+Gamma), u-14	1.Adv. Terrameter LS – 2D (6.5Y) 2.Adv Syscal R2 – 250 W (25 Y) 3. ERT system 4.Terrameter LS – 2D	
	Total Logger (17 No.), two spot logger with resistivity meter	1.Uptron - 14, (+8 gamma) (8 working) 2.OYO potable-1 (+gamma, flow & Temp). (Not working) 3.SAS log 200 – 2 (part of Terrameter), One is working	1. 33 working , 8 Repairable 2. Imaging System-2 no 3. TEM- 9 no	

Table 4.2 Region wise progress of Geophysical Activities for AAP 2021-22

	State/UT	VES		TEM		Total		
S.No.	State /UT	Target	Achievement	Target	Achievement	Target	Achievement	Achievement %
1	West Bengal	10	14	150	130	160	144	90
2	Sikkim	0	0	0	0	0	0	
3	Andaman and Nicobar Islands	0	0	0	0	0	0	
4	Bihar	25	0	150	152	175	152	87
5	Jharkhand	25	26	0	0	25	26	104
6	Chhattisgarh	50	50	300	304	350	354	101
7	NER	0	31	0	0	0	31	
8	Odisha	150	152	0	0	150	152	101
9	Uttar Pradesh	30	31	300	302	330	333	101
10	Uttarakhand	50	50	0	0	50	50	100
11	Delhi	0	0	0	0	0	0	
12	Maharashtra	75	76	0	0	75	76	101
13	Dadra and Nagar Haveli	0	0	0	0	0	0	
14	Tamilnadu	75	75	400	403	475	478	101
15	Puducherry	0	0	0	0	0	0	
16	Karnataka	70	74	210	214	280	288	103
17	Goa	0	0	0	0	0	0	
18	Kerala	25	25	250	253	275	278	101
19	Lakshadweep	0	0	0	0	0	0	
20	Andhra Pradesh	240	235	115	81	355	316	89
21	Telangana	0	6	35	80	35	86	246
22	Gujarat	0	20	0	0	0	20	
23	Daman and Diu	0	0	0	0	0	0	
24	Madhya Pradesh	75	56	0	0	75	56	75
25	Himachal Pradesh	0	0	0	0	0	0	
26	Jammu and Kashmir	75	53	0	0	75	53	71
27	Chandigarh	0	0	0	0	0	0	
28	Haryana	75	68	0	0	75	68	91
29	Punjab	75	83	0	0	75	83	111
30	Rajasthan	75	78	300	311	375	389	104
	Total	**1200**	**1203**	**2210**	**2230**	**3410**	**3433**	**101**

S.No.	Region	State /UT	Achevement (Logging)
1	ER, Kolkata	West Bengal	**3**
9	NR, Lucknow	Uttar Pradesh	**17**
14	SECR, Chennai	Tamilnadu	**1**
28	NWR, Chandigarh	Haryana	**5**
29	NWR, Chandigarh	Punjab	**3**
Grand Total			**29**

5.0 GROUND WATER LEVEL MONITORING

Introduction

Ground water regime monitoring is one of the key activities of Central Ground Water Board (CGWB). Monitoring of ground water regime is an effort to obtain information on ground water level and chemical quality through representative sampling. The primary objective of ground water monitoring by, is to record the response of various natural and anthropogenic stress on the groundwater regime which impacts the recharge and discharge parameters with reference to geology, climate, physiography, land use pattern and hydrologic characteristics on a regional scale. The key applications of the groundwater data acquired during regime monitoring in the country are

- The data is used for volumetric estimation of groundwater resources.
- To assess the impact of groundwater recharge and draft on long term basis.
- To categories area into overexploited, critical, semi-critical and safe based on long term water level trend analysis.
- In deciding the depth of water well drilling as well as depth of lowering the pumps.
- Help in design, implement and monitor the effectiveness of groundwater management, protection and conservation program
- To identify groundwater quality affected areas for taking remedial measures.
- To plan groundwater recharge interventions and study the impact in time and space.

At present, CGWB has a network of **23209** (Dug Wells: **16219**, Piezometers: **6338,** Hand Pump: **611,** spring: **41**) ground water observation wells throughout the Country**.** The state wise breakup is given below in table 5.1 .

Sl No	Name of the State/UTs	Number of GW Monitoring Stations (March 2022)				
		DW (Dug Well)	PZ (Piezometer)	HP (Handpump)	SP (Spring)	Total
1	Andhra Pradesh	656	316			**972**
2	Arunachal Pradesh	26	4			**30**
3	Assam	355	59			**414**
4	Bihar	710	40			**750**
5	Chhattisgarh	1122	173			**1295**
6	Delhi	23	106	**0**	0	**129**
7	Goa	88	44			**132**
8	Gujarat	836	264	0	0	**1100**
9	Haryana (CGWB Wells)	197	341	0	0	**538**
	Haryana (State Wells)	327	432			**759**
10	Himachal Pradesh	136	4	0	0	**140**
11	Jammu & Kashmir	287	14			**301**
12	Jharkhand	435	17			**452**
13	Karnataka	1413	262			**1675**
14	Kerala	1374	217			**1591**
15	Madhya Pradesh	1182	300			**1482**
16	Maharashtra	1716	207			**1923**
17	Manipur					
18	Meghalaya	60	10			**70**
19	Nagaland	22	8			**30**

Sl No	Name of the State/UTs	Number of GW Monitoring Stations (March 2022)				
		DW (Dug Well)	PZ (Piezometer)	HP (Handpump)	SP (Spring)	Total
20	Odisha	1501	122			**1623**
21	Punjab (CGWB wells)	146	342	0	0	**488**
	Punjab (State wells)	51	631	0	0	**682**
22	Rajasthan	708	558			**1266**
23	Tamil Nadu	793	586			**1379**
24	Telangana	275	610			**885**
25	Tripura	100	15			**115**
26	Uttar Pradesh	797	210			**1007**
27	Uttarakhand	38	12	126	41	**217**
28	West Bengal	686	397	485		**1568**
UT s						
1	Andaman & Nicobar	111	2			**113**
2	Chandigarh	1	25	0	0	**26**
3	Dadra & Nagar Haveli	17	0			**17**
4	Daman & Diu	21	3	0	0	**24**
5	Pondicherry	9	7			**16**
TOTAL		**16219**	**6338**	**611**	**41**	**23209**

The ground water levels are measured manually four times a year during the months of January, March/April/ May, August and November coinciding with crop season and onset of monsoon as well as to capture deepest and shallowest water level in the hydrological year. The ground water regime monitoring started in the year 1969 by Central Ground Water Board.

Ground water samples are collected once a year during the month of March/April/May to obtain background information of ground water quality changes on regional scale.

The database thus generated forms the basis for planning the ground water development and management programme. This data is used for assessment of ground water resources and changes in the regime consequent to various development and management activities.

5.1 Ground Water Scenario During Premonsoon 2021

The ground water level could not be carried out in time in the month of May 2021, due to the lockdown imposed in various parts of the country due to the outbreak of Covid-19 pandemic. In some states monitoring is done, but all stations are not covered. Due to insufficiency of data the water level analysis for the pre-monsoon season was not carried out.

5.2 Ground Water Scenario During August 2021

The ground water level data for August 2021**(Annexure-I)** indicates that, out of the 14198 wells analysed, 4722 (33.3 %) wells are showing water level less than 2 m bgl (metres below ground level), 4946(34.8%) wells are showing water level in the depth range of 2-5 m bgl, 2724 (19.2%) wells are showing water level in the depth range of 5-10 m bgl, 1108 (7.8%) wells are showing water level in the depth range of 10-20 m bgl, 474 (3.3%) wells are showing water level in the depth range of 20-40 m bgl and the remaining 277 (2%) wells are showing water level more than 40 m bgl,.The map of depth to water levels during August 2021 is shown in **Plate-VI**.

The general depth to water level of the country ranges from 0 to 5 m bgl. Almost 68 % of the wells analysed show water level in the range of 0- 5 m bgl.Very shallow water level of less than 2 m bgl is observed in considerable number of wells in almost all the states, such as except Chandigarh, Daman & Diu, Haryana, Punjab, and Tamil Nadu.Water level in the range of 2 –5 m bgl, is observed in almost all states except Haryana, Delhi, Punjab and Rajasthan, where less than 15 % of wells have water level of 2-5 mbgl..In major parts of north-western and western states, depth to water level is generally deeper and ranges from about 10- 40 m bgl. In parts Delhi, Haryana and Rajasthan, water level of more than 40 m bgl is also recorded.The maximum depth to water level of 116.93 m bgl is observed at Deshnokh in Bikaner dist. of Rajasthan, whereas the minimum is 1.04 m aglat Kanchannagar, in Surendranagar dist. of Gujarat.

Fig 5.1: No. of wells showing Depth to Water Level (mbgl), August 2021

5.3 Ground Water Scenario During November 2021

The ground water level data for November 2021 indicates that out of the total 14470 wells analysed, 4191 (29 %) wells are showing water level less than 2 m bgl,5755(39.8%) wells are showing water level in the depth range of 2-5 m bgl, 2923 (20.2%) wells are showing water level in the depth range of 5-10 m bgl, 1011 (7%) wells are showing water level in the depth range of 10-20 m bgl, 384 (2.7%) wells are showing water level in the depth range of 20-40 m bgl and the remaining 206(1.4 %) wells are showing water level more than 40 m bgl.

In general, depth to water level ranges from 2 to 5 m bgl as observed at about 69% of the monitoring stations. Very shallow water level of less than 2 m bgl is observed in all the states, except Chandigarh and Nagaland. Ground Water level in the range of 2-5 m bgl is predominant in the entire Country, as major part of the country shows water level in the range 2-5 m bgl. In parts of north-western and western states, especially in the states/UTs of Chandigarh, Delhi, Haryana, Punjab and Rajasthan, depth to water level is generally deeper and ranges from about 10m bgl to more than 40 m bglThe peninsular part of country recorded a water level in the range of 2 to 10 m bgl. The maximum depth to water level of 116.98 m bgl is observed in Bikaner district of Rajasthan whereas the minimum is less than 1 m bgl.**(Plate-VII, *Annexure- II*)**

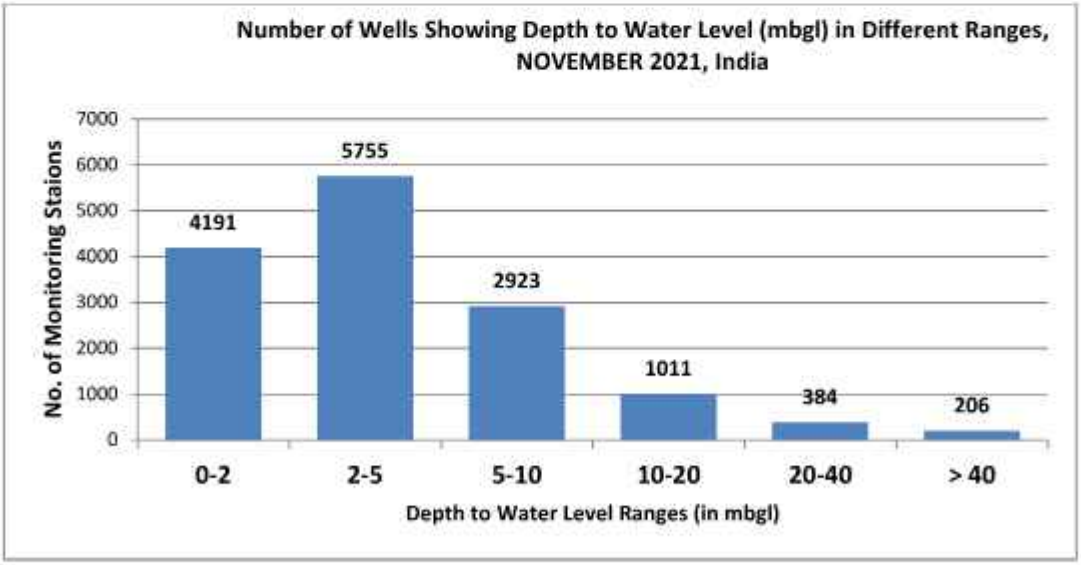


Fig 5.2: No. of wells showing Depth to Water Level (mbgl), November 2021

5.4 Ground Water Scenario During January 2022

The ground water level data for January 2022 indicates that out of the total 14249 wells analysed, 2605 (18.3%) wells are showing water level less than 2 m bgl (metres below ground level), 5891 (41.3%) wells are showing water level in the depth range of 2-5 m bgl, 3745 (26.3 %) wells are showing water level in the depth range of 5-10 m bgl, 1343 (9.4%) wells are showing water level in the depth range of 10-20 m bgl, 451 (3%) wells are showing water level in the depth range of 20-40 m bgl and the remaining 214 (1.5 %) wells are showing water level more than 40 m bgl. The maximum depth to water level of 129.2 m bgl is observed at Khara in Jodhpur district of Rajasthan, whereas the minimum is less than 1 m bgl.

The depth to water level map of January 2022for the country indicates that in general depth to water level ranges from 2 to 10 m bgl as observed at about 68% of the monitoring stations. Shallow water level of less than 2 m bgl is observed in very small and isolated pockets in the states of Assam, Odisha, Andhra Pradesh, Tamil Nadu, Konkan areas of Maharashtra and northern parts of Uttar Pradesh and Bihar.Water level in the range of 5 to 10 m bgl is prominent in all the states of Central and western parts of Karnataka and Kerala in Southern India. In major parts of north-western states depth to water level generally ranges from 10-40 m bgl. Water level of more than 40 m bgl is also prevalent in the north western part of the country. In some parts of Haryana, and Delhi and almost major parts of Rajasthan, water level of more than 40 m bgl is recorded.

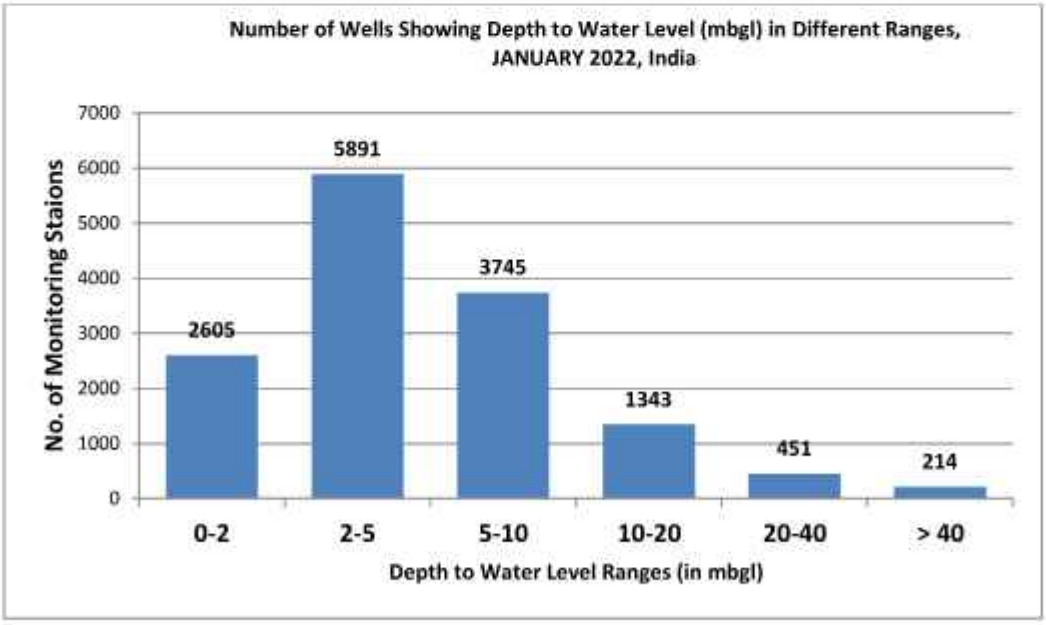
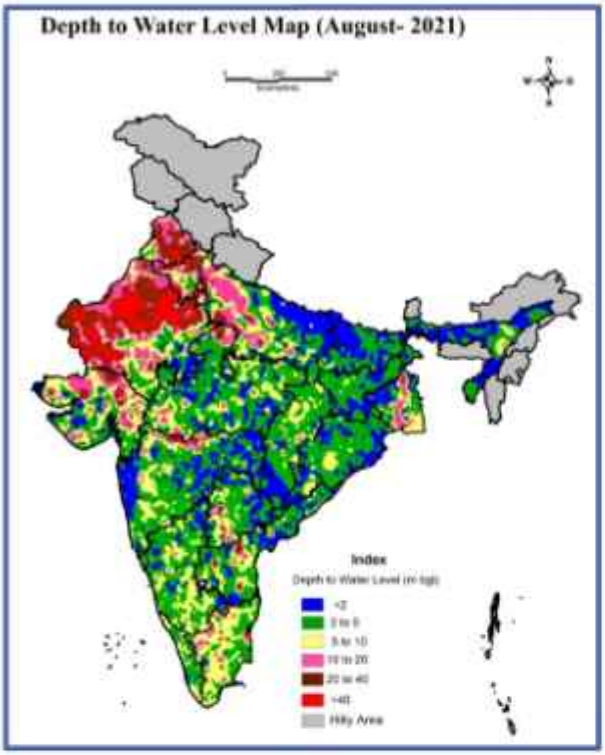


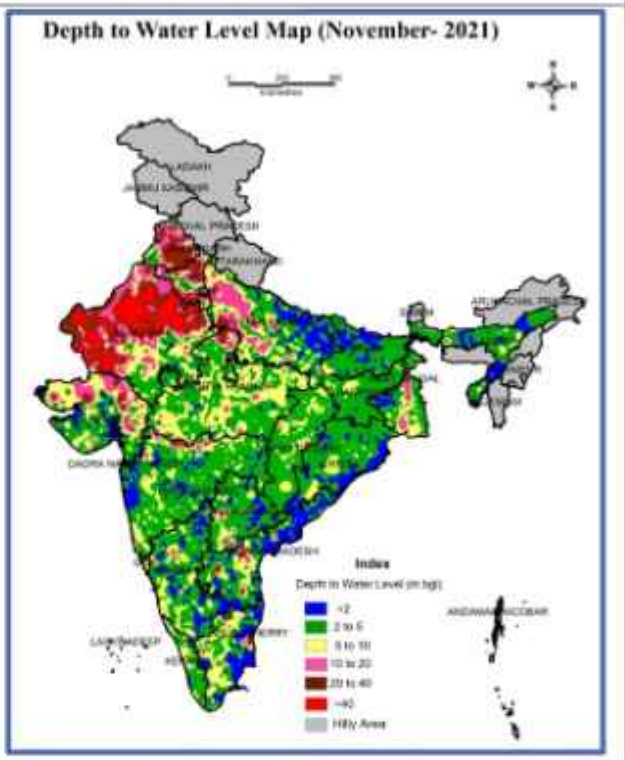
Fig 5.3: No. of wells showing Depth to Water Level (mbgl), January 2022

DEPTH TO WATER LEVEL AT A GLANCE

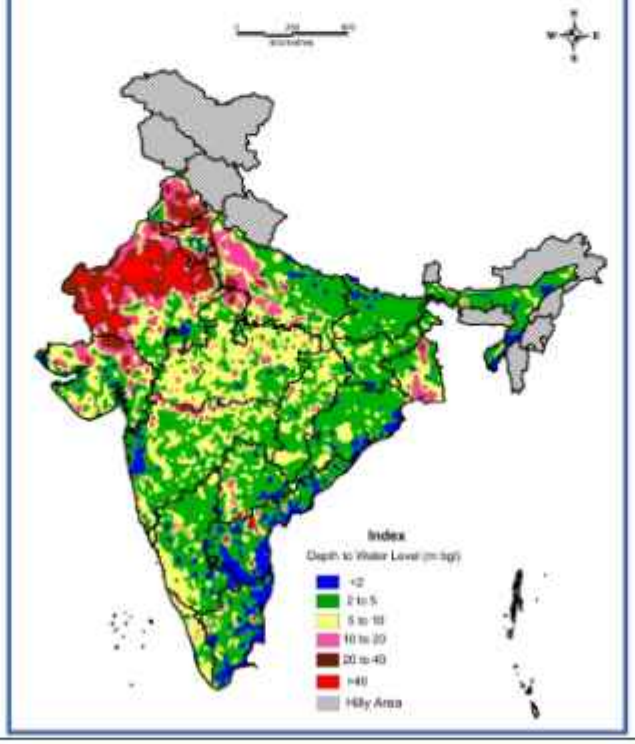
Depth to Water Level Map (August- 2021)



Depth to Water Level Map (November- 2021)



Depth to Water Level Map (January - 2022)



6.0 NATIONAL AQUIFER MAPPING & MANAGMEMENT PROGRAMME (NAQUIM)

6.1 Objectives

The National Aquifer Mapping and Management Programme (NAQUIM) was initiated and implemented by Central Ground Water Board under the scheme of Ground Water Management & Regulation during XII plan (2012-17). The Programme was designed as a follow-up of the recommendations of four Working Groups of the erstwhile Planning Commission namely "Sustainable Ground Water Development and Management", "Minor Irrigation and Watershed Development" , "Urban and Industrial Water Supply and Sanitation "and "Data Base Development and Management".

With an aim to support effective management of ground water resources in the country, the NAQUIM programme was taken up with the following objectives

1. Delineation of aquifers in terms of their horizontal and vertical extent.
2. Characterization of aquifers in terms of their hydrogeological roperties, resource availability and quality
3. Preparation of aquifer maps
4. Developing management strategies to ensure sustainability of ground water resources.

NAQUIM coverage

- Total Area: **33 lakh sq km**
- Total Mappable Area: ~ **25 lakh sq km**
- Target for Coverage till 31st March 2022: **20.83 lakh sq km**
- Achievement: **21.15 lakh sq km**
- Total States/UTs :**36**
- States/UTs covered: **20**

STATES/UTs FULLY COVERED	STATES/UTs TO BE COVERED IN 2022-23
Chandigarh UT	Andaman & Nicobar UT
Dadra & Nagar Haveli and Daman and Diu	Andhra Pradesh
Delhi	Arunachal Pradesh
Goa	Assam
Haryana	Bihar
Himachal Pradesh	Chhattisgarh
Jammu & Kashmir UT	Gujarat
Kerala	Jharkhand
Lakshdweep UT	Karnataka
Ladakh UT	Madhya Pradesh
Manipur	Maharashtra
Meghalaya	Odisha
Mizoram	Rajasthan
Nagaland	Sikkim
Pudducherry UT	Uttar Pradesh
Punjab	West Bengal
Tamil Nadu	
Telangana	
Tripura	
Uttarakhand	

NAQUIM COVERAGE till 20 Mar. 2022 (in Sq. Km.)						
Sl. No.	Name of the State/ UT	Total Area	Total Mappable area	Coverage till Mar. 2021	Since April 2021	Coverage till Mar. 2022
1	Andaman & Nicobar UT	8249	1774	800	396	1196
2	Andhra Pradesh	163900	141784	76285	26778	103063
3	Arunachal Pradesh	83743	4703	4048	0	4048
4	Assam	78438	61826	25752	13054	38806
5	Bihar	94163	90567	40625	21878	62503
6	Chandigarh UT	115	115	115	0	115
7	Chhattisgarh	136034	96000	54424.18	23000	77424
8	Dadra & Nagar Haveli	491	490	490	0	490
9	Daman & Diu UT	112	112	0	112	112
10	Delhi	1483	1483	1483	0	1483
11	Goa	3702	3702	3702	0	3702
12	Gujarat	196024	160978	104819.45	32259	137078
13	Haryana	44212	44179	44179	0	44179
14	Himachal Pradesh	55673	8020	8020	0	8020
15	Jammu & Kashmir UT	167396	9506	9506	0	9506
16	Jharkhand	79714	65797	41771	19117	60888
17	Karnataka	191808	191808	118993	36913	155906
18	Kerala	38863	28088	24426	3662	28088
19	Lakshdweep UT	32	32	32	0	32
20	Ladakh UT	54840	963	963	0	963
21	Madhya Pradesh	308000	265234	132163.48	67878	200041
22	Maharashtra	307713	256529	184194	33995	218189
23	Manipur	22327	2559	2559	0	2559
24	Meghalaya	22429	10645	9542	1103	10645
25	Mizoram	21081	700	700	0	700
26	Nagaland	16579	910	910	0	910
27	Odisha	155707	119636	61778	29821	91599
28	Pudducherry UT	479	454	454	0	454
29	Punjab	50368	50368	50368	0	50368
30	Rajasthan	342239	334152	230252.14	84452	314704
31	Sikkim	7096	1496	280	816	1096
32	Tamil Nadu	130058	105829	105829	0	105829
33	Telangana	111940	104824	81521	23303	104824
34	Tripura	10492	6757	6757		6198
35	Uttar Pradesh	241345	241345	161611.776	41306	202918
36	Uttarakhand	53484	11430	7811	3619	11430
37	West Bengal	88752	71947	36465	18036	54501
	Total	3289081	2496742	1633629	481497	2114567

Table: Staus of sharing of outputs with State Government

State	Total Number of Districts in the States	Mappable Districts	Districts covered under NAQUIM so far (including partially covered districts)	Number of Districts for which reports are placed on website	Number of Districts for which outputs are shared with SGWCC	Total Number of Districts for which outputs have been shared with DMDC	Total Number of Districts for which the outputs are shared through presentation
Andaman & Nicobar UT	3	3	2	1	1	1	1
Andhra Pradesh	13	13	9	7	7	7	2
Arunachal Pradesh	16	7	5	3	4	4	0
Assam	27	26	**15**	7	10	15	5
Bihar	**38**	**38**	**27**	**3**	**12**	**12**	**0**
Chandigarh UT	1	1	1	1	1	1	0
Chhattisgarh	27	21	19	10	16	15	3
Dadra & Nagar Haveli	1	1	1	1	0	0	0
Daman & Diu UT	2	2	2	0	0		
Delhi	9	9	9	9		9	9
Goa	2	2	2	2	2	2	0
Gujarat	33	33	29	6	20	4	4
Haryana	22	22	22	22	22	0	15
Himachal Pradesh	10	10	10	4	8	10	10
Jammu & Kashmir UT	20	20	20	17	17	17	17
Jharkhand	24	24	23	14	10	14	1
Karnataka	30	30	27	24	24	24	9
Kerala	14	14	14	7	11	10	10
Lakshdweep UT	1	1	1	1	1	1	1
Ladakh UT	2	2	2	0	0	0	0
Madhya Pradesh	52	52	37	11	21	21	7
Maharashtra	34	34	29	15	24	17	3
Manipur	9	9	9	1	9	9	0
Meghalaya	11	11	11	10	10	10	10
Mizoram	8	1	1	1		1	0
Nagaland	11	1	1	1	1	1	0
Odisha	30	30	25	12	13	14	6
Pudducherry UT	4	4	4	1	2	2	2
Punjab	22	22	22	22	22	22	12
Rajasthan	33	33	30	18	21	21	8
Sikkim	4	4	2	1	1	1	1
Tamil Nadu	38	38	38	24	36	28	16
Telangana	10	10	10	6	8	8	7
Tripura	4	4	4	4	4	4	4
Uttar Pradesh	75	75	64	24	34	34	34
Uttarakhand	13	6	6	2	2	3	3
West Bengal	23	23	19	5	5	10	1
	676	**636**	**552**	**297**	**379**	**352**	**201**

7. HELIBORN SURVEY -HIGH RESOLUTION AQUIFER MAPPING AND MANAGEMENT IN ARID REGIONS OF NORTH-WESTERN INDIA

Central Ground Water Board, Ministry of Jal Shakti and CSIR-NGRI, Hyderabad have signed a Memorandum of Agreement for High Resolution Aquifer Mapping and Management in Arid Regions of North Western India, by using advanced Heli-borne Geophysical Surveys and other scientific studies in parts of the States of Rajasthan, Gujarat Haryana, Punjab andHimachal Pradesh, covering an area of 3.88 lac sq/km under the National Aquifer Mapping Programme (NAQUIM) with financial outlay of Rs. 54 Cr for Phase-I. The Heliborne Surveys have been proposed in two phases. During the Phase-1, 1.01 lakh sq.km was proposed in 2020-22, out of which 19020 sq.km considered priority areas covering Sikar, Jaisalmer and Jodhpur districts of Rajasthan, Kurukshetra and Yamuna Nagar districts of Haryana and remaining areas are normal.

The heliborne survey data has been obtained through a total of 40323 flight line km, 86 km through Earth Resistivity Tomography (ERT) surveys and Electrical vector Resistivity Imaging (EVRI) have been used.The studies aimed at finding aquifer system with relatively fresh and saline zones, aquifer geometry of principal aquifer with demarcation of de-saturated and saturated aquifers, identification of paleochannels network and their linkage with the aquifer systems in the area, groundwater potential worthy sites for withdrawal of groundwater and water conservation structures through artificial or Managed Aquifer Recharge (MAR).Remaining area of 2.87 lakh. Sq.km is envisaged under Pahse-2 for 2022-24.

MoA Signed between CGWB & NGRI on 21 Dec. 2020

Heliborne Survey Inauguration at Jodhpur on 5th Oct. 2021

Heliborne Survey Inauguration at Jodhpur on 5th Oct. 2021

8 GROUND WATER QUALITY STUDIES

Central Ground Water Board has 16 Regional Chemical Laboratories to carry out Chemical Analysis of major and minor in organic constituents in water samples. The Chemical laboratories are well equipped to carryout Basic parameters, Trace Metaland Toxic elements determinations using so phisticated in struments like Atomic Absorption Spectrophotometer (AAS), Ion Chromatograph, Fluorimeter (Uranium Analyser), Digital PC based UV–VIS Spectrophotometer, Ion Meter, Flame Photometer, Conductivity meter, pHmeter, and Nephelometer. The laboratories are also provided with Electronic Monopan and Top loading Balances, Ultra -Pure Water System, Deionizer, Double Distillation Plant, Hot AirOven, Water Bath, Magnetic Stirrer and Hot Plates.The Chemical Laboratory at Lucknow and Chandigarh are additionally equipped with InductiveCoupled Plasma Spectrometer (ICPS) for sequential analysis of multiple toxic elements withhigh accuracy.Total Organic Carbon (TOC) analyzer is installed in the Regional ChemicalLaboratory at Chandigarh, Lucknow, Chennai and Kolkata. The Chemical Laboratories atChandigarh, Lucknow, Nagpur and Bengaluru are also equipped with Liquid Scintillation Counter (LSC) to undertake the analysis of radiation. The chemical analysis data generated by these laboratories is utilized for evaluating the groundwater quality in compliance withNational Standards (BIS 2012), to study the impact of anthropogenic activities on groundwater quality, demarcate areas of water quality deterioration and assess the point and non-point sources of ground water pollution so as to take necessary action for management ofground waterresources.

During 2021-22, a total of 42915 ground water samples had been analyzed out of which 19246 water samples were analyzed for determination of basic constituents, while 23331 water samples for heavy metals including U etc. Apart from these 117 samples were analysed for Total organic Carbon (22-Chandigarh, 95-Lucknow). 221 samples were analysed by Liquid scintillation counter for Alpha & Beta radiations. The details of water samples analyzed by different chemical laboratories during 2021-22are presented in table5.1

8.1 Location of Chemical Labs of CGWB

16 chemical laboratories-Bhopal, Raipur, Nagpur, Lucknow, Jammu, Chandigarh, Jaipur, Ahmadabad, Hyderabad, Bangalore, Chennai, Trivandrum, Patna, Kolkata, Guwahati & Bhubaneswar.

8.2 Number of Labs NABL accredited

9 Labs NABL Accredited- Bhopal, Bhubaneswar, Chandigarh, Guwahati, Hyderabad, Jaipur, Kolkata, Lucknow, Nagpur.

Table 8.1 STATUS OF INSTRUMENTS IN REGIONAL LABORATORY OF CGWB

S. No.	Instrument/ Equipment	No. of Instruments/Equipment in 16 Regional Chemical Laboratory of CGWB
1	pH Meter	1-2 in each Laboratory
2	EC Meter	1-2 in each Laboratory
3	Flame- Photometer	1-2 in each Laboratory
4	UV/Visible-Spectro-photometer	1 in each Laboratory
5	Atomic Absorption Spectro-photometer	1 in each Laboratory (2 each at Lucknow & Kolkata)
6	Ion Chromatograph	1 in each Laboratory
7	Ion meter	1 in each Laboratory
8	ICP-MS	1 each in Lucknow & Chandigarh laboratory
9	TOC	1 each in Lucknow, Chandigarh, Kolkata & Chennai laboratory
10	LSC	1 each in Lucknow, Chandigarh, Bengaluru, Nagpur laboratory
11	Ultra-Water Purification System	1 in each Laboratory
12.	Fluorimeter	1 in each Laboratory

Table 8.2 Region-wise Ground Water Samples analyzed during 2021-22

S. No.	RegionalOffices	NumberofAnalysedsamples	
		Basic	HeavyMetals
1.	NWHR, Jammu	645	700
2.	NWR,Chandigarh	1059	1286
3.	WR,Jaipur	1585	1838
4.	WCR,Ahmadabad	976	1524
5.	NCR,Bhopal	1504	1564
6.	NCCR,Raipur	1161	1136
7.	CR,Nagpur	1352	1529

S. No.	RegionalOffices	Number of Analysed samples	
		Basic	HeavyMetals
8.	NR, Lucknow	1151	1564
9.	MER,Patna	1510	1127
10.	ER,Kolkata	997	1581
11.	NER, Guwahati	1474	2051
12.	SER,Bhubaneswar	1526	1532
13.	SR,Hyderabad	1088	1098
14.	SWR,Bangalore	1312	2313
15.	SECR,Chennai	1328	1516
16.	KR,Trivandrum	578	972
	Total	19246	23331
GrandTotal(Basic+Heavy)		**42577**	

9.0 ASSESSMENT OF DYNAMIC GROUND WATER RESOURCES

Assessment of 'Dynamic Ground Water Resources of the country is being carried out periodically to assess the annual ground water recharge and utilization and its availability for future use. The assessment has been carried out for the reference years 1972, 1985, 1995, 2004, 2009, 2011, 2013, 2017 & 2020 (latest) till now. The methodology recommended by the Ground Water Estimation Committee- 1997(GEC- 97) was used for the assessment for the reference year 2004, 2009, 2011 & 2013 whereas the assessment for 2017 and 2020 were carried out following the methodology and norms recommended by Ground Water Estimation Committee- 2015 (GEC-2015).

As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction are assessed for each assessment unit (block/t aluka/ mandal/ tehsils/ firka etc.). The 'Stage of Ground Water Extraction' is then computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe (<= 70 %), Semi-Critical (>70 % and <=90 %), Critical (>90 % and <=100%) and Over-Exploited (>100 %).

A Central Level Expert Group (CLEG) was constituted by the Department of Water Resources, RD & GR, Ministry of Jal Shakti, Government of India for over-all supervision of the re-assessment of ground water resources in the entire country with reference year 2020. The groundwater resources assessment at the State/UT Levels was carried out jointly by Central Ground Water Board and State Nodal/Ground Water Departments under the guidance of respective State Level Committees (SLC) with over-all supervisionofCLEG.The National Compilation on Dynamic Ground Water Resources of India, 2020 has been published in June, 2021 (http://cgwb.gov.in/documents/2021-08-02-GWRA_India_2020.pdf). Ground Water Resource Assessment -2020 (GWRA-2020) has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad ((http://ingres.iith.ac.in).
As per latest assessment (2020), the total annual ground water recharge has been assessed as 436 bcm. Keeping an allocation for natural discharge, the annual extractable ground water resource works out as 398 bcm. The total annual ground water extraction has been assessed as 245 bcm. The average stage of ground water extraction for the country as a whole works out to be about 62 %. Out of the total 6965 assessment units (Blocks/ Districts/ Mandals/ Talukas/Firkas) in the country, 1114 units in various States (16 %) have been categorized as 'Over-Exploited' indicating ground water extraction exceeding the annual extractable ground water resources (availability). A total of 270 (4 %) assessment units have been categorized as 'Critical', where the stage of ground water extraction is between 90-100 %. There are 1057 'Semi-Critical' units (15 %), where the stage of ground water extraction is between 70 % and 90 % and 4427 (64 %) assessment units have been categorized as 'Safe' where the stage of Ground water extraction is less than 70 %. Apart from this, there are 97 assessment units (1 %), which have been categorized as 'Saline' as major part of the ground water in phreatic aquifers is brackish or saline. Similarly out of 24.33 lakh sq km recharge worthy area of the country, 4.09 lakh sq km (17 %) are under 'Over-Exploited', 0.86 lakh sq km (4 %) are under 'Critical', 3.4 lakh sq km (14 %) are under 'Semi-Critical', 15.67 lakh sq km (64 %) are under 'Safe' and 0.3 lakh sq km (1 %) are under 'Saline' category assessment units. Out of 397.62 bcm of Total Annual

Extractable Resources of the country, 50.54 bcm (13 %) are under 'Over-Exploited', 12.71 bcm (3 %) are under 'Critical', 54.11 bcm (14 %) are under 'Semi-Critical', 280.26 bcm (70 %) are under 'Safe' category assessment units.

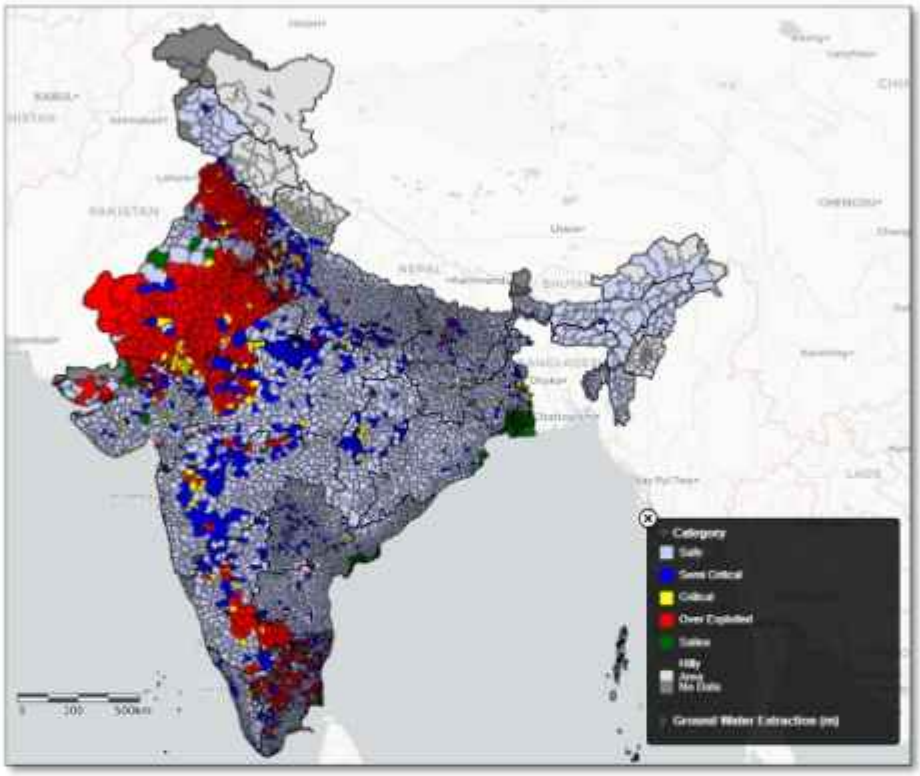


Fig-5: Categorization of Assessment Units

In comparison to 2017 assessment, the total number of assessment units in the country have increased from 6881 to 6965 with major contribution (in increase) from the State of Karnataka, Haryana and Punjab.The total annual ground water recharge has increased from 432 to 436 bcm, where major increse is noticed in the States of Uttar Pradesh, Andhra Pradesh, Karnataka, Telangana, Gujarat & Chhattisgarh. The changes are attributed mainly to changes in recharge from 'Other Sources'. Accordingly, the annual extractable resource of GW Resource Assessment, 2020 on comparison GW Resource Assessment, 2017 also shows a increase from 393 to 398 bcm. The ground water extraction has marginally decreased from 249 to 245 bcm. The overall stage of groundwater extraction has marginally decreased from 63 % to 62 %.

The state/UT wise details of ground water resource availability& categorization of assessment units as per the Assessment of Dynamic Ground Water Resources of India, 2020 are given in Annexures I & II respectively.

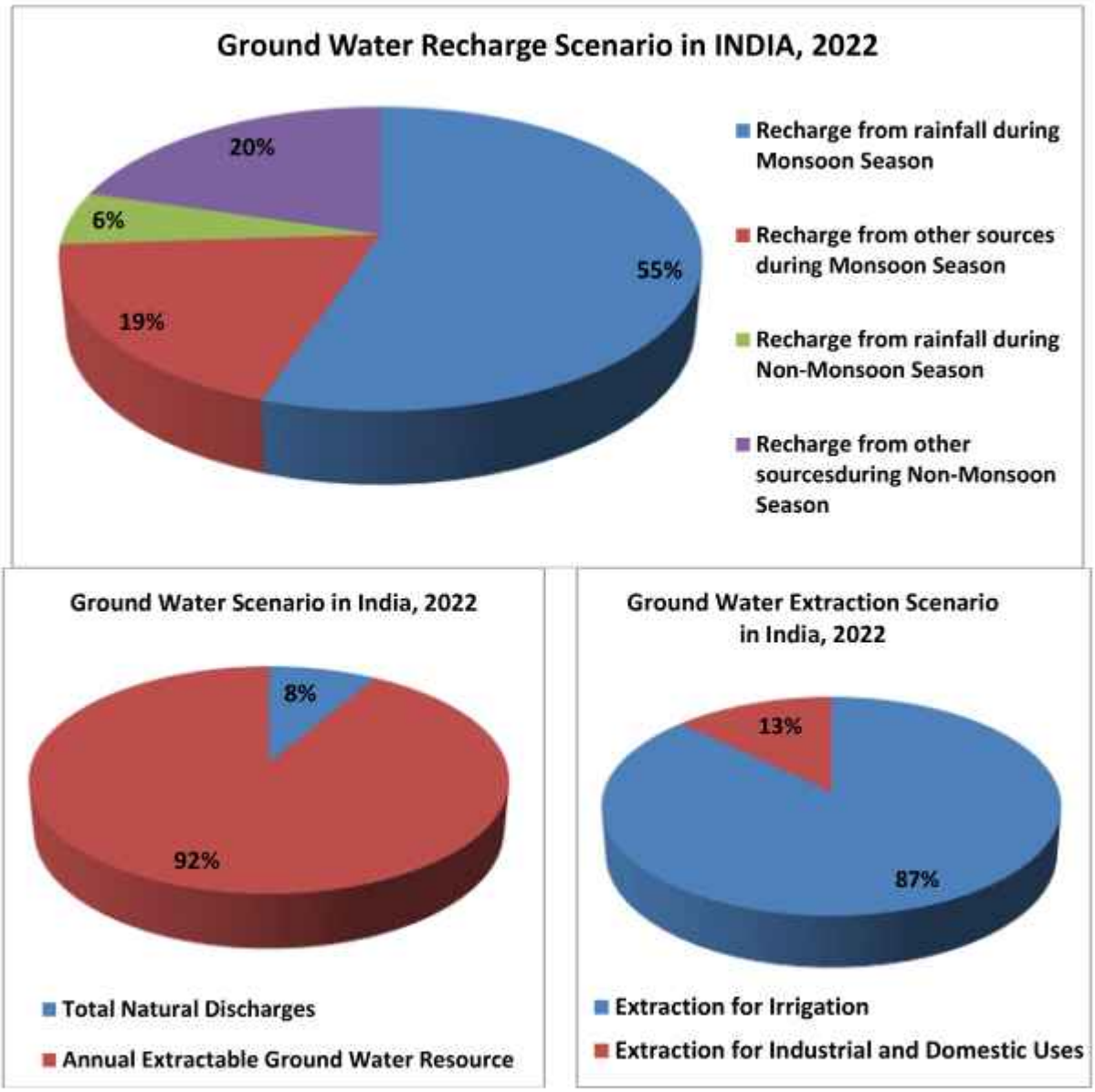


Fig-9.1: Ground Water Resources and Extraction Scenario in India, 2022

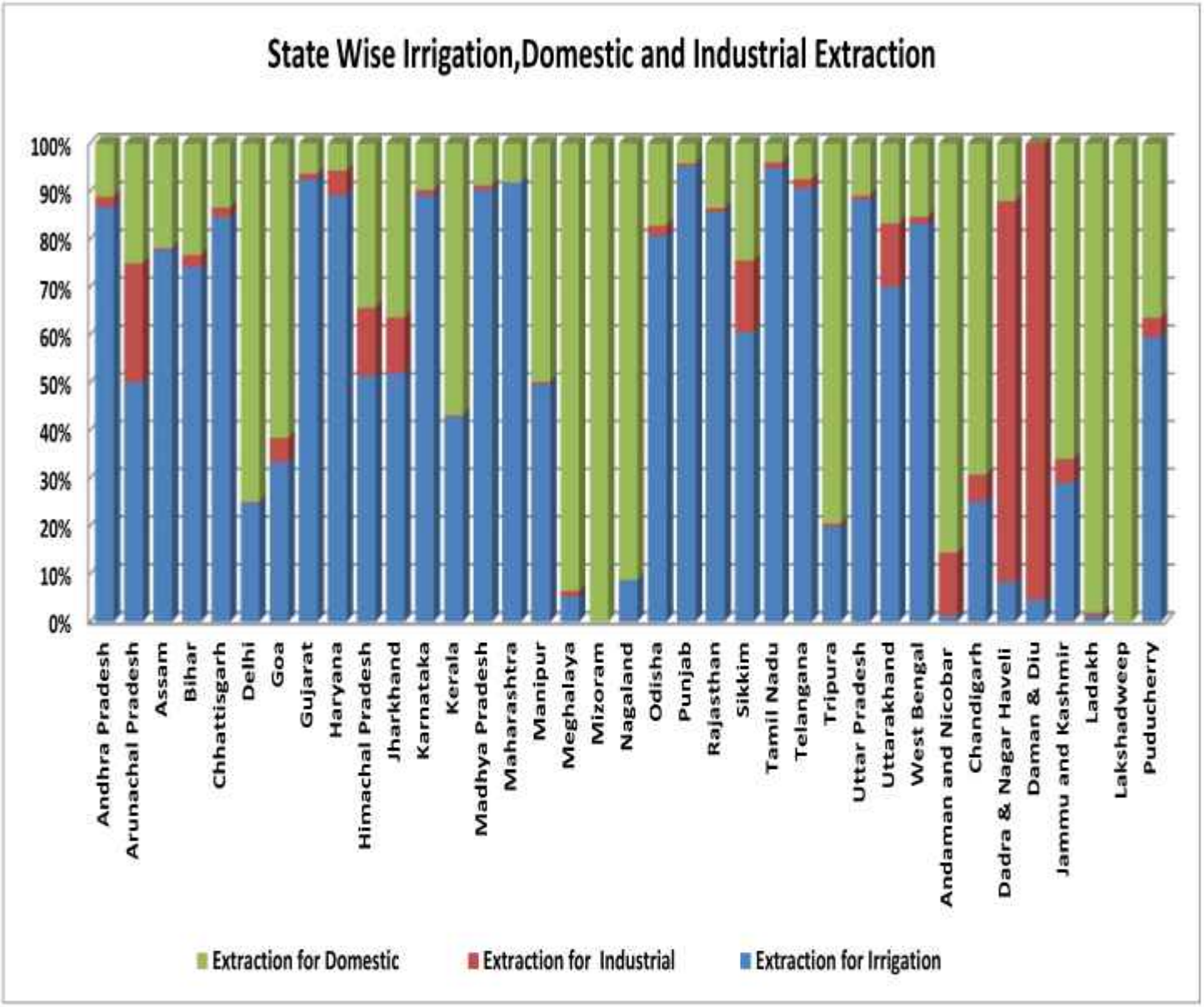


Fig.9.2: State wise Irrigation Draft *Vs* Domestic & Industrial

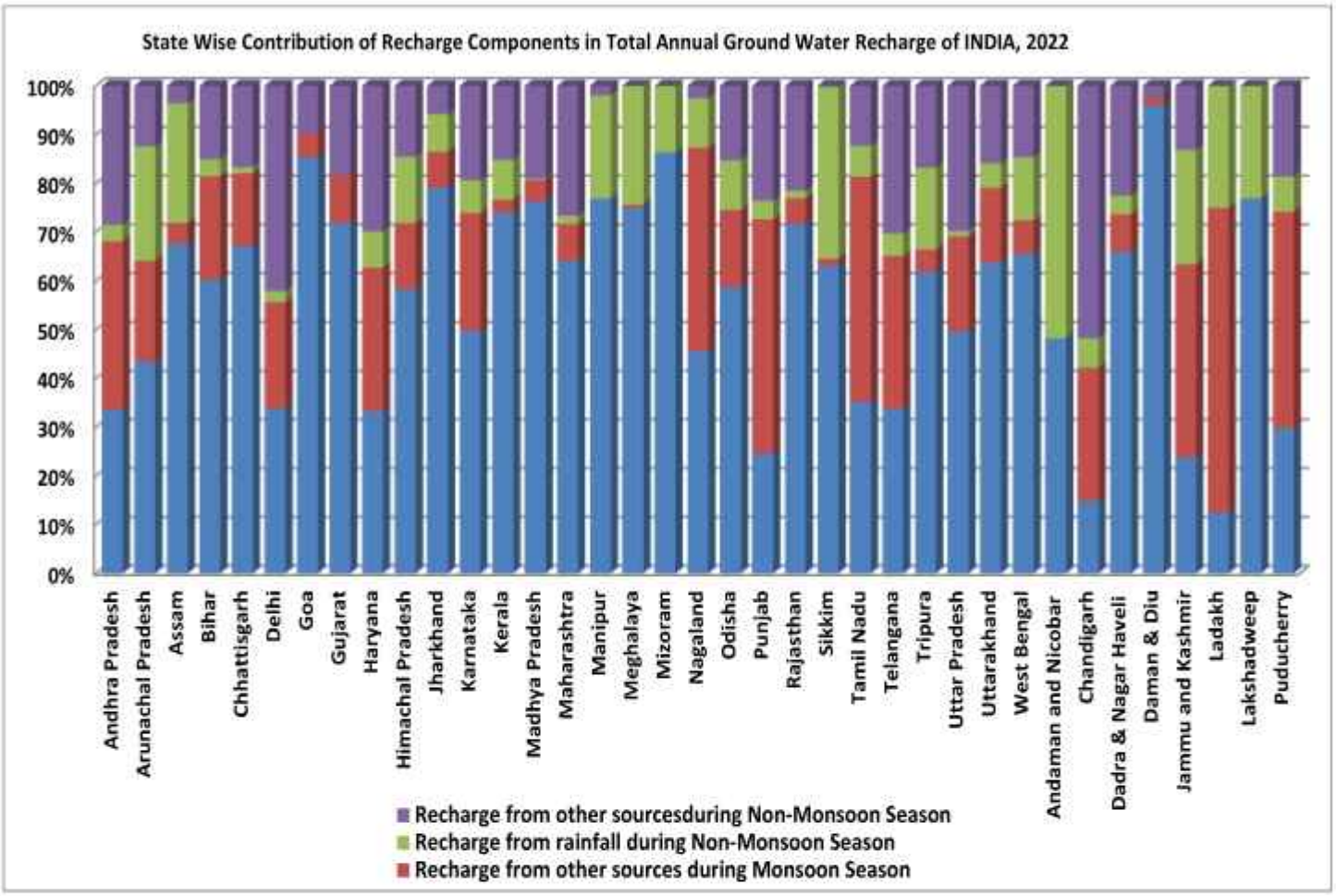


Fig-9.3: State wise contribution of recharge components in Total Annual Ground Water Recharge of India, 2022

Table: CATEGORIZATION OF BLOCKS/ MANDALS/ TALUKAS IN INDIA (2020)												
S.No.	State/Union Territories	Total No. of Assessed Units	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
	States		Nos.	%	Nos.	%	Nos.	%	Nos.	%	Nos.	%
1	Andhra Pradesh	667	551	82.61	40	6.00	15	2.25	23	3.45	38	5.70
2	Arunachal Pradesh	11	11	100.00								
3	Assam	28	28	100.00								
4	Bihar	534	471	88.20	51	9.55	5	0.94	7	1.31		
5	Chhattisgarh	146	110	75.34	27	18.49	9	6.16				
6	Delhi	34	3	8.82	7	20.59	7	20.59	17	50.00		
7	Goa	12	12	100.00								
8	Gujarat	248	182	73.39	24	9.68	4	1.61	25	10.08	13	5.24
9	Haryana	141	30	21.28	14	9.93	12	8.51	85	60.28		
10	Himachal Pradesh	10	10	100.00								
11	Jharkhand	259	244	94.21	10	3.86	2	0.77	3	1.16		
12	Karnataka	227	130	57.27	35	15.42	10	4.41	52	22.91		
13	Kerala	152	120	78.95	29	19.08	3	1.97				
14	Madhya Pradesh	317	233	73.50	50	15.77	8	2.52	26	8.21		
15	Maharashtra	353	271	76.77	63	17.85	8	2.27	10	2.83	1	0.28
16	Manipur	9	9	100.00								
17	Meghalaya	12	12	100.00								
18	Mizoram	26	26	100.00								
19	Nagaland	11	11	100.00								
20	Odisha	314	302	96.18	6	1.91					6	1.91
21	Punjab	150	17	11.33	10	6.67	6	4.00	117	78.00		
22	Rajasthan	295	37	12.54	29	9.83	23	7.80	203	68.81	3	1.02
23	Sikkim	4	4	100.00								
24	Tamil Nadu	1166	409	35.08	225	19.30	63	5.40	435	37.31	34	2.92
25	Telangana	589	321	54.50	180	30.56	44	7.47	44	7.47		
26	Tripura	59	59	100.00								
27	Uttar Pradesh	830	541	65.18	174	20.96	49	5.90	66	7.95		
28	Uttarakhand	18	14	77.78	4	22.22						
29	West Bengal*	268	191	71.27	76	28.36	1	0.37				
30	Andaman and Nicobar	36	35	97.22							1	2.78
31	Chandigarh	1			1	100.00						

S.No.	State/Union Territories	Total No. of Assessed Units	Safe		Semi-Critical		Critical		Over-Exploited		Saline	
	States		Nos.	%	Nos.	%	Nos.	%	Nos.	%	Nos.	%
32	Dadra & Nagar Haveli	1	1	100.00								
	Daman & Diu	2	1	50.00					1	50.00		
33	Jammu and Kashmir	20	20	100.00								
34	Ladakh	2	2	100.00								
35	Lakshadweep	9	7	77.78	2	22.22						
36	Puducherry	4	2	50.00			1	25.00			1	25.00
	Grand Total	6965	4427	63.56	1057	15.18	270	3.88	1114	15.99	97	1.39

Note

Blocks- Bihar, Chhatisgarh, Haryana, Jharkhand, Kerala, Madhya Pradesh, Manipur, Mizoram, Odisha, Punjab, Rajasthan , Tripura, Uttar Pradesh, Uttarakhand, West Bengal

Taluks- Goa, Gujarat, Karnataka, Maharashtra

Mandals- Andhra Pradesh, Telangana

District- Arunachal Pradesh, Assam, Meghalaya, Nagaland, Sikkim, Dadra & Nagar Haveli, Daman & Diu, Jammu & Kashmir, Ladakh

Valley- Himachal Pradesh,

Islands- Andaman & Nicobar, Lakshadweep

Firka- Tamil Nadu

Region- Puducherry

UT- Chandigarh

Tehsil- Delhi

***West Bengal- The Ground Water Resource Assessment as on 2013 has been considered for state of West Bengal.**

10. GROUND WATER CONSERVATION AND ARTIFICIAL RECHARGE

10.1Artificial Recharge work in Aspirational Districts.

Central Ground Water Board had taken up artificial recharge studies in aspirational districts in the States of Maharashtra, Andhra Pradesh & Telangana with the objective of harvesting the runoff water in streams to store at suitable locations for augmenting recharge groundwater. The work has been executed under the GWM&R scheme of CGWB in the year 2018 and have been completed in Andhra Pradesh, Maharashtra & Telangana States. The work includes construction of 77 check dams, 04 percolation tanks, 02 subsurface barrier, 46 recharge wells, 66 recharge shafts, and 41 piezometers with an outlay of 50.87 Crores and also covers the impact assessment studies for which piezometers with DWLRs have been provisioned.

S.No	State / District / Block	Details	Fund Allocated
1	Maharashtra / Osmanabad / Osmanabad	Constructed 55 Check Dams, 20 Piezometers & 46 Recharge wells.	Rs. 54.38 Cr.
2	Andhra Pradesh / YSR Kadapa / Pulivendula	Constructed 16 Check dams, 4 Percolation tanksSub-surfacebarriers and 36 Recharge shafts& 12 Piezometers.	
3	Telangana / Warangal / Bachennapet	Constructed 6 Check dams, 1 Sub-Subsurface barrier, and 31 Recharge shafts& 9 Piezometers.	

A total of 20 piezometers were constructed and water levels were monitored to know the impact of the recharge structures constructed in the Osmanabad blocks of Maharashtra. The study revealed an average rise in water level of 1.2 m in these piezometers. In Pulivendula block of YSR Kadapa district, Andhra Pradesh12 piezometers were constructed and their water levels were monitored. An average water level rise of 1.9 m has been noticed in these piezometers. In Bachannapet block of Jangaon district, Telangana state, 9 piezometers were constructed and monitored for water level. An average water level rise of about 4.0 has been recorded in these piezmeters.

Check Dam and Recharge shaft at Achavelli Village, Pulivendla Mandal, Cuddapah District, Andhra Pradesh

Check Dam at Kawlewadi, Osmanabad Mandal, Osmanabad District, Maharashtra

10.2 Construction of Bridge Cum Bandharain Maharashtra

CGWB has implemented construction of Bridge cum Bandhara (BCB)that serves a dual purpose of transportation as well as storage of water in the upstream side for drinking and irrigational needs besides other purposes such as ground water recharge asa pilot study. Projects were taken up in the eastern region of Maharashtra covering districts of Wardha and Amravati at five locations as follows:

1. Sarwadi village, Karanja Teshil, Wardha District
2. Deolivillage, Deoli Tehsil, Wardha District
3. Jamnivillage, Selu Tehsil, Wardha District
4. Tiwasa village, Tiwasa Tehsil, Amravati District
5. AjraPhata village, Samudrapur Tehsil, Wardha District

The areas are characterized by shallow aquifer which are formed by weathering of massive & amygdaloidal unit of basalt, down to a depth of 20m and semi-confined aquifers occur at slightly deeper levels which are overlain by massive Basalt and red boles from 20 to 40 m depth.

The project commenced in the year of 2018 and completed in 2020. Bandharas have been constructed by using state of the art technology at the cost of 30.30 Crores. Automatic radial gates are installed at Sarwadi, Tiwasa, Deoli&Jamni& Rubber dam at AjdaPhata to regulate the flow of water in the streams.

Ajraphata BCB project with Rubber Dam, Samudrapur Taluka, Wardha District, Maharashtra

Jamni BCB project with Automatic Gates (Radial), Selu taluka, Wardha District, Maharashtra

A total of 48 dug wells are being monitored in the area of influence at locations distant from each other in different minor basins. A rising pattern in depth to water level is invariably observed at most of the locations indicating impact of recharge. Water level rise is observed in the range of 1.2-2.5m in piezometers.

10.3 Convergence with MGNERGS in Water Stressed areas

CGWB is providing technical guidance in the site selection & design of artificial recharge structures and capacity building of MGNREGS officials in selected 09 Blocks of 08 States as a pilot study. The work has been initiated under the MGNREGS fund, so far completed in Andhra Pradesh, Telangana & Tamil Nadu (one block each) and the work is under progress in the rest of the blocks of the states.

10.4 Groundwater Augmentation through Artificial Recharge in water-stressed areas of Rajasthan (2021- continuing)

During 2021-22, CGWB has taken up artificial recharge for groundwater augmentation in water-stressed areas of Jaisalmer, Jodhpur & Sikar districts of Rajasthan.

Phase-I : Construction of Earthen Dam at Bastawa Mata (Block Balesar) and Indroka (Block Mandore) in Jodhpur district, Rajasthan with the objectives of artificial recharge to groundwater aquifer systems. The cost of the project is about Rs. 22.06 Cr, which includes Defect Liability Period of 01 Year after construction and 03 years of O&M by WAPCOS. The work is under progress.

Phase-II : Under this phase, it is proposed to construct Stone Masonary Check Dams (MCD), Anicuts, Concrete Check Dams (CCD & Recharge shafts) at **101** sites in parts of Balesar, Shekhla, Lohawat, Tinwari and Phalodi blocks of Jodhpur district and Sankra block of Jaisalmer and Shrimadhopur block of Sikar districts of Rajasthan. The district wise details of the Nos. of structures are to be constructed is given below and the work is under progress.

District	**Block**	**Types of WHS & Nos.**				**No. of Sites**
		MCD	Anicut	CCD	RS	
Jodhpur	Balesar	49	08	-	-	57
Jodhpur	Shekhla	06	03	-	-	09
	Lohawat	10	07	-	-	17
	Tinwari	02	04	-	-	06
	Phalodi	-	01	-	-	01
Jaisalmer	Sankra	-	-	04	-	04
Sikar	Shri Madhopur	02	02	-	03	07
Total		**69**	**25**	**4**	**3**	**101**

Progress of work carried out under MGNREGS during March-2022

State	Andhra Pradesh	Telangana	Karnataka	Tamil Nadu	Rajasthan	Rajasthan	Haryana	Madhya Pradesh	Maharashtra
District	Prakasam	Nallagonda	Chickballalpur	Krishnagiri	Bhilwara	Sirohi	Kurukshetra	Ujjain	Amravati
Block	Peddaravedu	Munugode	Gowribidanur	Mathur	Madalgarh	Reodhar	Babain	Badnagar	Warud
Sharing of Hydrogeological Information with State Agencies for Implementation	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete	Complete
Details of Training conducted (No & Dates on which training conducted)	5 Nos. (11.04.2018, 12.04.2018, 06.07.2018, 11.09.2018 & 12.09.2018)	3 Nos. (06.04.2018, 10.04.2018 & 26.06.2018)	3 Nos. (23.7.2018 to 25.7.2018, 22.2.2019 & 12.9.2019)	3 Nos. (01.05.2018 to 16.05.18, 28.06.18, & 28.06.2018)	nil	1 (in March 2019)	01 (11.9.2018)	1 (25.2.2019 to 27.2.2019)	1 (24.08.2018)
No of Awareness Programme conducted	2	2	2	2	nil	nil	nil	nil	1
Joint Visit (Date)	27.1 2019	28.1. 2019	23rd to 25th July 2019; 22.02.2019; 05.11.2019;	Sep 2018, July 2019 & March 2020	Sept. 2018	Oct-18	11 Sept. 18	June 2018; 25 to 27.10.2018; 24.3.2019; 4.11.2019 to5.11.2019	29.10.2018 30.10.2018, 4.12.2019
Commencement of Execution of Work	April 2018	April 2018	1st week of June 2019	24.07.18	19-Apr	19-Dec	11-Sep-18	18-Mar	18-Dec
No. of Structure Envisaged	1253	2174	103	20	56	126	15	436	1798

State	Andhra Pradesh	Telangana	Karnataka	Tamil Nadu	Rajasthan	Rajasthan	Haryana	Madhya Pradesh	Maharashtra
District	Prakasam	Nallagonda	Chickballalpur	Krishnagiri	Bhilwara	Sirohi	Kurukshetra	Ujjain	Amravati
Block	Peddaravedu	Munugode	Gowribidanur	Mathur	Madalgarh	Reodhar	Babain	Badnagar	Warud
No. of Structure Completed	1253	1713	24	20	47	101	9	316	120
No. of Structure In Progress	Nil	Nil	12	nil	1	4	1	227	3
Geo-tagging (Completed /in progress)	Completed	Completed	completed	Completed	Completed	Completed	In-progress	Completed	in progress
Initiation of Impact Assessment Study (Yes/No)	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Status of the Project	Completed	Completed	12 structures Work under progress .MACD construction is material intensive and need skilled labors .Hence the completion rate is slow	Work completed, State Govt. is advised to expedite the work of impact assessment	ALL 56 NOS SITES REALEASED, 47 Sites COMPLETED	All 126 sites relesed by CGWB & Financial approval for 50 sites awaited from state	As per latest data received from ABPO Babain, Total 15 No. of ponds are selected uner MNREGA insteadefo 16 Ponds. in Addition to this total 51 Nos. of recharge structures (recharge Pit =26 and Soakage Pits= 25) are completed.	Work in progress, Also working on some additional structures under MGNREGs	State Govt. is advised to expedite the work. Completion rate is slow due to the labour problem in covide issue.

11.0 Technical Assistance (SHORT TERM WATER SUPPLY INVESTIGATIONS)

Table 11. List of short-term investigations carried out region wise

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
1	NWR, Chandigarh	Feasibility study for construction of Tubewell for provision of Swimming Pool at Dhoelwal Military Complex, Ludhiana, Punjab	Dhoelwal Military Complex, Ludhiana, Punjab
2	NWR, Chandigarh	Feasibility study for construction of Tubewell in the area of Tibri Military Station, Gurdaspur, Punjab	Tibri Military Station, Gurdaspur, Punjab
	NWR, Chandigarh	Feasibility study for construction of tube-well at J-1 location under GE (EAST), Jalandhar, Punjab	GE (EAST), Jalandhar, Punjab
4	NWR, Chandigarh	Feasibility study for construction of Tubewell at Pump House No.8 (Hudson Line) Jalandhar Cantt. Punjab	MES, Jalandhar Cantt. Punjab
5	NWR, Chandigarh	Feasibility study for construction of Tubewell at Depot area of Air Force Station, Gurugram	Air Force Station, Gurugram
6	SER, Bhubaneswar	Short Term investigation for water supply for Indian Coast Guard Officer's Quarter	Indian Coast Guard
7	SER, Bhubaneswar	Hydrogeological Investigation Report for water supply for Dalkibeda Village, Chakdi GP, Jashipur Block, Mayurbhanj District, Odisha	VIP reference
	NCR	Water SupplyInvestigation for Central Proof Establishment in Defence areaItarsi, Hoshangabad district, MP	Ministry of Defence, Government of India
8	SECR, Chennai	Comprehensive rainwater harvesting system proposal	Air force station, Pudukkottai Road, Thanjavur
9	SECR, Chennai	Groundwater investigationat Sembium Campus	ESIC, Chennai
10	SECR, Chennai	Ground water investigation report for digging bore well	Board of Apprenticeship Training, Tharamani, Chennai, Tamil Nadu
11	SECR, Chennai	Ground water investigation report for digging of bore well	ESI Hospital, Ayanavaram Campus, Chennai
12	SECR, Chennai	Short Term Investigation for shallow well	Air force station, Tambaram
13	SWR	Groundwater survey for provision of borewell for augmentation of water supply of Baffle range at Air Force station Yelahanka, Bangalore.	Garrison Engineer (AF),
14			Military Engineer Services,
15			Yelahanka, Bangalore-63.

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
16	SWR	Groundwater survey for provision of borewell (1 nos.) for construction of catering building in ASC Centre and college at Agram post, Bangalore.	Garrison Engineer (P),
17			Military Engineer Services
18			
19	SWR	Groundwater survey for provision of borewell (3 nos.)at Aeronautical Development Agency Campus-II,SuranjanDas road,Bangalore	Project Management office,DRDO
20	SWR	Feasibility assessment of borewell installed at DIQA,Bangalore	Military Engineer Services
21			
22	SWR	Groundwater survey for provision of borewell for augmentation of water supply of Baffle range at CQAE(WE), Jalahalli, Bangalore	Military Engineer Services
23			
24	SWR	Rainwater harvesting project at IAF, Jalahalli station	Garrison Engineer (AF),
25			Military Engineer Services,
26			Jalahalli, Bangalore
27	SWR	Survey for provision of borewell at STC, BSF, Yelahanka campus and KLP, Karahalli campus	Border Security Force
28	SWR	Hydrogeological Investigations for construction of borewell at Naval base Karwar	Garrison Engineer Services, Karwar
29	SWR	Groundwater survey for provision of borewell (1 nos.) at HQGNA Gomantak (Naval area) at Vasco (Goa state)	Garrison Engineer,Gomantak
30	SWR	Groundwater survey for provision of borewell at ICGAE Goa	Military Engineer Services,
31			GE (I) (CG),Goa
32	SWR	Groundwater survey for provision of borewell at ASC Centre and college Bangalore	Garrison Engineer (Project),
33			Military Engineer Services
34	SWR	Groundwater survey for provision of borewell at Electronics and Radar Development Establishment, Ministry of Defence, CV Raman Nagar, Bangalore	DRDO
35	SWR	Hydrogeological investigation for provision of new borewell at Varuna Complex, Vimanapura, Bangalore	Military Engineer Services
36			
37	SWR	Short term investigation for provision of borewell location at he campus of Garrison Engineer (Project), Military Engineer Services,Ulsoor road	Military Engineer Services
38			

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
39 40	NCCR, Raipur	Short Term investigation around the NIBSM campus in Baronda, Raipur for groundwater development.	NIBSM campus in Baronda, Raipur
41	NHR	Investigation for Army area Pooh, Kinnaur, H.P.	Army
42	NHR	Investigation for Army area Karcham, Kinnaur	Army
43	NHR	Investigation for Army area Samdhu, Kinnaur	Army
44	NHR	Investigation for Army area Dogri, Kinnaur	Army
45	NHR	Investigation for R.W.H in Army area Subathu, Solan, H.P.	Army
46	UR	Feasible sites for construction of tube well	Army
47	UR	Feasible sites for construction and implementation of RWH	Army
48	UR	Possibilities of Artificial Recharge through Village pond in Raipur Industrial Area	Irrigation Department
49	UR	Feasible sites for construction and implementation of RWH	Army
50	UR	Geophysical investigation for construction of tube well	Nagar Panchayat Nayalgarh, Pauri
51	NWHR	Hydrogeological Investigation for construction of tubewell at Kunfer (Chandrakot), District Ramban, UT of J&K.	Defense
52	NWHR	Hydrogeological Investigation for construction of tubewell at Akhnoor Military Station, District Jammu, UT of J&K.	Defense
53	NWHR	Hydrogeological Investigation for construction of tubewell at Lower Narayana, District Jammu, UT of J&K.	Civil
54	NWHR	Hydrogeological Investigation for construction of tubewell at TirumalaTirupatiDevasthanama , Majeen, District Jammu, UT of J&K.	Civil
55	NWHR	Hydrogeological Investigation for construction of tubewell at Group Centre, CRPF Bantalab, District Jammu, UT of J&K.	Defense
56	NWHR	Hydrogeological Investigation for construction of Rain Water Harvesting Structure outside RBI, Jammu, District Jammu, UT of J&K.	Civil

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
57	NWHR	Hydrogeological Investigation for Water Supply at Upper Bhajwal, UT of J&K.	Civil
58	NWHR	Hydrogeological Investigation for construction of tubewell at 09, Campoo Platoon, Khumbthang, Kargil, Ut of Ladakh.	Defense
59	NWHR	Hydrogeological Investigation for construction of tubewell at Signal Regiment area, Khumbthang, Kargil, UT of Ladakh.	Defense
60	NWHR	Hydrogeological Investigation for construction of tubewell at Cangma, Leh, UT of Ladakh.	Defence
61	NWHR	Hydrogeological Investigation for construction of tubewell at 09, Adam Area, Lamayuru&Hanuthang, Kargil, UT of Ladakh.	Defense
62	NWHR	Hydrogeological Investigation for construction of tubewell at Field Workshop Area, Achinathang, Biamah&Batalik, Kargil, UT of Ladakh.	Defense
63	NWHR	Hydrogeological Investigation for construction of tubewell at Armoured Brigade Village, Muth, Nyoma, Leh, UT of Ladakh.	Defense
64	NWHR	Hydrogeological Investigation for construction of tubewell at R&O(In front of HQ AVN BDE), Leh Garrison, UT of Ladakh.	Defense
65	NWHR	Hydrogeological Investigation for construction of tubewell at Map Location B, Leh Garrison, UT of Ladakh.	Defense
66	NWHR	Hydrogeological Investigation for construction of tubewell at LSRC Nalah, Leh Garrison, UT of Ladakh.	Defense
67	NWHR	Hydrogeological Investigation for construction of tubewell at Airforce Unit, Taru Village, AFS Leh, UT of Ladakh.	Defense
68	NWHR	Hydrogeological Investigation for construction of tubewell at Spituk Parka, AFS Leh, UT of Ladakh.	Defense

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
69	NWHR	Hydrogeological Investigation for construction of tubewell at New ARU Area, Defense Institute of Himalayan Research, Dihar, Leh, UT of Ladakh.	Defense
70	NWHR	Hydrogeological Investigation for construction of tubewell at Horticulture Area, Defense Institute of Himalayan Research, Dihar, Leh, UT of Ladakh.	Defense
71	NWHR	Hydrogeological Investigation for construction of Tubewell at Company Bagh Area, Jammu, District Jammu, UT of J&K.	Civil
72	NWHR	Hydrogeological Investigation for construction of Tubewell at 98 Bn, BSF, BOP, Mukesh, UT of J&K.	Defense
73	NWHR	Hydrogeological Investigation for construction of Tubewell at High Density Fruit Plants Nursery, Chakroi, District Jammu, UT of J&K.	Civil
74	NWHR	Hydrogeological Investigation for construction of Tubewell at Sunjwan Military Station, District Jammu, UT of J&K.	Defense
75	NWHR	Hydrogeological Investigation for construction of Tubewell at Muncipal Park, NardarMohalla, ManhasBirdari and PHE, GujjarMohala, Janipur, District Jammu, UT of J&K.	Civil
76	NWHR	Hydrogeological Investigation for Calculation of discharge of Dugwell at Jambu Zoo, District Jammu, UT of J&K.	Civil
77	NWHR	Hydrogeological Investigation for construction of Tubewell at Kouri, Judda, Dugga&Sawlakote, District Reasi, UT of J&K.	Civil
78	NWHR	Hydrogeological Investigation for construction of Tubewell at Wazur, District Kulgam, UT of J&K.	Civil
79	NWHR	Hydrogeological Investigation for construction of Tubewell at Drugmulla District Kupwara, UT of J&K.	Civil
80	NWHR	Hydrogeological Investigation for construction of Tubewell at Naugam District Kupwara, UT of J&K.	Civil

SNo.	Region	Title of short term water supply investigation	Agency for which the study was taken up
81	NWHR	Hydrogeological Investigation for construction of Tubewell at Gulmarg, District Baramulla, UT of J&K.	Civil
82	NWHR	Hydrogeological Investigation for construction of Tubewell at Nanak Hills &Sangam Hills Defense Establishment, Leh, UT of Ladakh.	Defense
83	NWHR	Hydrogeological Investigation for construction of Tubewell at CRPF Camp, Roun Village, District Udhampur, UT of J&K.	Defense
84	NWHR	Hydrogeological Investigation for construction of Tubewell at CRPF Camp, Roun Village, District Udhampur, UT of J&K.	Defense
85	NWHR	Hydrogeological Investigation for construction of Tubewell at IIM Jammu, District Jammu, UT of J&K.	Civil
86	NWHR	Hydrogeological Investigation for water supply/ feasibility report construction of Tubewell at CRPF Camp, Kootah, Hiranagar, District Kathua, UT of J&K.	Defense
87	NWHR	Hydrogeological Investigation for construction of Tubewell at Signal Regiment of Northern Command, District Udhampur, UT of J&K.	Defense
88	NWHR	Hydrogeological Investigation for construction of Tubewell at Left and Right Embankment of Tawi River, District Jammu, UT of J&K.	Civil
89	CGWB,CR ,NAGPUR	Feasibility Study for Rain Water Harvesting and Artificial Recharge in Laxminarayan Institute of Technology (LIT), Nagpur City, Nagpur District, Maharashtra	Laxminarayan Institute of Technology (LIT), Nagpur City, Nagpur
90	CGWB,CR ,NAGPUR	Water Supply Investigation Report for the Possibilities of Rain Water Harvesting in Indian Army's Project of OTM Accession for Hq 41 Artillery Division and its Units at Wadgaonsheri, Pune District, Maharashtra	Indian Army's Project of OTM Accession for Hq 41 Artillery Division and its Units at Wadgaonsheri, Pune
91	CGWB,CR ,NAGPUR	Feasibility Study of Rainwater Harvesting and Artificial Recharge at Headquarter Maintenance command, IAF, Nagpur	Headquarter Maintenance command, IAF, Nagpur

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
92	CGWB,CR ,NAGPUR	WSI report for INS Hamla, Navy Base, Mumbai District, Maharashtra	INS Hamla, Navy Base, Mumbai
93	SUO, Delhi	Site selection report for construction of tube wells at new & old pinto parks, airforce station palam, delhi cantonment	Defense Establishments
94	NR	Serviceability (Yield) Test report for existing borewells.	Military Engineer Services, Garrison Engineer (West), New Cantt. Allahabad-211001
95	NR	Provision of Rain Water Harvesting System of AF STN Memaura, Asstt. Garrison Engineer (I)(AF) Memaura, Lucknow	Asstt. Garrison Engineer B/R (AF), Military Engineer Services, Memaura, Lucknow-900293
96	NR	JawaharNavodayaVidyalay, Sonbhadra Campus	JawaharNavodayaVidyalay , Sonbhadra Campus
97	NR	Feasibility of New Borewell Near PH No-26	Garrison Engineer (East)
98			Shahjahanpur Road, Bareilly Cantt.
99	NR	Feasibility Report on development of two borewell at Faizabad Cantt. Area	Ministry of Defence (GOI), Military Engineer Services, Garrison Engineer, Faizabad
100	NR	Provn of OTM ACCN and VEH Dispersal area for 44 COY (SUP) at Lucknow cantt.	Garrison Engineer (Project)
101			Military Engineer Services
102			229, MG Road
103			Lucknow-226002
104	NR	Rooftop Rain Water Harvesting Structure at Air Force Station, Memaura, Lucknow	Air Force Station, Memaura, P.O. Banthra Lucknow-226401
105	NR	Provision of Rain Water Harvesting system in KendriyaVidyalaya at AF Stn Memaura	AF StnMemaura
106	NR	Request for feasibility report for one TW in Kanpur Cant area	Garrison Engineer (I)
107			Military Engineer Services
108			Cariappa Road, Kanpur Cantt-04
109	NR	Request for feasibility report for TW at Airforce station Memaura	Airforce station Memaura
110	NR	Request for feasibility report for TW at UP State Institute of Forensic Science	UP State Institute of Forensic Science
111	NR	Installation of RTRWH at Ordinance Clothing Factory, Shahjahanpur, UP	Ordinance Clothing Factory, Shahjahanpur, UP

SNo	Region	Title of short term water supply investigation	Agency for which the study was taken up
112	NR	Feasibility report for construction of New Borewell at New Cantt. Allahabad	New Cantt. Allahabad
113	NR	Construction of New Borewell	Garrison Engineer E/M
114			Military Engineer Services
115			PIN-900450
116	NR	Report on the selection of New Tubewell site at Roberts Line, Garrison Engineer E/M, Cantt Lucknow	Roberts Line, Garrison Engineer E/M, Cantt Lucknow
117	NR	Hydro-geophysical investigation report of CRPF campus, Chakia, District Chandauli, U.P.	CRPF campus, Chakia, Chandauli district, Uttar Pradesh
118	WCR, Ahmedabad	HYDROGEOLOGICAL INVESTIGATION FOR WATER SUPPLY TO COAST GUARD STATION LEKAWADA, TALUKA & DISTRICT – GANDHINAGAR, GUJARAT	Coast Guard,Porbandar, Gujarat
119	ER, KOLKATA	Short Term Water Supply Investigation report on "Selection of tube well sites for provision of OTM Accommodation PH-II of INF BN No.3 at Bengdubi Military Station area, Darjeeling district, West Bengal" submitted and issued to MES.	Bengdubi Military Station area, Darjeeling district, West Bengal.
120	WR, Jaipur	Feasibility study for construction of tubewell at Kishangarh Military station, Sam Block, Jaisalmer District, Rajasthan	DefenceKishangarh Military Station, Jaisalmer
121	WR, Jaipur	Feasibility study for construction of tubewell at Sonar Environmental Park & Training Area, Located in Military station Jaisalmer, Rajasthan	Defence ,Sonar Environmental Park & Training Area, Military station Jaisalmer, Rajasthan
	WR, Jaipur	Feasibility Study for installation of borewell at South West Command HQ office complex, Military Cantonment, Jaipur District, Rajasthan	Defence ,South West Command HQ office complex, Military Cantonment, Jaipur District, Rajasthan
122	WR, Jaipur	Feasibility Study for installation of borewell in Queens Line (MG's House) at Military Cantonment, Jaipur District, Rajasthan	Defence, Military Cantonment, Jaipur District, Rajasthan
123	WR, Jaipur	Provision of 1 Nos. Borewell at Sagat Singh Stadium at Jaisalmer Military Station, Jaisalmer	Defence ,Jaisalmer Military Station, Jaisalmer

12.0 TECHNICAL REPORTS AND PUBLICATIONS

Sr No	Region	Title of Reports Published
1	NWR, Chandigarh	Ground Water Year Book 2021-22 (Haryana)
2	NWR, Chandigarh	Ground Water Year Book 2021-22 (Punjab & Chandigarh UT)
3	SER, Bhubaneswar	NAQUIM Report of Keonjhar District
4	SER, Bhubaneswar	NAQUIM Report of Ganjam District
5	SER, Bhubaneswar	NAQUIM Report of Khurda District
6	SER, Bhubaneswar	NAQUIM Report of Jharsuguda District
7	SER, Bhubaneswar	NAQUIM Report of Gajapati District
8	SER, Bhubaneswar	NAQUIM Report of Cuttack District
9	SER, Bhubaneswar	NAQUIM Report of Subarnapur District
10	SER, Bhubaneswar	NAQUIM Report of Angul District
11	SER, Bhubaneswar	Master Plan of Artificial Recharge -2020
12	SER, Bhubaneswar	Dynamic Ground Water Resources of Odisha-2020
13	SER, Bhubaneswar	Ground Water Year Book -2020
14	NCR	National Aquifer Mapping Reports- 9 Nos
15	NCR	Basic data Reports - 42
16	NCR	Special Study Report - Draft
17	NCR	Ground Water Year Book
18	NCR	Dynamic Ground water Resource Book
19	SECR, Chennai	Ground Water Year Book 2020 – 21 - 1 No.
20	SECR, Chennai	GW Bulletin – 4 Nos. (4 measurements)
21	SECR, Chennai	Aquifer Mapping and Management Plan Gundar, Thambraparni Aquifer Systems
22	SECR, Chennai	Basic Data Reports of Exploratory Wells drilled during 2021-22 in Tamil Nadu - 60 nos.
23	SWR	AMP Taluk Report Conventional format: 5
24	SWR	AMP Taluk Report AIMS format: 68
25	SWR	Micro level Report: 11
26	SWR	NAQUIM District Report: 2
27	SWR	Periodical monitoring Report (Karnataka & Goa): 4
28	SWR	Water supply investigation: 9
29	SWR	Ground Water Year Book 2020-21: 1 no Karnataka &1 no Goa
30	SWR	Geophysical report: 1
31	SWR	Hydro-chemical Report: 1
32	SWR	BDRs: 227
33	SWR	GEC Reports_2020: 2
34		1. Karnataka
35		2. Goa

Sr No	Region	Title of Reports Published
36	NCCR, Raipur	Ground water yearbook, Chhattisgarh (2020-21)
37	NCCR, Raipur	Annual General report 2021-22
38	NCCR, Raipur	Aquifer Mapping and Management Plan of Mungeli district
39	NCCR, Raipur	Aquifer Mapping and Management Plan of Baloda Bazar district
40	NCCR, Raipur	Aquifer Mapping and Management Plan of Surajpur district
41	NCCR, Raipur	Aquifer Mapping and Management Plan of Jangir Chapa district
42	NCCR, Raipur	Aquifer Mapping and Management Plan of Balod district
43	NCCR, Raipur	Aquifer Mapping and Management Plan of Koriya district
44	NCCR, Raipur	Short Term investigation around the NIBSM campus in Baronda, Raipur for groundwater development.
45	NCCR, Raipur	Basic Data Report of Kerajhariya, Beliya, Badwar, Bhalaur, Bidritola, mero&PatadandExploratory Wells Drilled in Koriya District, Chhattisgarh state.
46	NCCR, Raipur	Basic Data Report of Khusuropali villageExploratory Well Drilled in Gariyaband District, Chhattisgarh state.
47	NCCR, Raipur	Basic Data Report of Amartal, Bhilouni, kesla, Khokra, Kirit, Mudapar, VyasnagarExploratory Well Drilled in Jangir Chapa District, Chhattisgarh state
48	NCCR, Raipur	Basic Data Report of Accholi, Bhatagaon, Sirrabhata, Tamora, Tatenga, Khamtarai village Exploratory Well Drilled in Balod District, Chhattisgarh state
49	NCCR, Raipur	Basic Data Report of Bharda, Telki, Bawli, Lalpur, Umaria, Sendri village Exploratory Well Drilled in Mungeli District, Chhattisgarh state
50	NCCR, Raipur	Basic Data Report Sitapar, Bitkuli, Hirmi, Datan and Sanda village Exploratory Well Drilled in Baloda Bazar District, Chhattisgarh state
51	NHR	Ground Water Year Book
52	NHR	Ground Water Exploration Report
53	UR	Ground Water Year Book
54	UR	Ground Water Resource Estimation
55	NWHR	Ground Water Year Book
56	CGWB,CR,NAGPUR	Aquifer Mapping & Management Plan Report of Nanded
57	CGWB,CR,NAGPUR	Aquifer Mapping & Management Plan Report of Solapur
58	CGWB,CR,NAGPUR	Aquifer Mapping & Management Plan Report of Nandurbar

Sr No	Region	Title of Reports Published
59	CGWB,CR,NAGPUR	Report on Electrical Resistivity Surveys for Aquifer Mapping (NAQUIM) in Parts of Maharashtra State
60	CGWB,CR,NAGPUR	Groundwater Yearbook 2020-2021
61	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Bhadhurni, MulTaluka,Chandrapur District, Maharashtra
62	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Dongargaon, MulTaluka,Chandrapur District, Maharashtra
63	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Awali, PanhalaTaluka,Kolhapur District,Maharashtra
64	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Mendhki, BrahmapuriTaluka,Chandrapur District, Maharashtra
65	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Bhanapur,
		SaoliTaluka, Chandrapur District, Maharashtra
66	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Sushi Dabgaon,
		MulTaluka, Chandrapur District, Maharashtra
67	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Ladbori,
		SindewahiTaluka, Chandrapur District, Maharashtra
68	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Neri, ChimurTaluka,Chandrapur District, Maharashtra
69	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Chamorshi, ChamorshiTaluka,Gadchiroli District, Maharashtra
70	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Konsari, ChamorshiTaluka,Gadchiroli District, Maharashtra
71	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Navegaon, ChamorshiTaluka,Gadchiroli District, Maharashtra
72	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Kotoli, ShahuwadiTaluka,Kolhapur District, Maharashtra
73	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Welldrilled at Pedrewadi, AjraTaluka,Kolhapur District, Maharashtra
74	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at KasbaSangaon, KagalTaluka,Kolhapur District, Maharashtra
75	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Shingnapur, KarvirTaluka,Kolhapur District, Maharashtra

Sr No	Region	Title of Reports Published
76	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at ShiraleTarfMalkapur, ShahuwadiTaluka,
		Kolhapur District, Maharashtra
77	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Anuskura, ShahuwadiTaluka,Kolhapur District,Maharashtra
78	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Gaganbawda, GaganbawdaTaluka,
		Kolhapur District,Maharashtra
79	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Hamidwada, KagalTaluka,Kolhapur District,Maharashtra
80	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Hattivade, AjraTaluka, Kolhapur District, Maharashtra
81	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Kadgaon, GadhinglajTaluka, Kolhapur District, Maharashtra
82	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well and Observation Well at Haladgaon, HinganaTaluka,
		Nagpur District, Maharashtra
83	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Ambapur, SakriTaluka, Dhule District, Maharashtra
84	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Ghoti, RamtekTaluka,Nagpur District, Maharashtra
85	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well and Observation Well Drilled at Kondamedi, MaudaTaluka,
		Nagpur District, Maharashtra
86	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Nandagomukh, SavnerTaluka,Nagpur District, Maharashtra
87	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well and Observation Well Drilled at Wadamba, RamtekTaluka,
		Nagpur District, Maharashtra
88	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Nawargaon, MaudaTaluka,Nagpur District, Maharashtra
99	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Nayakund, ParseoniTaluka,Nagpur District, Maharashtra
100	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well and Observation Well Drilled at Tarsha, MaudaTaluka,
		Nagpur District, Maharashtra
101	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well and Observation Well Drilled at Tekadi, ParseoniTaluka,
		Nagpur District, Maharashtra

Sr No	Region	Title of Reports Published
102	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Kelwad, SavnerTaluka,Nagpur District, Maharashtra
103	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Rewral, MaudaTaluka,Nagpur District, Maharashtra
104	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Warud, ShindkhedaTaluka, Dhule District, Maharashtra
105	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Boradi, ShirpurTaluka, Dhule District, Maharashtra
106	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Chillare, ShirpurTaluka, Dhule District, Maharashtra
107	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Dussane, SakriTaluka, District Dhule, Maharashtra
108	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Junawane, DhuleTaluka, Dhule District, Maharashtra
109	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Kawathi, DhuleTaluka, Dhule District, Maharashtra
110	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Rajkot, SakriTaluka, Dhule District, Maharashtra
111	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Waki, SakriTaluka, Dhule District, Maharashtra
112	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Kondigre, ShirolTaluka, Kolhapur District, Maharashtra
113	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Champa, UmredTaluka, Nagpur District, Maharashtra
114	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well and Observation Well at Akluj, MalshirasTaluka, Solapur District, Maharashtra
115	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Aundh, KhatavTaluka, Satara District, Maharashtra
116	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at PendhriMukta,
117		SaoliTaluka, Chandrapur District, Maharashtra
118	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well Drilled at Bhoyani, ManoraTaluka,Washim District, Maharashtra
119	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Dhanoli, KuhiTaluka, Nagpur District, Maharashtra
120	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Mandwa, BhiwapurTaluka, Nagpur District, Maharashtra
121	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well atSalwa, KuhiTaluka, Nagpur District, Maharashtra
122	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Manori, Umred, Nagpur District, Maharashtra

Sr No	Region	Title of Reports Published
123	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well atSukaliGharapure, HingnaTaluka, Nagpur District, Maharashtra
124	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Salai (Godhani), Nagpur Rural, Nagpur District, Maharashtra
125	CGWB,CR,NAGPUR	Basic Data Report of Exploratory Well at Malkhambi, MalshirasTaluka, Solapur District, Maharashtra
126	SUO, Delhi	Ground Water Year Book 2020-21
127	SUO, Delhi	Ground Water Resource Estimation 2020
128	NR	National Aquifer Mapping Reports- 13Nos (Aurraiya, Basti, Etawah, Ayodhya, Gorakhpur, Hardoi, Jaunpur, Kannauj, Mahrajganj, Mainpuri, Mirzapur, Philibhit and Unnao district)
129	NR	Basic data Reports – 26
130	NR	Ground Water Year Book
131	NR	Dynamic Ground water Resource Book as on 31st March 2020.
132	NR	State Report- Geophysics Volume – I (Surface Geophysics)
133	NR	Report on Ground Transient Electromagnetic (GTEM) survey in parts of Sitapur, Barabanki and Lucknow district of Uttar Pradesh.
134	WCR, Ahmedabad	Basic Data Reports of Exploratory Wells drilled during 2021-22 in Gujarat - 10 nos.
135	WCR, Ahmedabad	Aquifer Mapping and Management Plan of Surendranagar District, Gujarat
136	WCR, Ahmedabad	Aquifer Mapping and Management Plan of Botad District, Gujarat
137	WCR, Ahmedabad	Aquifer Mapping and Management Plan of DevbhumiDwarka, Gujarat
138	WCR, Ahmedabad	Aquifer Mapping and Management Plan of Bhavnagar, Gujarat
139	WCR, Ahmedabad	Aquifer Mapping and Management Plan of Amreli, Gujarat
140	WCR, Ahmedabad	Aquifer Mapping and Management Plan of Valsad, Gujarat
141	WCR, Ahmedabad	Gujarat state Ground water Year Book 2021-22
142	WCR, Ahmedabad	Dynamic Ground water Resource Book as on 31st March 2020
143	CGWB, ER, KOLKATA	Ground Water Year Book 2020 – 2021 for West Bengal and Andaman & Nicobar Islands

Sr No	Region	Title of Reports Published
144	- do -	Aquifer Mapping studies in parts of Murshidabad, Birbhum&Bardhaman districts, West Bengal
145	- do -	Report on special studies entitled, "Hydro meteorological impact on the ground water regime of West Bengal",
146	- do -	NAQUIM Report entitled "Aquifer mapping and Aquifer Management Plan of DakshinDinajpur district"
147	WR, Jaipur	Rajasthan State Ground water Year Book 2021-22
148	WR, Jaipur	Dynamic Ground water Resource Book as on 31st March 2020
149	WR, Jaipur	State Hydrogeological Report-Rajasthan-2020-2021
150	WR, Jaipur	Aquifer Mapping & Management Plan Report- Tonk, Rajasthan
151	WR, Jaipur	Aquifer Mapping & Management Plan Report - Chittorgarh Rajasthan
152	WR, Jaipur	Aquifer Mapping & Management Plan Report-Bundi, Rajasthan
153	WR, Jaipur	Aquifer Mapping & Management Plan Report- Rajsamand ,Rajasthan
154	WR, Jaipur	Aquifer Mapping & Management Plan Report- Pratapgarh ,Rajasthan
155	WR, Jaipur	Aquifer Mapping & Management Plan Report -Churu. Rajasthan
156	WR, Jaipur	Aquifer Mapping & Management Plan Report - Jaipur,Rajasthan
157	WR, Jaipur	Aquifer Mapping & Management Plan Report- Udaipur ,Rajasthan
158	WR, Jaipur	Basic Data Reports of Exploratory Wells drilled during 2021-22 in Rajasthan
159	WR, Jaipur	Ground Water Quality of Rajasthan
160	WR, Jaipur	Groundwater Pollution Control Studies in Industrial Cluster Of Jodhpur, Jodhpur, Rajasthan
161	WR, Jaipur	Report on Geophysical Data Gap Analysis for Aquifer Mapping Study and findings of previous Geophysical survey- Tonk District, Rajasthan
162	WR, Jaipur	Report on Geophysical Data Gap Analysis for Aquifer Mapping Study and findings of previous Geophysical survey- Pratapgarh District, Rajasthan

13.0 SPECIAL STUDIES

SNo	Region	Title of the Special Study	Collaborating Agency, if any	Duration of the study	Completed/continuing
1	NWR, Chandigarh	Selenium Contamination in Nawansheahr	In house		ongoing
2	NWR, Chandigarh	Causes and Remediation of Uranium contamination in Punjab & Haryana	In house		Ongoing
3	NWR, Chandigarh	IAEACRA Project	BARC	5 years	Ongoing
4	NWR, Chandigarh	High Resolution Aquifer Mapping and Management in arid regions of NW India (Collaborative study of CGWB-NGRI)	NGRI	1 year	Report compilation under progress
5	NCR	Hydro Geo-Chemical Investigation, Kumedi Village, Sanwer Block, District Indore M.P	-	2021-22	-
6	NCR	Tapi River Basin Management	India European Water Partnership Phase-2 (IEWP)	2021-2022	Continuing
7	SECR,Chennai	Hydro chemical studies in Industrial clusters: in Industrial clusters areas of Manali, Cuddalore, Vellore, Tiruppur, Erode and Coimbatore Districts.	-	1 Year	Completed
8	SECR, Chennai	Micro level mapping of Uranium hotspot areas and its relation with Palaeo channel	-	1 Year	Completed
9	SECR, Chennai	Study of ground water regime in Tiruvadanai Aquifers: in Ramanathapuram District		1 Year	Completed

SNo	Region	Title of the Special Study	Collaborating Agency, if any	Duration of the study	Completed/continuing
10	SECR, Chennai	Urban Hydrogeological study: in Tiruchirappalli District.	-	1 Year	Completed
11	UR	Assessment of dissolved radon concentration in groundwater of Haridwar & Dehradun Districts of Uttarakhand	NIH	2021-23	Continuing
12	SUO, Delhi	Rise in Water Table & Seepage in basements of Greater Kailash-II, New Delhi	DJB & DMRC	1Year	Completed
13	NR	Data Generation for aquifer mapping with focus on paleo channels in parts of Ganga Yamuna Doab in Kaushambi-Kanpur stretch, U.P.	NGRI, Hyderabad	2018 onwords	Study completed.
14	WCR, Ahmedabad	Ground water contamination around Viswamitri river stetches" at Vadodara,Gujarat.	Inhouse	1 Year	Completed
15	WCR, Ahmedabad	"An Appraisal on ground water quality of drinking water and irrigation purposes covering a stretch of Bharuch – Garudeshweralong Narmada river, Gujarat "	Inhouse	1 Year	Completed
16	WCR, Ahmedabad	Ground water contamination in and around polluted river stretches of Khari river,Gujarat"	Inhouse	1 Year	Completed
17	ER, KOLKATA	"Hydro meteorological impact on the ground water regime of West Bengal".	NIL	1 year	Completed.
18	WR, Jaipur	Groundwater Pollution Control Studies in Industrial Cluster Of Jodhpur, Jodhpur, Rajasthan	None	1 year	Completed.

14.0 REGULATION AND CONTOL OF GROUND WATER EXTRACTION

Central Ground Water Authority, constituted under Section 3 (3) of the Environment (Protection) Act, 1986 has been regulating ground water development in the country The Authority regulates in 20 States/ UTs (14 States/ UTs. Have their own Acts and 2 States are regulating under some GO). The Authority issues No Objection Certificate for ground water abstraction.

- Exercise of powers under section 5 of the Environment (Protection) Act, 1986 for issuing directions and taking such measures in respect of all the matters referred to in sub-section (2) of section 3 of the said Act.
- To resort to penal provisions contained in sections 15 to 21 of the said Act.
- To regulate and control, management and development of ground water in the country and to issue necessary regulatory directions for the purpose.
- Exercise of powers under section 4 of the Environment (Protection) Act, 1986 for the appointment of officers.

Important activities of CGWA during the period mentioned are given below:

14.1 Guidelines For Ground Water Regulation

Ministry of Jal Shakti notified the revised guidelines on 24.09.2020 for regulation and control of ground water development in the country. These guidelines have Pan India applicability. The salient features of guidelines are as follows:

- Individual domestic consumers (in both rural and urban areas) drawing ground water for drinking/ domestic use, Rural drinking water supply schemes, Armed Forces and Central Armed Police Forces Establishments, Agricultural users and Micro and Small Enterprises drawing < 10 KLD of groundwater are exempted from seeking NOC.
- No new industries except those MSME based on ground water are permitted in Over-exploited assessment units.
- Expansion of existing industries except MSME located in Over-exploited assessment units involving increase in quantum of ground water abstraction is not permitted
- Industries drawing >100 KLD of ground water in OE, Critical and Semi-critical assessment units are required to submit impact assessment reports along with the application.
- Industries drawing > 100 KLD of ground water in OE, Critical and Semi-critical assessment units are required to get annual water audit done and reduce fresh groundwater consumption by 20% over the next three years.
- No NOC is granted for groundwater extraction for water parks, theme parks, amusement parks in OE assessment units
- No ban on ground water extraction in Over-exploited units by infrastructure and mining projects, as these are location specific.
- All ground water users seeking NOC are required to pay ground water abstraction charges in Critical, Semi-critical and Safe assessment units and ground water restoration charges in OE assessment units. The charges so collected are to be utilized by States/ UTs on specific water conservation measures, specific supply/ demand side interventions.

- No charges are to be paid by users drawing saline ground water.
- Users drawing groundwater illegally are liable to pay Environmental Compensation.
- Penal provision has been kept for ensuring compliance of various conditions of NOC.
- Bulk water suppliers have also been brought in the ambit of NOC.
- Installation of Sewage Treatment Plants is mandatory for all new residential apartments/ Group Housing Societies where ground water requirement is more than 20 m3 /day. STP treated water is to be used by such users for flushing, green belt etc.
- All users seeking NOC are required to install tamper proof flow meters along with telemetry on all the ground water abstraction structures.
- In over-exploited assessment units, use of ground water for construction activity is permitted only if no treated sewage water is available within 10 km radius of the site.
- Committee for operationalization of

14.1.1 Processing Of Applications For Grant / Renewal Of No Objection Certificate (Noc) For Ground Water Withdrawal

CGWA evaluated applications from Industries/ Infrastructure Units / Mining Projects for grant of NOC for ground water withdrawal as per provisions of the Notified guidelines dated 24.09.2020. A total of **2616**nos. of new NOCs were **issued**and **592** nos. of NOCs **renewed** during the 2021-22. In addition, **exemption** was given in respect of **3977**Nos. **new** applications and **373 Nos renewal** cases. As compared to previous year, there has been an increase of **107%** in issuance of new NOC/ exemption and that of **170%** in case of renewals.Presently,NOCs are also being issued in Over-exploited, Critical and Semi-critical areasas per revisedguidelines dated 24.09.2020, which were earlier held up owing to observation made by Hon'ble NGT in its various orders. All such applications have since been processed and pendency cleared.

14.2 Monitoring Of Compliance Of Conditions Stipulated In The Noc

CGWA has developed online self-compliance format for the project proponent to upload the compliance conditions as stipulated in the NOC and the same is being monitored by the Regional Offices for ensuring the compliance conditions by the project proponent. Mobile app has also been developed for this purpose with the provision of verification by CGWA officers.

Show-cause Notices are also being issued to units, which have been found notto havefully complied with the NOC conditions. Orders for sealing of bore/tube wells and/ or disconnection of electricity supply through the concerned DCs/ DMs were also issued in respect of units, which did not give satisfactory replies to the show-cause notice.

While processing applications for renewal of NOC, compliance of NOC conditions by the proponents was ensured. Penalty of Rs. 1 lakh under Section 15 of Environment (Protection) Act of 1986 was imposed on the violators. NOC was issued after the compliance was completed and penalty was deposited by such offenders to the concerned Regional Office of CGWB.

14.3 46thMeeting of Central Ground Water Authority

The 46th meeting of Central Ground Water Authority (CGWA) was held on **08.06.2021** under the Chairmanship of ShriG C Pati, Chairman, CGWA through Video Conferencing.Member, CGWA apprised in brief about the developments that had taken place since the last meeting of the Authority held on 21.01.2021. Regional Director, CGWA made a brief presentation on the agenda items and the following decisions were taken during the meeting.

- The members agreed to extend the date of submission of Impact assessment Report by Industries, Comprehensive Hydrogeological Report by Mining projects and Water Audit up to 31/03/2022 without Environmental Compensation (EC). Firms who have already paid EC same shall be adjusted in the Abstraction/ restoration charges of the firm for subsequent years.
- Members agreed to a yearly advance payment of the abstraction charges, which would be adjusted in succeeding year bases on actual ground water withdrawal. It was suggested that a separate committee may be constituted to explore the possibilities of levying minimum charges on the industries in case of nonwithdrawal.
- Members agreed that timeline may be extended for installation of Temper proof Digital Water Meter and temper and Temper proof Digital Water Flow meter with Telemetry System by 31.03.2022, or as specified in NOC, whichever is later.
- Members agreed to constitute committee to fix the criteria for empanelment of Water Audit Agency/ Expert so as to have uniformity/ transparency in empanelment.
- The revised policy for accreditation was approved by the members.
- The committee did not agree to the proposal for allowing third party water audit to determine the quantity of ground water abstracted for the abstraction charges till the installation of digital flow meter with telemetry for all the mines.
- It has been agreed by the members that applicants can seek reimbursement of water charges / cess from CGWA in the subsequent years paid to State Governments limited to the water abstraction/ restoration charges levied by CGWA.

14.4 On Site Inspection By CGWB

On-site inspections were carried out by the Regional Offices of CGWB to check the compliance of NOCs accorded by CGWA before recommending the renewal applications to CGWA, New Delhi. Necessary show- cause notices were issued to the project proponents who have not complied with the conditions of the NOC issued by CGWA.

14.5 Developemt Of Advanced Nocap Software

Development of upgraded version of NOCAP has been initiated.

15.0 INTERNATIONAL COLLABORATION

Participating Country	Participating organization/University India and Foreign	Area of Study	Nodal Officer / alternate officer	Activities
Australia	CGWB, CHQ & NWR, Haryana GW Dept Australia – CSIRO	Insitu Salinity Remediation Jhaggar, Haryana	Nodal - Sh Anoop Nagar, Member (HQ) Supporting- Dr M.Senthilkumar, Scientist D	Finalization of study area is under approval with Chief Minister. State has submitted draft study area report.
EU	CGWB, CHQ, NR & UR, GIZ, IHE –Delft	Understanding Recharge dynamics in Ram Ganga Basin	Nodal - Sh Anoop Nagar, Member (HQ) Supporting- Sh S G Bhartariya, , RD Shr S N Dwivedi, , Scientist D	Sampling for stable isotopes to be taken up during May 2022. Discussion on the sampling locations to be finalized after discussion with IHE expert Sharing of report and lithologs
Netherlands	CGWB, CHQ & CGWB, SECR	Recharge measure to arrest Saline Water intrusion North of Chennai City Tamilnadu	Nodal – Sh Anoop Nagar, Member (HQ) & Dr Suresh, Scientist E Supporting- Dr M.Senthilkumar, Scientist D	Study area finalized north of Chennai City. Identification of state department Preliminary discussion held with Chennai Metro Water & Sewerage Board Further, meeting with Chennai Metro Water is being planned.
Hungary	CGWB, RGI	Vulnerability studies in Karst Aquifer system of Central Chhattisgarh.	Dr Ranjan Kumar Ray, , Scientist E	Study area finalized in Chattisgarh Exchange program and joint field investigaations

Participating Country	Participating organization/University India and Foreign	Area of Study	Nodal Officer / alternate officer	Activities
Morocco	CGWB, RGI	Ground Water Management	Nodal –Dr Utpal Gogoi, Member (E) &Dr Poonam Sharma., Scientist E	Preparatory meeting with CWC & NIC held under the Chairmanship of Dr. B. R. K. Pillai, Commissioner (CAD), Department of Water Resources, RD&GR on 30 june,21 in which the Board was asked to submit the agenda for the JWG meeting which held in july,21. The input of Board as discussed in the meeting is attached. PFA relevant details of MoU by MoJS for information. The minutes of the meeting has not been received as yet
Israel	CGWB, CHQ & CGWB NR	Ground Water Management	Nodal – Sh Anoop Nagar, Member (HQ) & Dr Suresh, Scientist E	Activities are being finalised
Tanzania	CGWB, RGI	Ground Water Management	Nodal – Sh Anoop Nagar, Member (HQ) RD RGI	Proposal submitted for capacity building of officers of Govt of Tanzania 2 proposal – Capacity building in Tanzania Capacity building in India
Japan	CGWB,	Ground Water Management	Nodal – Member (HQ) Supporting- Dr M.Senthilkumar, Scientist D	Land subsidence due to groundwater exploitation study is being taken up Groundwater quality changes due to irrigation water

16.0 RAJIV GANDHI NATIONAL GROUND WATER TRAINING AND RESEARCH INSTITUTE

16.1. Introduction

Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI) located at Raipur, Chhattisgarh caters to the training requirements of Central Ground Water Board (CGWB) and also many State Govt. Organizations, Public Sector Undertakings (PSUs), Academic Institutions, NGOs etc. in the field of ground water. Since the XII Plan, RGNGWTRI under HRD and Capacity Building Scheme of Ministry of Water Resources, River Development and Ganga Rejuvenation has been implementing a three-tier training programme keeping in view the requirements of the National Aquifer Mapping Project (NAQUIM). These training courses enable creation of a trained workforce for implementation of NAQUIM and overall sustainable development of the ground water resources of the country.

16.2 . Human Resource Development

It has been the earnest endeavor of CGWB to keep its technical personnel abreast with the latest developments in all aspects related to ground water development and management across the globe. Besides Officers of the Board, trainees from State Departments, PSUs, NGOs and candidates from abroad also participated in the training courses being organized by CGWB/RGNGWTRI.
During the AAP 2021-2022, the training program was envisaged in three different tiers i.e. Tier-I (National level), Tier-II (State level) and Tier-III (Block level). Total of 117 numbers of Training Courses (47-Tier I, 20-Tier II and 50- TierIII) were conducted by RGNGWTRI in which a total of 7962 trainees (1257- Tier I, 1086-Tier II and 5619- Tier-III) were imparted training including 2916 female participants.

Against the targeted 60 Tier-I training programme 47 trainings could be achieved due to restrictions on gathering to avoid the spread of COVID-19. Overall, although the target of trainings could not be achieved by 13 numbers, however, target in terms of number of trainees is exceeded by 362 participants. Out of 47 tier-I trainings only three (03) has been conducted in physical/offline mode, while rest of 44 were conducted in online mode due pandemic situation.

During the AAP 2021-2022, the training program was envisaged in three different tiers i.e. Tier-I (National level), Tier-II (State level) and Tier-III (Block level). The training calendar of RGNGWTRI for the year 2021-2022 is presented in the **Table 2**. The courses have been designed to impart state-of-the-art technologies aimed at the capacity building of the trainees to meet the emerging demand in ground water. Against the targeted 60 Tier-I training programme 47 trainings could be achieved due to restrictions on gathering to avoid the spread of COVID-19.Inspite of short fall of 13 no of trainings , trainees participation is exceeded by 362 participants. Out of 47 tier-I trainings only three (03) has been conducted in physical/offline modeat RGNGWTRI, Raipur, while rest of 44 were conducted in online mode due to pandemic situation.

The State and Block Level training programmes (Tier-II and Tier-III) conducted both physical and *online using MS TEAM / Google Meet platform* on various sub-themes covering various aspects of aquifer potential, development and management of aquifers, water quality and its suitabilityby the respective Regional Offices of CGWB. Out of 50 Tier-III program, 10 training were organised in online mode while 40 trainings were organised in physical/offline mode.All20 tier-II training programmes were conducted *online using MS TEAM / Google Meet platform.*

Table 16.1 Summary ofTargets& Achievements of training courses in AAP 2021-2022

Training Courses	Targets Training (Nos.)	Targets Trainees (Nos.)	Achievements (Nos.)	No. of Participants	Male	Female
TIER – I (National Level)	60	1200	47	1257	972	285
TIER – II (State Level)*	20	400	20	1086	691	395
TIER – III (Block Level)*	50	6000	50	5619	3383	2236
	130	**7600**	**117**	**7962**	**5046**	**2916**

*Tier – II and Tier – III training courses are conducted through CGWB's Regional Offices.

16.3 . Research and Development Projects

Apart from regular trainings, RGNGWTRI has taken up various research projects in collaboration with various prestigious institutes in India. The project wise progress of Research & Development Projects of RGNGWTRI, CGWB, and Raipur as on 31st March, 2022 is given in table 16.2 below:

Sl. No	Title of the project	Collaborating Organisations	Name of faculty	Duration	Progress as on 31.03.2022
1.	Sustainability of aquifer systems underlying Bemetara Block, District Bemetara, Chhattisgarh - a mathematical modelling approach.	RGNGWTRI, CGWB, Raipur & CGWB, NCCR, Raipur	Sh. Prabir Naik, Scientist D	May 2015-March 2019	Report published.
2.	Stable isotope investigations to study surface water – ground water interaction in a hard rock terrain in parts of Durg District, Chhattisgarh.	RGNGWTRI, CGWB, Raipur & PRL, Ahmedabad	Sh. B. K. Sahoo, Scientist D	December 2015 – March 2019	Report published.
3.	Impact of the Climate Change on the Ground Water Regime in Punjab and Haryana States.	RGNGWTRI	Dr. P. K. Naik, Scientist D		Data being processed
4.	Occurrence and Movement of Arsenic in ground water, Rajanandgaon, Chhattisgarh – An ground water modelling approach	RGNGWTRI	Dr. AnandiGayen, Scientist D	June 2017-March 2020	Draft Report submitted, under scrutiny

Sl. No	Title of the project	Collaborating Organisations	Name of faculty	Duration	Progress as on 31.03.2022
5.	Spring Flow Modelling in Sikkim Himalayas	RGNGWTRI, Raipur & ER, Kolkata	Dr. AVSS Anand, Scientist D	June 2019-March 2020	Collected basic data from Eastern Region, Kolkata
6.	Ground Water Flow Modelling Androth island, Lakshadweep	RGNGWTRI	Dr. AVSS Anand, Scientist D	June 2019-March 2021	Draft Report submitted, under scrutiny
7.	Sea water intrusion in coastal area of the Sundarbans : a study in and around Ganga Sagar Islands, West Bengal	RGNGWTRI	Dr. AnandiGayen, Scientist D	June 2019-March 2021	Draft Report submitted, under scrutiny
8.	Impact of Climate Change on Coastal Aquifers in parts of Tamil Nadu	RGNGWTRI	Sh. Sivakumar, Scientist D	2020-2022	Consultation of literature is in progress
9.	Impact of Jawaharnagar dumping site on Surface and Ground Water Quality in Mechel-Malkajgiri District, Telangana State	RGNGWTRI	Sh. J. Siddharthkumar, Scientist D	2020-2022	Consultation of literature is in progress

16.4 . Accreditation of Ground Water Professionals

As per new Notified Guidelines of Central Ground Water Authority, all projects extracting/proposing to extract ground water in excess of 100 m^3/day in Over-exploited, Critical and Semi-critical areas and mining projects shall have to submit mandatorily the impact assessment report of existing/ proposed ground water withdrawal on the ground water regime and also the report on the socio-economic impacts prepared by accredited consultants. Accreditation is a process of instilling an assurance and confidence amongst all the stake holders in acceptable standards of the reports/conclusions submitted by them for various purposes. Accreditation is a tool to assess and evaluate the standards and quality of the work done or to be done by an individual or an organization. The Applicants who seek permission for Water Extraction are confidant in hiring the services of the accredited individuals or institutions for conducting a hydrogeological survey and prepare a report on the ground water scenario and the impact of the proposed extraction on the ground water reservoir, which is mandatory for issuance of No Objection Certificate (NOC) from Central Ground Water Authority.

RGNGWTRI is entrusted for screening and scrutiny of applications received from individuals and Institutes for accreditation. During the year 2021-22, 38 Individuals and 13 Institutes are finalised and awarded certificate to work as 'Accredited individuals / Institutes' as be

16.5. Bhujal News

Bhujal News is a quarterly Journal of CGWB. From the year 2020, the publication of Bhujal News has now been started from Rajiv Gandhi National Ground Water Training and Research Institute (RGNGWTRI), Naya Raipur.

The work assignments in CGWB are mostly field-based and the output is measured by the submission of reports based on field studies. But, because the reports have limited circulation, the results of these field studies and ideas developed thereof do not reach the scientific community. Even CGWB's Officers are deprived of this information. Bhujal News fulfills this gap and serves as a conduit between CGWB and the outside world by highlighting CGWB's contributions in the field of ground water to the general public.

RGNGWTRI, Naya Raipur has published volume Bhujal News, 31 (1-4) 2021 in the month of March 2022. The same is uploaded and available for reference under the link

http://cgwb.gov.in/documents/Bhujal%20News_%20Vol-31_Final.pdf

It includes eight papers that cover Aquifer mapping, Geophysical, Hydrogeological and Chemical aspects of ground water. These papers are

Table. 5

Sl. No.	Title of Paper	Author(s)
	Occurrence of High Uranium in Shallow Ground Water in India	Sudhir K Srivastava & G C PaΘ
	Rejuvenaton and management of springs in Uttarakhand Himalaya	Anurag Khanna and Debasish Bagchi
	Assessment of selected Potenally Toxic Trace Elements contaminaΘon and groundwater chemistry of the Kolkata Megacity, West Bengal, East India	SuparnaDaΣa, Sonam, Anirvan Choudhury, Atalanta N Chowdhury, KeishamRadhapyari
	Decadal Groundwater Level Analysis of Lucknow District and Effect of Urbanizaon on Groundwater	Madhukar Singh, Sujatro Ray Chowdhury, Prashant Rai, VikasRanjan, Pramod Kr. Tripathi
	Seawater ingress, aquifer salinizaΘon and groundwater security in Puri Urban and adjoining areas area in parts of the coastal Odisha	S. Sahu, P. K. Mohapatra1, N.C.Nayak, A.Choudhury, C Mohanty, D.N. Mandal
	Decadal variaΘon in ground water Quality: A case Study of Jodhpur district, Rajasthan	Dr. S. K. Jain, Ms Balinder P. Singh and Dr. Jaipal Garg
	Assessment of Ground Water Quality in Industrial Zone of Dewas (MP) Central India	V. K. Kulshreshtha and Sourabh Gupta
	Study on ground water quality due to release of waste product from Chada Sugar and Industrial Pvt. Ltd., Village – Kiri Afgana, District- Gurudaspur, Punjab State	Tarun Mishra and Kiran Lale

16.6. Research papers published by the faculties

1. Dr. AnadiGayen, Scientist E has published a book chapter as a co-author, entitled "Review on ground water scenario of seven sisters along with Sikkim state in the northeastern regions of India" in Book Water Resources: Quality and Management with international ISBN published online in February 2022. Authors- Dr. AnadiGayen and Joji, V S.
2. Dr. AnadiGayen, Scientist E has published a book chapter as a co-author, entitled "Evolution of fresh water ecosystem in the very small tropical carbonate island in the Arabian Sea and its management" in Book Water Resources: Quality and Management with international ISBN published online in February 2022. Authors- Joji, V S. and Dr. Anadi Gayen.
3. Sh. TBN Singh, Scientist D has published a research paper as a co-author, entitled "Assessment of Potentially Toxic elements contamination on the fertile agricultural soils within Fluoride affected areas of Jamui district, Indo-Gangetic alluvium plain, India" in the Springer Journal- Water, Air, Soil Pollution (2022) 233:39, published online in January 2022. Authors- Suresh Kumar, Rambabu Singh, A.S.Venkatesh, G.Udaybhanu, and T.B.N.Singh.
4. Dr. AnadiGayen, Scientist D and Sh. YadaoraoMawale has published a book chapter on 'Review on the adoption of Sustainable Management Strategies through Monitoring and Analysis of Long-term behavior of Groundwater Regime on the Chhattisgarh State, India', in the Book title "Groundwater Management in Arid and Semi-Arid Regions of Hard Rock Terrains' edited by Sh. B. Venkateswara Rao, by BS Publication.
5. Dr. AnadiGayen, Scientist D, published online an article India one of the most Climate - Vulnerable countries in WATER@GOV, thewaterdigest.com.
6. Sh. Rajeev Ranjan Shukla, Scientist D as a co-author has published one research paper entitled 'Isotope and hydrochemical systematics of groundwater from a multi-tiered aquifer in the central parts of Indo-Gangetic Plains, India – Implications for groundwater sustainability and security' by TirumaleshKeesari, Uday Kumar Sinha, Dipankar Saha, S.N. Dwivedi, Rajeev Ranjan Shukla, Hemant Mohokar, Annadasankar Roy in Journal of Science of The Total Environment (Elsevier) Volume 789, 1 October 2021, 147860, 2021.
7. Dr. AnadiGayen, Scientist-D authored and published paper entitled "Prospective of Evaluation of Training (EoT) for Enhancing Professional Competence of the Hydrogeologists and Strengthening Participation of the Stakeholders- A Case Study" in International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), Volume 10, Issue 1, October 2021, pp 7-41.
8. Dr. AnadiGayen, Scientist D, presented paper entitled "Water Conservation in the hilly terrains: Ancient Practices and Innovative Perspectives" in the International Conference on Ground Water, 2021 on 29.10.2021.
9. Sh. Rajeev Ranjan Shukla, Scientist-D published paper entitled " Ground Water Potential of Bihar and insight into its sustainable development and Management" authored by A, K, Agrawal, S. K. Samanta, Indranil Roy and R. R. Shukla in Journal of Geological Society, Spec. Publ. Geol. Soc. India, No.9, 2021, pp 98-111.

10. Sh. K. C. Mondal, Scientist B, as a first author along with D. C. Jhariyal and H. S. Mandal published a paper on "Geoelectric Imaging to Assess Aquifer Conditions in Raipur City, Chhattisgarh, India, Using Schlumberger Method", at Journal Geological Society of India, Vol.97, August 2021, pp.943-950.
11. Dr. AnandiGayen, Scientist D, has published a research paper with the title "Scientific Approach towards Knowledge Enhancement and Capacity Building for the Professionals Working in Water Sector through Training Needs Analysis (TNA) Tool –A Case Study" inInternational Journal of Advanced Research in Science, Communication and Technology (IJARSCT) Volume 6, Issue 2, June2021, pp. 2581-9429.
12. Dr. AnandiGayen, Scientist D, has published a Book Chapter "Arsenic in Groundwater: Occurrence, Implications and Mitigation strategies" in 'Groundwater Quality – Assessment and Environmental Impact' edited by Rafael M. Vick, ISBN: 978-1-53618-807-3 published by Nova Science Publishers, Inc., pp. 1-52.
13. Sh. TBN Singh, Scientist D, as a co-author, has published one research paper on "Ground Water Resources of Jharkhand State and its Development Perspective", in Special publication of the 'Journal of Geological Society of India', No. 9, 2021.
14. Sh. Rajeev Ranjan Shukla, Scientist D as a co-author has published one research paper entitled 'Isotope and hydrochemical systematics of groundwater from a multi-tiered aquifer in the central parts of Indo-Gangetic Plains, India – Implications for groundwater sustainability and security' byTirumaleshKeesari, Uday Kumar Sinha, Dipankar Saha, S.N. Dwivedi, Rajeev Ranjan Shukla, Hemant Mohokar, Annadasankar Roy in Journal of Science of The Total Environment (Elsevier) Volume 789, 1 October 2021, 147860,2021.
15. Sh. Biswarup Mohapatra, Scientist C, as a co-author has published one research paper on 'Identifying Fluoride Endemic Areas and Exposer Pathways for Assessment of Non-Carcenogenic Human Health Risk Associated with Ground Water Fluoride for Gujrat State, India' in Journal of Environmental Science and Pollution Research by Springer, Published online: 05 May 2021. (Environ Sci Pollut Res (2021). https://doi.org/10.1007/s11356-021-14156-2).
16. Sh. M Siva Kumar, Scientist E, as a co-author, has published one research paper on "Climate Change Variability assessment on water resources by SWAT model: A review", in publication of the Journal of "Environmental Science", 13 March 2021.

16.7. OTHER Activities of RGNGWTRI

16.7.1 Awards and Recognition

Dr.AnandiGayenScientist D

Connecting Scholars Since 2014

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InSc Awards 2021

Certificate

Research Excellence Award 2021

awarded to

Dr. Anadi Gayen

M.Sc. (Applied Geology), Ph.D. (Hydrogeochemistry),
Post Doctoral Fellow (Hydrogeology)
Scientist-D and Senior Faculty
Department of Water Water Resources, River Development and Ganga Rejuvenation,
Ministry of Jal Shakti, Government of India.
Rajiv Gandhi National Ground Water Training and Research Institute
Chhattisgarh.

For the work with following details:

Publication Type: Journal
Paper Title: Geo- Spatial Distribution of Arsenic Contamination of Groundwater Resources in Intricate Crystalline Aquifer System of Central India: Arsenic Toxicity Manifestation and Health Risk Assessment. Published the Paper as a Corresponding Author.
Volume: 27
Issue No.: 6
Month of Publication: January
Year: 2021
Page No.: 1 - 25
ISSN: 1000-7619

Nanjesh Benner
Director, Iscbc

Best Paper Award

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5.731

INTERNATIONAL JOURNAL OF ADVANCED RESEARCH IN SCIENCE, COMMUNICATION AND TECHNOLOGY

IJARSCT

CERTIFICATE

OF APPRECIATION

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Anadi Gayen

Rajiv Gandhi National Ground Water Training and Research Institute, Naya Raipur, Chhattisgarh

FOR PUBLISHING THE RESEARCH PAPER ENTITLED

Sustainable Management of Springs for Water Security in Parts of the Sikkim Himalaya Region – An Overview

IN IJARSCT, VOLUME 9, ISSUE 1, SEPTEMBER 2021

Certificate No: BP-2021-0288

www.ijarsct.co.in

ISSN

Editor-in-Chief

1. **In Sc Award in category of Research Excellence Award 2021 :**

   Dr.AnandiGayen, Scientist D, has received Research Excellence Award 2021 for this paper **"Geo-spatial Distribution of Arsenic Contamination of Groundwater Resources in Intricate Crystalline Aquifer System of Central India: Arsenic Toxicity Manifestation and Health Risk Assessment"**, vol27(6), 2021, pp.1-25.

2. **Best Paper Award**

   Dr.AnandiGayen, Scientist D, has received Beast Paper Award for his paper '**Sustainable Management of Springs for Water Security in Parts of the Sikkim Himalaya Region – An Overview'** in IJARSCT, Volume 9, Issue 1, September 2021. The award was given by International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)

   Dr.K C Mondal, Scientist-B

   A proud moment for RGNGWTRI, CGWB as Dr. K C Mondal, Scientist B, RGNGWTRI, Central Ground Water Board was awarded PhD from NIT Raipur for the research topic

16.8. Nation Webinar on Smart Irrigation and Participatory Irrigation Practices

RGNGWTRI has successfully organized **Nation Webinar on Smart Irrigation and Participatory Irrigation Practices** on 12th Nov 2021. In which about 157 delegates, farmers, officers/officials from Krishi Vidyapeeth, NGO, Central and State Govt. organizations participated. Presentations were given by Sun Moksha Power Pvt. Ltd., Bangalore; Jain Irrigation Systems Ltd., Jalgaon, Maharashtra and Niagara Drip and Smart Irrigation Automation Systems, Tamilnadu on different smart irrigation techniques. The National Seminar was coordinated by Dr. Bhushan R. Lamsoge, Scientist D and Sh. Biswarup Mohapatra, Scientist C, RGNGWTRI.

Selected Photographs of activities at conducted by RGNGWTRI during 2021-2022

Photographs of the new building of RGNGWTRI, Naya Raipur

Main entrance gate

Inauguration stone

Admin & Academic Building

Hostel & Guest House Building

Admin & Academic entrance

Hostel & Guest House entrance

VC/Conference Hall with an interactive panel

Seminar / Assembly Hall

Training Hall

Training Hall

Smart Podium in Training Hall

Smart Podium in Training Hall

GIS & Computer Lab with Smart Podium

Smart Podium in Training Hall

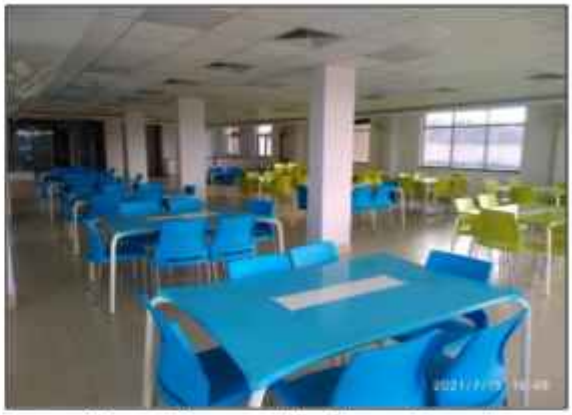
Regional Director's room

Faculty's room

A corridor between Admin & Hostel building

A corridor between Admin & Hostel building

Mess / Canteen facility at Hostel



Mess / Canteen facility at Hostel

Drinking-Water Facility

Visiters' waiting lobby in front of RD room

Guest faculty room

Hostel rooms for trainees

Lobby at Hostel

Lobby at Admin building

17.0 NATIONAL HYDROLOGY PROJECT (NHP)

National Hydrology Project (NHP) is a continuation of Hydrology Project (HP)- I and HP- II, a central sector scheme of Department of Water Resources, RD&GR, Ministry of Jal Shakti, Government of India with a total outlay of Rs. 3679.7674 Crore (Rs. 3,640 Crore for NHP and Rs. 39.7674 Crore for establishment of National Water Informatics Centre (NWIC), a repository of Nation-wide Water Resources data for a period of 08 years extending from 2016-17 to 2023-24. Out of the total outlay, 50% of the amount has to be repaid by Central Government to the Word Bank (WB) and the remaining 50% would be Central Assistance. The project will be implemented by 49 agencies involving 10 Central and 39 States agencies.Since CGWB is one of the implementing agencies of the NHP project, a budget of Rs 41.72 Crore has been allocated for the purpose.

17.1 Objective of NHP

To improve the extent, quality, and accessibility of water resources information, decision support system for floods and basin level resource assessment/ planning and to strengthen the capacity of targeted water resources professionals and management institutions in India.

17.2 The components of NHP are:

- **Component A:** Water Resources Monitoring systems
- **Component B:** Water Resources information systems
- **Component C:** Water Resources Operation and Planning systems
- **Component D:** Institutional Capacity Enhancement

NHP will improve and expand hydrology data and information systems, strengthen water resources operation and planning systems, and enhance institutional capacity for water resources management. The project will strengthen the information base and institutional capacity for effective decision making in water resources planning and operational management at the basin level across the country using the latest available technology and tools. NHP will also help in contributing to the Digital India initiative by integrating water resources information across state and central agencies.

17.3 Role of CGWB in NHP

CGWB being the Nodal Agency for Ground Water, NHP cell in all the CGWB, Regional Offices have been created for providing Technical Guidance & Support to States in the implementation of NHP.

17.4 The proposed major activities of CGWB under NHP

- **Real time monitoring of Water Level &Water Quality in coastal aquifers of Tamil Nadu & UT of Puducherry**through construction of 60 piezometers & installation of 60 DWLRs.
- **Establishment of Center of Excellence for groundwater modeling** for providing technical support to the Central/ State Agencies in Ground Water Flow Model Studies.
- **Real time monitoring of Water Level through Installation of 3400 additional DWLRs (Phase-1) in the existing Piezometers**located in the Overexploited, Critical and Semi-Critical Blocks of the Country.
- **Real time monitoring of Water Level through Installation of 1800 additional DWLRs(Phase 2) in the existing Piezometers**located in the Overexploited, Critical and Semi-Critical Blocks of the Country.
- **Capacity Building** of the officers from State / Central implementing agencies through National and International level trainings every year.
- **Awareness Campaigns** are being conducted at six places every year on state specific groundwater issues.
- Organizing **National Level Workshops** on need based Topics

17.5 The Status of various activities of CGWB under NHP is given below:

- The Activity Construction of 60 Piezometersin coastal aquifers of Tamil Nadu & UT of Puducherry for Real time monitoring of Water Level & Water Quality has been completed in the month of Oct-2021.

- Organized 03 Nos Domain Specific Training for 54 Nos officers from State/Central Implementing Agencies under NHP through RGI, Raipur during the months Sep 2022, Dec 2022 & Jan 2023

- Young Water Professional Training Program is an innovative professional training course that is led by Western Sydney University and Indian Institute of Technology, Guwahati on behalf in the Australia India Water Centre (AIWC). The program is collaboration between Australian Water Partnership and the National Hydrology Project (NHP), Ministry of Jal Shakti, Government of India in the field of Water Resources Management. The duration of the programme is for 1 Year (Nov 2021 – Oct 2022) and classes were conducted through Virtual mode. 03 No of Officers from CGWB have been nominated to attend the Young Water Professional Training Program.

- Procurement of 60 DWLRs with Water Quality Probe and Telemetry for installation in the 60 Piezometer constructed in the coastal aquifers of Tamil Nadu & UT of Puducherry for Real time monitoring of Water Level & Water Quality, for which Contract has been awarded in the Month of January 2022.

18.0 MODELLING TALK SERIES -CENTRE OF EXCELLENCE FOR GROUND WATER MODELLING

Centre of Excellence (CoE) has been set up with its headquarters in Faridabad and four regional Hubs each at NR, Lucknow, SECR, Chennai, SER, Bhubaneswar& WCR, Ahmedabad who are further looking after the regional offices as listed below.

Regional Hub	Location	Regions
North	NR, Lucknow	NWR, Chandigarh/ NWHR, Jammu/ NHR, Dharamshala/ UR, Dehradun/ SUO, Delhi
South	SECR, Chennai	SR, Hyderabad/ SWR, Bangalore/ KR, Trivandrum
East	SER, Bhubaneswar	ER, Kolkata/ MER, Patna/ NER, Guwahati
West	WCR, Ahmedabad	WR, Jaipur/ NCR, Bhopal/ NCCR, Raipur/ CR, Nagpur

The core activities of Centre of Excellence includes

- Providing technical expertise and hands-on support on ground water modelling.
- Coordinate the activities related to Ground Water Modelling with the officers concerned and Mentors.
- Progress monitoring on Modelling assignments to CGWB officers
- Organize regular presentations for exchange of knowledge within CGWB and with other institutions/departments/organizations.
- Standardization of methods and procedures for modelling and preparation of protocols in consultation with TAMC
- Procurement, distribution and maintenance of modelling as well as various GIS software in CGWB
- Coordination with NHP modelling team.
- Inventory of availability of important Software in CHQ as well as at the regional offices of CGWB is completed.
- Groundwater modelling framework document has been prepared and provided to all the modelers of CGWB.

In addition to the regular activities, CoE has also undertaken following activities:

- Compilation of a compendium of important Ground water modelling studies taken up in CGWB since inception is in advanced stage.
- A google group and whats-app group has been created for quick interaction and sharing of ground water modelling related issues faced by CGWB officers during modelling.

- Modelling study for an area of 64,500 sq.kms have been completed and report are under finalization.
- Four modelling studies have been taken up under the CoE, they include
    - Udam Singh Nagar, Uttarakhand
    - Bhiwadi Industrial cluster, Rajasthan
    - Neemrana Industrial cluster, Rajasthan
    - Ghaziabad Industrial cluster, NCR & UP

18.0 GROUNDWATER MODELLING TALK SERIES

CoE also organized a series of external and internal talks to which enabled Scientists of CGWB gained knowledge on the best practices followed around the world in data collection, application of groundwater modelling for problem solving and advanced data analysis. Scheduled was planned with one fortnight talk from external expert, followed by other fortnightly talk from the internal resources of CGWB on the modeling work completed. The details of the external lectures is placed blow;

Sl.No	Name of the expert delivering talk	Topic of the GW Modelling talk	Date
1	Prof Phanindra, Asst Professor IIT Hyderabad	**Application of GRACE data in Water Resources Management**	07/01/2022
2	**Prof ElgaSalvadore** **IHE Delft**	**Use of satellite data and other global datasets for water resources assessment, management and water accounting**	04/02/2022
3	Sh S K Sinha, Advisor, & Scientist (Retd), Atal Jal	**Ground Water Modelling Past Experience and Way Forward**	04/03/2022
4	Dr S Suresh, Scientist D, CGWB Faridabad	**Groundwater Modelling Tips for budding modelers with a case study**	04/03/2022

19.0 Pradhan Mantri Krishi Sinchayee Yojana- Har Khet Ko Pani - Ground Water (PMKSY-HKKP-GW)

The Government has formulated Pradhan Mantri Krishi Sinchayi Yojana *'Har Khet Ko Pani'* Ground Water Irrigation scheme with the vision of extending the coverage of irrigation in a focused manner. The scheme of PMKSY was approved in 2015-16 for creating additional irrigation from ground water resources. Ground water component aims utilizing ground water for irrigation purpose in areas, where ground water is sufficiently available. Further, to enhance small and marginal farmer's income in such areas by providing assured irrigation facility under the scheme. The Operational guidelines were issued by the Department of Water Resources, RD & GR, Ministry of Jal Shakti vide OM no P-15011/24/2016-SPR/2037-2105 dated July 2016. However, keeping in view of various requirements to implement the scheme and feedback received from State Governments, guidelines have been revised on 8.11.2018 and 28.05.2019. The scheme effectively launched in 2019-20 after revision of guidelines

Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti has sanctioned 13 projects under PMKSY-HKKP-GW scheme since 2019 in 10 States namely Assam, Arunachal Pradesh, Gujarat, Nagaland, Manipur, Mizoram, Tripura, Tamil Nadu, Uttar Pradesh, Uttarakhand.

The total cost of these 13 ongoing projects are Rs. 1020.53 Crore and Central Assistance is Rs.852.64 Crore of which **700.33** crore has already been released as on October'22 by the Ministry of Jal Shakti.

Presently, seven project in Arunachal Pradesh, Assam, Manipur, Nagaland, Tripura and Gujarat have been completed with the help of combined effort of CGWB, DoWR, RD& GR, Ministry of Jal Shakti and concerned State Implementing Agency.

As on November 2022, 29229 wells have been constructed, 77123 Ha command area have been created and 66440 number of small & marginal farmers have been taken the benefit of PMKSY-HKKP-GW scheme so far.

The details & present status of these projects are summarized in the following table:

Status of Fund Allocation in last 3 year under PMKSY-HKKP-GW (as on 31 March' 22)

Fund release to the State in FY- 2019-20, 2020-21, and 2021-22 under PMKSY-HKKP-GW Scheme							
Sl. No	**State**	**Projects**	**Project cost in crore**	**CA Releasd (as on 31 March'22)**	**CA released in 2019-20**	**CA released in 2020-21**	**CA released in 2021-22**
1	Assam	Assam Phase-I	246.07	435.96	132.87	13	37.8
2		Assam Phase-II	292.01		0	10	242.29
3	Arunachal Pradesh	Arunachal Pradesh Pahse-I	45.3	79.89	24.46	10	5.985
4		Arunachal Pradesh Pahse-II	44.95		24.15	0	15.295
5	Gujarat	Gujarat	163.29	34.5	6	0	28.5
6	Nagaland	Nagaland	18.15	15.6	9.75	0	5.85
7	Manipur	Manipur	61.68	54.396	0	33.306	21.09
8	Mizoram	Mizoram	16.04	8.66	0	8.66	0
9	Tamil Nadu	Tamil Nadu	9.13	5.28	0	3.67	1.61
10	Tripura	Tripura Phase-I	13.31	35.632	7.15	0	2.38
11		Tripura Phase-II	48.34		0	0	26.1
12	Uttarakhand	Uttarakhand	15.89	13.72	0	1.36	12.36
13	Uttar Pradesh	Uttar Pradesh	46.37	16.69	16.69	0	0
	Total		1020.53	700.33	221.07	80.0	399.26

Status of PMKSY-HKKP-GW As on 31st March'22				
S. No	**Projects**	**Wells to be constructed Target/ Achievement Nos**	**Project Command Target/ Achievement Ha**	**Beneficiaries Target/ Achievement Nos**
1	Assam-Ph-I	4779 / 4779	19116 / 19116	19643 / 19643
2	Assam Phase-II	4916/ 4753	19680/ 12240	17216/ 12117
3	Arunachal Pradesh-Ph-I	473 / 473	1785/ 1785	3350/ 3350
4	Arunachal Pradesh Ph –II	519/ 519	1957/ 1957	3633/ 3637
5	Nagaland	262/ 262	667/ 667	264/ 264
6	Tripura Ph-I	231/ 231	339/ 339	851/ 844
7	Tripura Ph-II	890/ 778	2670/ 265	1639/ 269
8	Manipur	550/ 550	2057/ 2057	1445/ 1445
9	Mizoram	209/ 96	553/ 188	411/ 191
10	U.P.	14752/ 13361	28085/ 26613	15252 13849
11	Uttarakhand	206/ 204	1030/ 860	1075/ 859
12	Gujarat	2512/ 841	3768/ 879	3655/ 1104
13	Tamil Nadu	166/ 147	610/ 374	1233/ 866
	Total	**30465/ 26994**	**82317/ 67340**	**69667/ 58438**

20.0 SEMINARS/MEETINGS/WORKSHOPS/PAPER CLIPS

20.1 Seminars/Workshops/

- Ms. Sayelli Tembhurne, Sc-B gave presentation on "Ground Water Resources Management Strategy in Eastern part of Rajasthan with special reference to Sawai Madopur District, Rajasthan, India" in International Virtual Conference on Ground Water organized by department of Geology, Mohan Lal Sukhadiya, Udaipur.

- Sh. P.K. Jain, Regional Director, CGWB, NCR, Bhopal presented detail, the new CGWA guidelines as a resource person in Virtual conference for Understanding & Re-Engineering Water & Wastewater Management organised by Confederation of Indian Industry(CII) on 27.08.2021.

- "Nadi Utsav"- 'Celebrating River of India' under Azadi Ka Amrit Mahotsavon 17th December 2021 at Burhanpur Cleanliness Drive at Tapti river bank- Rajghat in Burhanpurandawareness programme on KundiBhandara at KundiBhandara, Burhanpur.

- Sh P. K.Jain , Regional Director, Dr. Rakesh Singh Sc-D Sh. Kamlesh OjhaSc-B, Dr. K.Paramasivam AHG, Sh.K.L.Pradeep AHG, attended the one day reflection workshop under Atal Bhujal Yojana on 06/12/2021.

- Dr. Rakesh Singh Sc-D, attended and presented National water convention Gwalior 2021. On 10/12/2021.
- Workshop on Hindi was organized through video conferencing on 13th August 2021 at SECR, Chennai. Shri. Rakesh Gupta, Deputy Director(Rajbhasha) graced the occasion and during his speech he briefed about incentives and remuneration offered for the work done in Hindi. Officers and staff of SECR and Division IV participated in the workshop.

- State level workshop on "Springshed Management in Uttarakhand" was organized by CGWB, UR, Dehradun in association with Peoples Science Institute (PSI) at Dehradun on 02/09/2021 (Through online and offline-hybrid mode).

- The Monthly Review meeting was held on 21.10.2021 at Jilla Seva Sadan, Modasa, District Aravalli under Chairmanship of Additional District CollectorShri Pradipsinh Rathod, Aravalli, Modasa for information and guidelines regarding implementation of CGWA guidelines. As a member of meeting, Dr.K.M.Nayak, Scientist-B, CGWB, Ahmedabad described all the details of CGWA guidelines to committee member

- Under Azadi ka Amrit Mahotsav (AKAM) to Celebrate Rivers of India.River Utsav was celebrated by Central Ground Water Board, Ahmedabad as an awareness program on bans of SaraswatiRiver at Rani Ki Vav, Patan on 21.12.2021. Total 52 participants (25 females, 27 males) attended the program.

20.3 Public Interaction Programmes

Public Interaction Programmes (PIP) Report (till Mar. 2022)									
	2018 - 21	2021-22				2018-22			
Sl. No.	State	Target 2021-22	Total PIPs conducted (Since 15 Mar. 2021)	Total participants	Female Participants	Cumul. Target 2018-22	PIPs conducted	Participants since April 2018	Female Participants since April 2018
1	Andaman & Nicobar Islands					9	2	158	43
2	Andhra Pradesh	8	8	665	159	40	26	1561	329
3	Arunachal Pradesh	4	0	0	0	4	0	0	0
4	Assam	10	14	577	224	54	52	2176	860
5	Bihar	20	17	1580	454	48	42	4007	1474
6	Chandigarh	1	1	38	9	3	7	468	158
7	Chattisgarh	19	20	1979	896	76	61	5628	2567
8	Goa		2	97	31	0	2	97	31
9	Gujarat	25	22	805	146	84	51	3193	715
10	Haryana	15	17	1345	593	62	31	2560	1114
11	Himachal Pradesh	10	10	637	279	43	44	3529	1721
12	J. & Kashmir	10	10	376	35	24	16	807	113
13	Jharkhand	20	10	697	299	42	32	3062	1457
14	Karnataka	23	20	1725	474	90	72	5077	1927
15	Kerala	10	11	1066	459	62	58	4588	2958
16	Ladakh		2	41	1	0	2	41	1
17	Mad. Pradesh	35	20	1036	139	82	56	5152	891
18	Maharashtra	22	24	2410	628	68	63	6359	1723
19	Manipur	1	1	33	5	1	1	33	5
20	Meghalaya	6	7	701	359	17	16	1297	607
21	Mizoram		0	0	0	2	2	317	154
22	Nagaland	1	0	0	0	2	1	11	0
23	Odisha	20	18	2472	1003	66	73	6360	2603
24	Punjab	10	8	504	184	64	21	3519	1337
25	Rajasthan	20	21	898	212	65	37	2471	460
26	Sikkim					5	1	80	31
27	SUO- Delhi	8	10	696	230	20	17	1071	315
28	Tamil Nadu	30	29	2930	1268	80	69	7522	4138
29	Telangana	8	7	755	295	41	52	2462	898
30	Tripura	4	4	235	110	5	10	542	238
31	Uttar Pradesh	42	56	4946	1553	92	99	9285	2249
32	Uttarakhand	6	9	417	138	21	17	1001	220
33	West Bengal	13	14	947	285	57	59	4668	1293
	Total	**401**	**392**	**30608**	**10468**	**1329**	**1092**	**89102**	**32630**

20.4 Visit Of Hon'ble Ministers

- Sh. Pralhad Singh Patel, Hon'ble Union Minister of State, MoJS, visited Nagpur at Dr. Babasaheb Ambedkar International Airport, Nagpur on 25th August 2021.

- Hon'able Union Minister Sh. Nitinjee Gadkari, Mo RT & H, Sh. NarendrajeeTomarMoA& FW, Dr. Aswath Narayan, Hon'ble IT minister Karnataka State and Sh. Sanjay Agrawal, Secreatry, MoA& FW, released a booklet on "Water Conservation & Artificial Recharge –A success Story Tamaswada Pattern Wardha, Maharashtra" during inaugural function of Agrovision 2021 held at Nagpur.

- Sh. Gajendra Singh Sekhawat, Hon'ble Union Minister of Ministry of Jal Shakti, Sh. Pankaj Kumar, IAS Secretary,(DoWR, RD &GR) MoJs and Sh. SubodhYadav, IAS, Joint Secretary, (Admin, IC& GW, DoWR, RD &GR) MoJs visited Guwahati and Tezpur, Assam for review of PMKSY Projects, Brahmaputra & Barak Valley Flood Control &erosion related issues, conference of PHE Ministers of NE states for JalJeevan Mission etc. from 16.09.2021 to 17.09.2021.

- Hon'ble Minister of State, Sh. Bishweswar Tudu Ministry of Jal Shakti and Tribal Development, Govt. of India, chaired a meeting on 'Introductory meeting of officers of Ministry of Jal Shakti, Govt. of India' at State Guest House, Bhubaneswar on 15.09.2021. Sh. P.K. Mohapatra, the Regional Director along with senior officers of SER, attended the meeting.

- Hon'ble Minister MOJS Sh Gajendra Singh Shekhawat chaired a review meeting of JJM and SBM of Southern States by, at Vidhana Soudha, Bangalore. Smt T.S. AnithaShyam, HOO, Sh. N. Jyothi Kumar, Sc- E, & Sh.H.P.Jayaprakash,Sc-D attended the The meeting was attended by Chief Minister of Karnataka and the Water Resources/Rural Development Minister from Southern States on 05.03.2022.

20.5 Parliamentary Committee Visits

- Parliamentary Standing Committee on Water Resources visit to Ganga Sagar on 25.08.2021. The main purpose of the visit was to understand the disposition of the fresh ground water aquifer system in the Island.

Parliamentary Standing Committee (Water Resources) visit to Ganga Sagar on 25.08.2021.

- Meeting of the Parliamentary Standing Committee on Water Resources with the Chief Secretary, Govt. of West Bengal, and other dignitaries on 26.08.2021 at the Banquet Hall of Lalit Great Eastern Hotel, Kolkata. Dr. P. Nandakumaran, Chairman, CGWB, Dr. S. K. Samanta, Regional Director, CGWB, ER, Kolkata, and officers of ER attended the meeting.

Meeting of the Parliamentary Standing Committee on Water Resources with the Chief Secretary, Govt. of West Bengal, and other dignitaries on 26.08.2021 at the Banquet Hall of Lalit Great Eastern Hotel, Kolkata.

- Study visit of the Parliamentary Standing Committee on Water Resources to Kolkata from 24.08,2021 to 27.08.2021.

- Parliamentary Committee on Official Language inspected the Central Ground Water Board, DivisioXV, office in Kolkata on 21.11.2021. The committee reviewed the work of Hindi in the office and gave instructions as per the requirement. Top officials of the Ministry and Department were present on the occasion. Dr. N Nandkumaran P., Chairman, CGWB, represented CGWB. Mrs.S Yadav, Sc-D & H.O.O, Sh. A. Jayachandran, EE, Div. XV, Sh. H. K. Behera, AEE, Div.XV and Hindi officers from CHQ attended the meeting.

Inspection meeting on Official Language

- Committee of Parliament on Official Language (second Sub- Committee) visited Guwahati and inspected Progress of use of Hindi in CGWB, Division-VII on 20/11/2011.

- माननीय संसदीय राजभाषा समिति की दूसरी उपसमिति द्वारा प्रो. **रीताबी जोशी जी, सांसद, प्रयागराज की अध्यक्षता मे श्री मनोज तिवारी जी, सांसद, उत्तरी-पूर्वी दिल्ली, श्री बाळूभाऊधानोरकरजी, सांसद, चंद्रपुर एवं श्रीमती. रंजनबेनभट्टजी, सांसद, वडोदरा** ने केंद्रीय भूमि जल बोर्ड, प्रभाग6 ,नागपुर, महाराष्ट्र के साथ 27thDecember2021 को निरीक्षण बैठक की।

- Committee of Parliament on Official Language (second Sub- Committee) visited Guwahati and inspected Progress of use of Hindi in CGWB, Division-VII on 20/11/2011.

- Committee of Parliament on Official Language (second Sub- Committee) visited Guwahati and inspected Progress of use of Hindi in CGWB, Division-VII on 20/11/2011.

- The Parliamentary Standing Committee on Water Resources visited on Chandigarh 13-14 July, 2021. The committee reviewed the progress of various ongoing Schemes such as Jal Jeevan Mission, Swacch Bharat Mission and Atal BhuJalYojana and their implementation by various state departments of Punjab and Haryana. The discussion were directed towards arresting ground water depletion and contamination with an aim to provide potable drinking water to people.

- Meeting of Parliament Committee on Official Language held an inspection meeting of second sub committee at Hyderabad on 27.2.2022.

- The 2nd sub-committee of parliament standing committee on implementation of official language reviewed the progress of work under Hindi of CGWB, SWR on 19/04/2021

during the visit of the committee to review the work in central Govt. offices located in Bangalore from 15.04.2021 to 19.04.2021. The progress of work in Hidi of SWR was presented to the Committee by Sh.V.kunhambu, Regional Director, SWR in the presence of Sh. G.C.Pati,

- Chairman, CGWB, and Smt. Sangita P.B Hindi Officer, SWR, Sh Gopal Prasad Assistant Director(OL), CGWB & Officers from Ministry of Jal Shakti.
- (Photographs from the 2nd sub-committee of parliament standing committee visit in Bangalore)

20.6 Visit of JS/chairman/ Members of CGWB

- Joint Secretary (Admin, IC & GW),DOWR RD &GR, Member(NW), CGWB and Director (GW), DOWR RD&GR visited Kolkata from 01.11.2021 to 03.11.2021, in connection with review meetings with the beneficiaries of NE States pertaining to NHP and PMKSY-HKKP-GW.

Photo: NHP Meeting morning session

Photo: PMKSY HKKP Meeting Afternoon Session

- Dr Sunil Kumar Jain, Member (South), CGWB, visited Central Ground Water Board, Central Region and Division VI, Nagpur on 28th December 2021 to review the progress of various activities of the current Annual Action Plan. Member (South), CGWB, along with Dr. P. K. Jain, Regional Director and all the officers of CGWB CR, & Executive Engineer, AEE, Division VI, discussed in detail the achievements till December 2021 against the Annual Target of all the activities. He specifically asked about the important issues and the status of action taken in this regard.
- Shri Subodh Yadav,IAS Joint Secretary (GW),DoWR,RD&GR, Ministry of Jal Shakti convened meeting with concerned State departments during his visit to Bangalore on 14.7.2021 regarding the ongoing schemes of MoJS, Atal Jal/ implementation of Master

Plan for Artificial Recharge in pilot districts etc& issues pending with State Government for further the road map in these matters. CGWB, SWR also attended the meeting and participated in the discussions.

- Dr. N Nandkumaran P., Chairman, CGWB, visited Central Ground Water Board, Eastern Region, Kolkata on 23/11/2021 to review the progress of various activities of the current Annual Action Plan.
- Dr.Utpal Gogoi, Member (East), CGWB, visited Central Ground Water Board, Eastern Region, Kolkata on 15/12/2021 to review the Quarterly progress of various activities of the current Annual Action Plan.
- Dr.Sunil Kumar, Jain Member (South). Dr. Ratikanta Nayak, Scientist-E, Sh. D.N. Mandal, Scientist-D Officers from CGWB, CHQ, Officers from CGWB, SR, Hyderabad, AP State Ground Water & Water Audit Dept., Telangana State Ground Water Dept and representatives from IIT-Hyderabad held a meeting ar SR, Hyderabad on Issues related to IN-GRES Software for Ground Water Resource Assessment 2022 on 22.02.2022.
- Meeting taken by Shr. Ashish Kumar, Director (GW&Vig.), Ministry of Jal Shakti, DoWR,RD&GRwith the Regional Diector and officers regarding Regarding progress and finalization of DPR for PIB proposal on 21.03.2022.

20.8 Press Clippings

News Paper Name: Lokmat

लोकमत

पी. के. जैन : पवनार येथील धाम नदी पूजन कार्यक्रम

प्रत्येक नदी प्रदूषणमुक्त राहावी यासाठी विज्ञानयुगात प्रयत्न व्हायलाच हवेत

लोकमत न्यूज नेटवर्क

पवनार : प्रत्येक नदी ही गंगा असून, केवळ गंगा नदीच नाही तर देशातील प्रत्येक नदी प्रदूषित होण्यापासून वाचविण्यासाठी प्रत्येक नागरिकाने पुढाकार घेतला पाहिजे. यासाठी शासन, प्रशासन तसेच स्थानिक नागरिकांनी आपल्या भागातील नद्यांचे महत्व इतरांना पटवून दिले पाहिजे. शिवाय नदी पूजन केले पाहिजे, असे प्रतिपादन केंद्रीय भूमी-जल बोर्डाचे क्षेत्रीय निदेशक पी. के. जैन यांनी केले.

महात्मा गांधी अमृत महोत्सवाचे औचित्य साधून केंद्रीय भूमी-जल बोर्ड, सद्भावना ग्रामीण विकास संस्था तसेच धुर्वे उद्योग समूहाच्या वतीने पवनार येथे धाम नदीच्या तीरावर नदी पूजन कार्यक्रम पार पडला. याप्रसंगी ते मार्गदर्शन करताना बोलत होते. व्यासपीठावर सरपंच शालिनी आदमने, जलतज्ज्ञ माधव कोटस्थाने, बजाज फाउंडेशनचे विनेश कावरे, सद्भावना ग्रामीण विकास संस्थेचे अध्यक्ष गिरिश भगत, अटल पांडे आदींची उपस्थिती होती.

मनोगत व्यक्त करताना भगत यांनी पाणी आणि नदीचे महत्व पटवून दिले. शिवाय सद्भावना ग्रामीण विकास संस्थेच्या वतीने तसेच जमनालाल नेहरू पीस ट्रस्ट मुंबईच्या मदतीने तामसवाडा पेंटर्वाडा आधारित महाकाळ, सटोडा नाला, धानोरा भगवंतेडा नाला, मोती नाला हे तीन प्रकल्प धाम नदीवर राबविल्याचे सांगत या प्रकल्पाला राष्ट्रीय पुरस्कार मिळाल्याचेही त्यांनी स्पष्ट केले. तसुल रोडे यांनी भूजल चाळणी कशी खालावत आहे, याची माहिती सांगोडी उपस्थितांना दिली. मूर्ती व निर्माल्य विसर्जन आदींमुळे नदी प्रदूषित होत असल्याने न्यायालय व शासनाच्या सूचनांची प्रभावी अंमलबजावणी प्रत्येक नदी काठावरील गावांमार्फत होणे गरजेचे आहे. शिवाय नदी प्रदूषणमुक्त राहावी यासाठी करावयाच्या उपाययोजनांसाठी नदी काठावरील ग्रामपंचायतींना शासनानेही काही विशेष निधी दिला पाहिजे, असे सरपंच शालिनी आदमने यांनी सांगितले. कार्यक्रमाला ग्रामविकास संस्थेचे गिरीश पंडित, सचिन तांबुलकर, लोकेश रंगारी, अंकुश चारोकी, चरण आगलावे, दिगंबर उमाळे, आशिष आटे, एल. एस. नागपुरे आदींनी सहकार्य केले.

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News Paper Name: Navrastra

नवराष्ट्र

उपक्रम स्वातंत्र्याच्या अमृत महोत्सवानिमित्त आयोजन

पवनार येथे धाम नदीच्या पाण्याचे पूजन

वर्धा ब्युरो. भारतीय स्वातंत्र्याच्या अमृत महोत्सवाच्या निमित्त भारत सरकारच्या जलशक्ती मंत्रालयाच्या अंतर्गत येत असलेल्या केंद्रीय भूजल बोर्ड मध्य क्षेत्र नागपूर कार्यालयातर्फे सदभावना ग्रामीण विकास संस्था व पूर्ती सिंचन समृद्धी कल्याणकारी संस्था यांच्या सहकार्याने शुक्रवारी 17 डिसेंबर रोजी पवनार येथील धाम नदीच्या पाण्याचे पूजन करण्यात आले.

जल पूजन कार्यक्रमाच्या अध्यक्षस्थानी केंद्रीय भूजल बोर्ड मध्यक्षेत्र नागपूर कार्यालयाचे निदेशक डॉ. प्रभात जैन होते. प्रमुख अतिथी म्हणून सदभावना ग्रामीण विकास संस्थेचे मिलिंद भगत, पूर्ती सिंचन समृद्धी कल्याणकारी संस्थेचे माधव कोटस्थाने, बजाज फाउंडेशनचे विनेश काकडे, पवनारच्या सरपंच शालिनी आटमने, विश्व हिंदू परिषदेचे अटल पांडे यांची उपस्थिती होती. भागवताचार्य हेमंतशास्त्री पाचखेडे, मुकुंद देशपांडे, वैभव मिश्रा व उपस्थित मान्यवरांच्या हस्ते धाम नदीच्या पाण्याचे पाणलोट आधारित पूजन करण्यात आले. भूजल कार्यालय नागपूरचे वैज्ञानिक राहुल शेंडे यांनी पॉवर पॉइंट प्रेझेंटेशनच्या माध्यमातून भूजल व्यवस्थापनावर माहिती दिली. प्रास्ताविक सदभावना ग्रामीण विकास संस्थेचे मिलिंद भगत यांनी केले. त्यांनी जेएमपीटी व सीएसआरच्या सहकार्याने धाम नदीच्या परिसरात राबविण्यात आलेल्या प्रकल्पाची माहिती दिली. आभार दिनांबर उमक यांनी मानले. आयोजनाकरिता लोकेश रंगारी, अविनाश आटे, किसन कामडे, लीलाधर नवघरे, सत्यजीत जांभुळकर, गितीश पडोळे, सचिन तांदुळकर, सुबोध लाभे, आकाश बावणे, मनीष सरडवार, किशोर कोहाडे, अजय सोरदे, संजय भगत यांनी सहकार्य केले.

प्रत्येक नदीला गंगेचा दर्जा

अध्यक्ष डॉ. प्रभात जैन म्हणाले की, देशातील प्रत्येक नदीला गंगेचा दर्जा देण्यात आला आहे. प्रत्येकाने आपल्या गावातील नदीला देवी मानत नदी प्रदूषित न करण्याचे आवाहन त्यांनी केले. पाणी हेच जीवन असून पाण्याचे संवर्धन करावे, असेही त्यांनी सांगितले. उपस्थित अन्य मान्यवरांनीदेखील पाणी संवर्धन व संरक्षण यावर समयोचित विचार व्यक्त केले.

नदी उत्सवाचे आयोजन

देशातील नद्यांना पौराणिक प्रतिष्ठा मिळवून देण्यासोबतच पुनर्स्थापना करणे, नद्यांचे शोषण थांबविणे, प्रदूषणापासून मुक्त करणे, जीवनदायी नद्यांना सुरक्षित ठेवण्याचे उद्देशाने भारत सरकारच्या वतीने 17 ते 23 डिसेंबरदरम्यान नदी उत्सवाचे आयोजन करण्यात येत आहे. त्या अनुषंगाने या कार्यक्रमाचे आयोजन करण्यात आले.

News Paper Name: Tarun Bharat

तरुण भारत
Paschim Vidarbha | 2021-12-18 | Page- 12
epaper.tarunbharat.net

जलपूजनला उपस्थित माधव कोटस्थाने, डॉ. जैन, पांडे, पाचखेडे, काकडे व अन्य

पवनार येथे धाम नदीवर जलपूजन

◆ वर्धा, १७ डिसेंबर

केंद्रीय भूजल बोर्ड मध्य क्षेत्र नागपूर कार्यालयाच्या वतीने पूर्ती सिंचन समृद्धी कल्याणकारी संस्था व सद्भावना ग्रामीण विकास संस्था वर्धा यांच्या सहकार्याने १७ डिसेंबर रोजी पवनार येथील धामनदीच्या पाण्याचे पूजन करण्यात आले.

अध्यक्षस्थानी केंद्रीय भूजल बोर्ड मध्य क्षेत्र नागपूर कार्यालयाचे निदेशक डॉ. प्रभात जैन होते तर प्रमुख अतिथी म्हणून पूर्ती सिंचन समृद्धी कल्याणकारी संस्थेचे माधव कोटस्थाने, सद्भावना ग्रामीण विकास संस्थेचे मिलिंद भगत, बजाज फाऊंडेशनचे विनेश काकडे, पवनारच्या सरपंच शालिनी आदमने, विश्व हिंदू परिषदेचे विदर्भ प्रांत संघटनमंत्री अटल पांडे यांची उपस्थिती होती.

हेमंतशास्त्री पाचखेडे, मुकुंद देशपांडे, वैभव मिश्रा यांच्या हस्ते धाम नदीच्या पाण्याचे पाणलोट आधारित पूजन करण्यात आले. यावेळी डॉ. प्रभात जैन म्हणाले, देशातील प्रत्येक नदीला गंगेचा दर्जा देण्यात आला आहे. प्रत्येकाने आपल्या गावातील नदीला देवी मानावे व नदी प्रदूषित न करण्याचे आवाहन त्यांनी केले. पाणी हेच जीवन असून पाण्याचे संवर्धन करावे, असेही त्यांनी सांगितले.

जलतज्ज्ञ माधव कोटस्थाने यांनी जलसिंचनाची गरज या विषयावर मार्गदर्शन केले. पूर्ती सिंचनच्या वतीने तयार करण्यात आलेला तामसवाडा प्रकल्प सिंचनासाठी पथदर्शी ठरत असून या प्रकल्पामुळे सिंचनासोबत रोजगारही उभा होत असल्याचे सांगून कामाची माहिती त्यांनी दिली. मान्यवरांनीदेखील पाणी संवर्धन व संरक्षणावर समायोचित विचार व्यक्त केले. भूजल कार्यालय नागपूरचे वैज्ञानिक राहुल शेंडे यांनी पॉवर पॉईंट प्रेझेंटेशनच्या माध्यमातून भूजल व्यवस्थापनावर माहिती दिली.

देशातील नद्यांना पौराणिक प्रतिष्ठा मिळवून देण्यासोबतच पुनर्स्थापना करणे, नद्यांचे शोषण थांबविणे, प्रदूषणमुक्त करणे, जीवनदायी नद्यांना सुरक्षित ठेवण्याच्या उद्देशाने केंद्र सरकारच्या वतीने १७ ते २३ डिसेंबरदरम्यान नदी उत्सवाचे आयोजन करण्यात येत आहे.

प्रास्ताविक मिलिंद भगत यांनी केले. पशस्वितेकरिता लोकेश रंगारी, अश्विन आटे, बिसेन कामडे, लीलाधर नवघरे, सत्यजित जांभूळकर, गिरीश पडोळे, सचिन तांदूळकर, सुबोध लाभे, आकाश बावणे, मनीष नरडवार, किशोर कोहाडे, अजय सोरदे, संजय भगत यांनी सहकार्य केले. ◀(तभा वृत्तसेवा)

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पवनार येथे धाम नदीच्या पाण्याचे पूजन

देशोन्नती वृत्तसंकलन...

[illegible]

News Paper Name: Punya Nagri

पुण्य नगरी

पवनार येथे धाम नदीच्या पाण्याचे पूजन

पुण्य नगरी / प्रतिनिधी

वर्धा : भारतीय स्वातंत्र्याच्या अमृत महोत्सवानिमित्त भारत सरकारच्या जलशक्ती मंत्रालयांतर्गत केंद्रीय भूजल बोर्ड मध्य क्षेत्र नागपूर कार्यालयाच्या वतीने सद्भावना ग्रामीण विकास संस्था व पूर्वी सिंचन समृद्धी कल्याणकारी संस्था यांच्या सहकार्याने १७ डिसेंबर रोजी पवनार येथील धाम नदीच्या पाण्याचे पूजन करण्यात आले.

या जलपूजन कार्यक्रमाच्या अध्यक्षस्थानी केंद्रीय भूजल बोर्ड कार्यालयाचे निदेशक डॉ. प्रभात जैन तर प्रमुख अतिथी म्हणून मिलींद भगत, माधव कोटस्थाने, वज्रज फाउंडेशनचे विनेश काकडे, पवनारच्या सरपंच शालिनी आत्रामने, विश्व हिंदू परिषदेचे अटल पांडे यांची उपस्थिती होती. भागवताचार्य हेमंतशास्त्री पाचखेडे, मुकुंद देशपांडे, वैभव मिश्रा व मान्यवरांच्या हस्ते धाम नदीच्या पाण्याचे पाजलोट आधारित पूजन करण्यात आले.

यावेळी डॉ. प्रभात जैन यांनी मार्गदर्शन केले. भूजल कार्यालय नागपूरचे वैज्ञानिक राहुल शेंडे यांनी पॉवर पॉईंट प्रेझेंटेशनच्या माध्यमातून भूजल व्यवस्थापनावर माहिती दिली. देशातील नद्यांना पौराणिक प्रतिष्ठा मिळवून देण्यासोबतच पुर्नस्थापना करणे, नद्यांचे शोषण थांबविणे, प्रदुषणापासून मुक्त करणे आणि जीवनदायी नद्यांना सुरक्षित ठेवण्याचे उद्देशाने भारत सरकारच्या वतीने १७ ते २३ डिसेंबर दरम्यान नदी उत्सवाचे आयोजन करण्यात आले आहे

प्रास्ताविक मिलींद भगत यांनी केले. आभार दिगंबर उमक यांनी मानले. यशस्वीतेकरिता लोकेश रंगारी, अश्विन आटे, बिसेन कामडे, तिलाधर नवघरे, सत्यजीत जांभुळकर, गिरीष पडोळे, सचिन तांदुळकर, सुबोध लाभे, आकाश बावणे, मनीष नरडवार, किशोर कोहाडे, अजय सोरदे, संजय भगत आदींनी सहकार्य केले.

Smart Wardha

Page No.4 December 19, 2021

News Paper Name: Dainik Bhaskar

पवनार की धाम नदी में हुआ जलपूजन

बड़ी | संवाददाता

भारतीय स्वतंत्रता के अमृत महोत्सव के अवसर पर केंद्रीय भूजल बोर्ड मध्य क्षेत्र नागपुर कार्यालय की ओर से जल संसाधन मंत्रालय, भारत सरकार के तहत सद्भावना ग्रामीण विकास संस्था और पूर्ति सिंचन समृद्धि कल्याणकारी संस्था वर्धा के सहयोग से 17 दिसंबर को पवनार में धाम नदी की पूजा की गई। जलपूजन कार्यक्रम के अध्यक्षस्थान पर केंद्रीय भूजल बोर्ड मध्यक्षेत्र नागपुर कार्यालय के निदेशक डॉ. प्रभात जैन और प्रमुख अतिथि के रूप में सद्भावना ग्रामीण विकास संस्था के मिलिंद भगत, पूर्ति सिंचन समृद्धि कल्याणकारी संस्था वर्धा के माधव कोटस्थाने, बालाजी फाउंडेशन के विनोद काकड़े, पवनार की सरपंच शालिनी आदमने और विश्व हिंदू परिषद के अहान घोड़े उपस्थित थे।

भगवताचार्य हेमंतशास्त्री चाफलेकर, मुकुंद देशपांडे, वैभव मिश्रा और अन्य व्यक्तियों ने धाम नदी के जलकुंड पर आयोजित पूजन किया। अध्यक्ष डॉ. प्रभात जैन ने कहा कि देश की हर नदी को गंगा का दर्जा दिया गया है। उन्होंने सभी से अपील की कि वे अपने गांव की नदी को देवी मानकर

नदी को प्रदूषित न करें। उन्होंने कहा कि जल ही जीवन है और जल का संरक्षण किया जाना चाहिए। उपस्थित अन्य व्यक्तियों ने भी जल संरक्षण पर अपने विचार व्यक्त किए। भूजल कार्यालय नागपुर के वैज्ञानिक राहुल शेंडे ने पावर प्वाइंट प्रेजेंटेशन के माध्यम से भूजल प्रबंधन की जानकारी दी। पौराणिक प्रतिष्ठा देने के साथ-साथ देश की नदियों का जीर्णोद्धार करना। नदियों को प्रदूषण से निजात दिलाने जीवन रक्षक नदियों को सुरक्षित रखने के उद्देश्य से भारत सरकार द्वारा 17 से 23 दिसंबर तक नदी उत्सव का आयोजन किया जा रहा है। इसी सिलसिले में कार्यक्रम का आयोजन किया गया। सद्भावना ग्रामीण विकास संस्था के मिलिंद भगत ने परिचयात्मक भाषण दिया। उन्होंने जेएनयूसी और मीडियाज के सहयोग से धाम नदी के क्षेत्र में लघु परियोजना की जानकारी दी। दिनेश रावत ने धन्यवाद माना। आयोजन के लिए लोकेश रंगारी, अविनाश आठे, विनोद काकड़े, सौरभ नाकवे, सत्यजीत जांभुलकर, गिरीश पठोने, सचिन लोहुलकर, सुबोध लाभे, आकाश बावने, मनीष गरड़वार, किशोर कोहाड़े, अजय मोरे और मंगल भगत आदि ने सहयोग किया।

Sun, 19 December 2021
https://epaper.bhaskarhindi.com/c/65857741

Hindi Diwas

The Sentinel
of this land, for its people

Five-day training on 'Springshead' management ends

Stress laid on rejuvenation of spring water bodies & sources

STAFF REPORTER

GUWAHATI, Dec 17: The NEHARI (North Eastern Hydraulic & Allied Research Institute) – an arm of the 'Brahmaputra Board' organized a five-day training workshop on 'Scientific Data Collection and Processing Techniques for Springshed Management and Rejuvenation' in association with the NIH (National Institute of Hydrology) and the CGWB (Central Ground Water Board) during December 13-17, 2021.

The NEHARI HQs (headquarters) is at Radnoowar in North Guwahati.

Secretary, DoWR (Department of Water Resources), Ministry of Jal Shakti, Pankaj Kumar inaugurated the training workshop online on December 13 in the presence of Rajiv Yadav, Chairman of the Brahmaputra Board and Dr I.V. Tyagi, Director, NIH.

Pankaj Kumar highlighted the importance of the springs rejuvenation programme in the northeastern States to safeguard the water resources for source sustainability.

B.K. Kalita, Vice-Chairman of the Brahmaputra Board; Dr Pradip Kumar Bora, Director, NERWALM (North Eastern Regional Institute of Water & Land Management); Dr Indra Kumar, NIH; Dr A.K. Lohani, NIH were also present during the online inauguration.

The NERWALM was established at Tezpur, Assam by North Eastern Council (NEC), in December 1989 when NEC was under the Ministry of Home Affairs, Govt. of India.

The course of the programme is tailored for junior to mid-level officers of Government Departments related to the water resources sector serving in the NER (North East Region).

This training workshop was designed to inculcate knowledge on 'Springshed Management' to help in better conservation and rejuvenation of spring water.

Ranjit Deka, Director of NEHARI appealed that 25 government officials serving in the North Eastern States of Arunachal Pradesh, Assam, Manipur, Mizoram, Meghalaya, Nagaland, Tripura including Siliguri, West Bengal have participated in the training workshop.

A total of 18 scientists – five each from NIH and CGWB imparted trainings on various topics pertaining to 'Springshed Management'.

Dr D.S. Bisht (NIH) and Dr K. Radhapyari (CGWB) presented various hands-on exercises for the officer trainees for data collection and in-situ water testing. Course coordinator from NIH, Dr S.S. Rawat stressed the need for a comprehensive inventory of springs for the Indian Himalayan States.

The importance of the geological investigation to the identification of the recharge area of the springs was elaborated on by Tapas Chakraborty, course coordinator of CGWB.

ଜଳର ସୁବିନିଯୋଗକୁ ଗୁରୁତ୍ୱ

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f Flartm.

Demonstration and protest in front of the Regional Council of Secondary Education Bam.

Berhampur(A.O. Bureau): The Student-Teacher-Parent-Intellectual Coalition has staged a protest in front of the Berhampur branch office of the Odisha Secondary Education Council. A large number of teachers, students and parents from Ganjam, Gajapati, Kandhamal and South districts joined in the protest. The 20-point memorandum was sent to the Minister of Mass Education and the Secretary of the Council for

Secondary Education through the Ganjam district commissioner. It was demanded that the textbooks in the textbooks be rectified and that the old test procedures, such as the board test and the board test, be addressed immediately. Ganjam Osar president Hemanshu Kumar Pattanayak and Purnachandra Acharya, editor of the newly funded school, coordinated the event today, while state executive member Ramchandra Polai, Kandhamal district secretary Ranjan Mohanty and Gajapati district executive president Chittaranjan Panda were prominently present.In view of the protesters and a broad protection was made from the pave. The security arrangement conducted by Gopalpur IIC Madhabanadan Nayak and ASI Mahendra Kumar Behera for the protection of protests personal along with official.

Public orientation program on ground water management held at MalKangiri college

MalKangiri AO Bureau : a one day Public orientation program on ground water management held at MalKangiri College by Central ground water board south eastern region Bhubaneswar department of Water resources river development and Ganga rejuvenation ministry of jal shakti Govt of India it was organised to commemorement 75th Independence of india celebrating azadi ki amrit mahastav exactly this year to mark this occasion. In the event Additional District Magistrate MalKangiri Mr

Biranen Pradhan joined as Chief guest and Mr Ashok Kumar sarangi Principal of MalKangiri College presided over the meeting the central ground water board Bhubaneswar senior scientist Mr Rajeev Kumar Tripathi and Mr Sudhanshu Mohanty senior water scientist participated in the program as guest of honour they explained the status of ground water it's quality management and rivers water conservation it's proper use and utilisation to the students.Dr Ranjan kumar swain HoD commerce department conducted the program. many lectures teaching staff, non teaching staff's and about 200 students and social workers of Srusti organisation participated in the one day ground Water resources orientation program Finally Mr Goutam Kumar Nayak senior Lecturer thanked all the participants.

Atanu Bhowmik Chosen As RSP Director In-Charge

Rourkela, (AO Bureau): The Public Enterprises Selection Board has selected SAIL Executive Director Atanu Bhowmik for the post of Director In-Charge of the Rourkela Steel Plant (RSP). After finalizing Bhowmik's name, the selection board recommended

Workshop on employee resource management through online

Rayagada: (AO Bureau): Celebrating the 75th Independence Day of the Government of India on the occasion of Ajadika Amrit Mohotsvab. An online workshop on Employee Provision through the Regional Office of Berhampur has been organized in the hall of the Human Resources

TRINITY MIRROR

THURSDAY 24-3-2022

Scarcity of water could be overcome by saving rain-water, students advised

Madurai, Mar.24:

Water scarcity could be easily overcome by making the Rain Water Harvest, addressed the Principal of the Mano College Abdul Khadir at World Water Day celebrations organised by the Sankaran Kovil Manonmaniam Sundaranar University.

South East Coastal Scientist and Doctor Sakthi Murugan, was the chief guest of the function organised jointly by the Union Ministry of Jal Sakthi and Ground Water Board. The Chief Guest briefed the causes for the depletion of the ground-water level and also gave tips to increase the groundwater level.

The principal of the Mano College Abdul Khadir in his presidential address hoped that world water day has been celebrated every year on March 22, to create awareness among the general public the ways and means of improving the groundwater level.

He underlined that there are plenty of industries to manufacture various products but no industry in the country has been functioning to produce water, instead the country has been equipped with Reverse Osmosis industries.

People in the country are solely depending upon rains and the groundwater for their day-to-day living. Hence, the people should make thrift-spending of water and should not contaminate the water available to us, appealed the Principal.

The College Professor Senthilkumar welcomed the gathering and the Physical Director Ganesan felicitated the occasion. Prof Menaka compared the programme. NSS Officer of the College Muthulakshmi gave out her felicitations at the function.

The girl students of the college who have excellently answered to the questions relating to the Water Day celebrations were distributed prizes. More than 200 students and 25 faculty members of the college participated in the function.

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necessary action ately for their ies.

y Mayor t, Corporation commr Mujibur t, Town Engineer haralli, Executive ro P.Sivapalham, ersan, Deputy commr for Rorak Zone mithi, Councillors zhi and Jaffer d various other were present.

భూగర్భ జలాల పెంపుకు కృషి చేద్దాం:ఆర్‌డి

విశాఖాంధ్ర-యర్రగుంట్ల : నీటిలో కురిసే వర్షపునీటిని ఒడిసిపట్టడం ద్వారా భూగర్భ జలాలను పెంపొందించుకొని అవసరాలకు సద్వినియోగం అయ్యేలా చూడాల్సిన బాధ్యత ప్రతి ఒక్కరిపై ఉందని కేంద్ర భూగర్భ జల సంస్థ సీనియర్ సైంటిస్ట్ (ఆర్‌డి) సుబ్బారావు సూచించారు. శనివారం స్థానిక మార్కెట్ యార్డు ఆవరణలో రైతులతో జరిగిన జిల్లాస్థాయి కార్యక్రమంలో ఆయన పాల్గొని నీటి పొదుపుపై వివరించారు. చెక్‌డ్యాంలు, అనకట్టల నిర్మాణాలతో పాటు ఇంకుడు గుంతలు ఏర్పాటు చేయడం ద్వారా భూగర్భజలాల పెంపుకు కృషి చేసినట్లు అవుతుందన్నారు. నీటిని పొదుపుగా వినియోగించడం ద్వారా భవిష్యత్‌లో నీటి కొరత రాకుండా చూడాల్సి ఉందన్నారు. ఈ కార్యక్రమంలో ఎగ్జిక్యూటివ్ ఇంజనీర్ రంధ్ర మోహన్‌సింగ్, ఎంపీడీవో శివారెడ్డి, మండల వ్యవసాయ అధికారి లక్ష్మీకుమార్, మార్కెట్ యార్డు సెక్రటరీ రత్నబాబుతో పాటు కమిటీ చైర్మన్లు గోపాలరెడ్డి, రైతులు పాల్గొన్నారు.

భూగర్భజలాల పరిరక్షణ అందరి బాధ్యత

మాట్లాడుతున్న కార్పొరేట్ మాధవి

దుగ్గిరాల మండలం (మంగళగిరి గ్రామీణం), మన్యంటుడే : భూగర్భ జలాలను పరిరక్షించాల్సిన బాధ్యత అందరిపై ఉందని కేంద్రీయ భూగర్భజల బోర్డు శాస్త్రవేత్త తాటి మాధవి అన్నారు. శనివారం మండలంలోని దుగ్గిరాలలో పైలరాబాదులోని భారత అంకెక్ట్ మంత్రిత్వ శాఖ వారు భూగర్భజలాల అవగాహనపై అవగాహన సదస్సు నిర్వహించి మాట్లాడారు. భూగర్భజలాల సంరక్షణ విధానంపై గ్రామస్థులకు వివరించారు. నీరు వృథా కాకుండా జాగ్రత్తగా వాడుకోవడం, చెక్‌డ్యాంలను నిర్మించుకోవాలన్నారు. ప్రతి నీటిబొట్టును కాపాడాల్సిన బాధ్యత అందరిపై ఉందన్నారు. భూగర్భజలాలు పెంపొందిస్తే భవిష్యత్తరాలకు నీటి కొరత ఏర్పడదన్నారు. కార్యక్రమంలో మండలాధ్యక్షుడు సర్పంచి లక్ష్మణ్, ఎంపీటీసీ సభ్యుడు వెంకటయ్య, మండల వ్యవసాయ శాఖ, గ్రామ పెద్దలు, రైతులు పాల్గొన్నారు.

"GROUNDWATER ISSUES AND MANAGEMENT IN KARIMNAGAR DISTRICT".

Huzurabad, Singapuram, 25, March, **Delhi to Galli** : Civil engineers have a responsibility to ensure that groundwater supply in purest form ground water is our national resource and that the community that uses it effectively, can live a civilized life, build a community free of hunger with good crops, we should protect groundwater effectively , provide pure water to the public and protect their health. Shri J Siddartha Kumar, Regional Director, CGWB Central Ground Water Board, Hyderabad, said.He was the chief guest at the inaugural session of a three-day training program on "Groundwater Issues and Management in Karimnagar District" which began in KITS Singapore today.Shri J Siddartha Kumar explained that the Department of Civil Engineering teaches civilization to human beings, who provide food, housing, roads and other basic infrastructure for human beings engineering students to have a sense of dedication to society and to do projects that are necessary for the people.The program was attended by Dr. Sudhir Kumar, Senior Scientist, Scientists Carolina louis, T.Madhav , technical operator M.Hemanth of Central Ground Water Board, Hyderabad; Kiran Kumar, Jagan assistant professors of KITSS acted as coordinator for this program. The event was attended by Dr. Anil Kumar, Head, Department of Civil Engineering, KITS Singapore, faculty and students.

సాక్షి 31-3-2022

ఉపాధి పనులు సద్వినియోగం చేసుకోండి

- ప్రతి వర్షపు చుక్కను ఒడిసిపట్టాలి
- భూగర్భజలాలు పెంచేందుకు కృషి చేయాలి

అదనపు సదస్సులో మాట్లాడుతున్న కలెక్టర్ పాల్గొన్నారు

శాస్త్రీయ పద్ధతుల్లో భూగర్భ జలాల సంరక్షణ

కొత్త పల్లి వద్ద భూమిలు రీచార్జ్‌ను పరిశీలిస్తున్న కలెక్టర్ రాగాలక్ష్మి, జేసీ పి.వి

• జిల్లా కలెక్టర్ రాగాలక్ష్మి పిలుపునిచ్చారు

అనంతపురం అర్బన్: భూగర్భజలాలను సంరక్షించుకోవడం అత్యవసరమని కలెక్టర్ రాగాలక్ష్మి పిలుపునిచ్చారు. భూగర్భజల శాఖ ఆధ్వర్యంలో శుక్రవారం కలెక్టరేట్‌లో వివిధ కార్యక్రమాలు నిర్వహించారు. 'అందరం సమాన- భూగర్భజల ప్రణాళిక' అనే అంశం సమావేశం నిర్వహించారు. ఈ సందర్భంగా కలెక్టర్ మాట్లాడుతూ జిల్లాలో 2021లో 6 నెలల్లో 4 నీటి సంవత్సరాలు గడిచాయి. ఈ సమయంలో 482 నీటి సంవత్సరాలు గడిచాయి. 6 నెలల్లో సాధారణం కంటే 48 శాతం వర్షపాతం నమోదైంది. 2 నీటి సంవత్సరాలు గడిచాయి. 2 నీటి సంవత్సరాలు గడిచాయి. 2 నీటి సంవత్సరాలు గడిచాయి

21. OUT REACH ACTIVITIES

21.1 Social Media
Facebook

Cleanliness drive Special campaign 2.0 carried out by CGWB, State Unit Office, Delhi at West Block, R K Puram, Delhi

Sh. Pankaj Kumar, Secretary (DoWR, RD&GR), along with Sh. Subodh Yadav, Joint Secretary, Ministry of Jal Shakti interacted with the Regional Directors and officers of Central Ground Water Board.

Sh Subodh Yadav, Joint Secretary (Admin IC & GW) Ministry of Jal Shakti visited Regional office of Central Ground Water Board on 19.09.2022 and reviewed the progress of various activities of the Region

Cleanliness drive Special campaign 2.0 carried out by CGWB, State Unit Office, Vishakhapatnam at Teneti Park Beach, Visakhapatnam, Andhra Pradesh

Central Ground Water Board welcomes Sh. Pankaj Kumar, Secretary (DoWR, RD&GR), Ministry of Jal Shakti at CHQ, Bhujal Bhawan, Faridabad.

#SpecialCampaign2.0 Central Ground Water Board , Lucknow carried out external cleanliness drive at Kudia Ghat on the bank of river Gomti, Lucknow

Scientist from CGWB, NCR, Bhopal shared NAQUIM report of Balaghat district, Madhya Pradesh with district administration, Balaghat

#SpecialCampaign2 #SwachhBharat #swachhbharatswasthbharat Cleanliness Drive Special Campaign 2.0 by Central Ground Water Board, Jaipur

Monitoring of Piezometric Head at Khowai district, Tripura

तिरुवनंतपुरममेंसंसदीयराजभाषासमितिनेकेंद्रीयभूमिजलबोर्ड, तिरुवनंतपुरम कार्यालय के साथ निरीक्षण बैठक की। इस दौरान समिति ने मंत्रालय एवं विभाग के वरिष्ठ अधिकारियों की उपस्थिति में हो रहे राजभाषा हिंदी के कार्यों का अवलोकन किया।

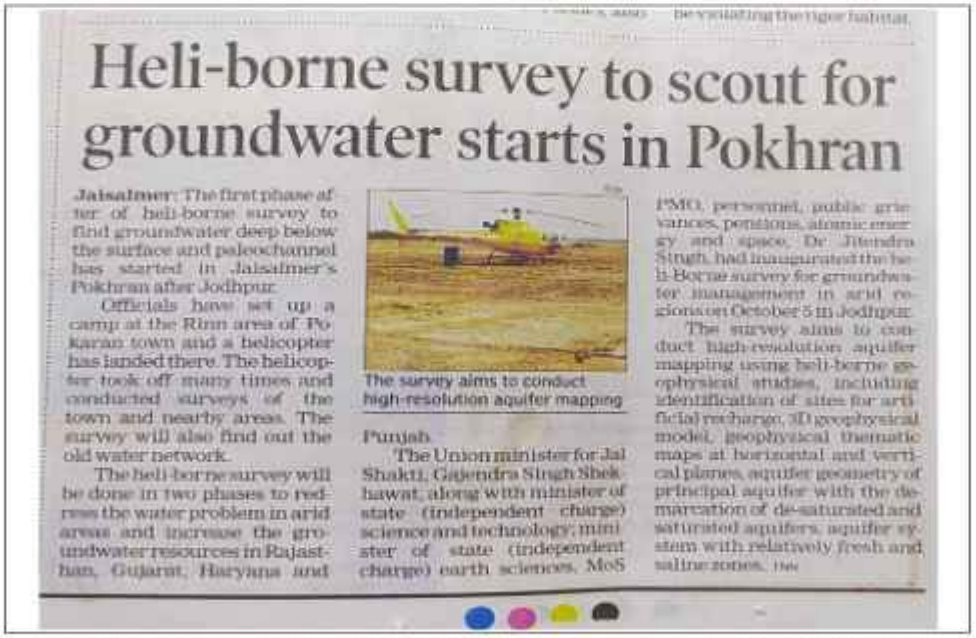
21.2 TWITTER

CGWB, CR, NAGPUR conducted PIP on "Aquifer Mapping and Ground Water Management " for Nagpur district, Maharashtra on virtual mode under the Banner @75 Years of India's Independence. Dr. Swati K. Dharmadhikari, Principal, Tirpude College of Social Work, Nagpur was the Chief guest.

Dugwell constructed for irrigation in Ahwa, Dang district, Gujarat under PMKSY-HKKP-GW scheme. In an area facing water shortage for irrigation, small and marginal farmers are getting benefitted from such interventions.

High resolution aquifer mapping using State-of-the-art Heli-borne survey in Ganga-Yamuna Doab, U.P. undertaken as a part of a tripartite agreement involving CGWB, NMCG and CSIR-NGRI.

Ground water recharge through abandoned Dugwell at Nakipura block Loharu district Bhiwani Haryana.



Heli-borne survey to scout for groundwater starts in Pokhran

Jaisalmer: The first phase after of heli-borne survey to find groundwater deep below the surface and paleochannel has started in Jaisalmer's Pokhran after Jodhpur.

Officials have set up a camp at the Rinn area of Pokaran town and a helicopter has landed there. The helicopter took off many times and conducted surveys of the town and nearby areas. The survey will also find out the old water network.

The heli-borne survey will be done in two phases to redress the water problem in arid areas and increase the groundwater resources in Rajasthan, Gujarat, Haryana and Punjab.

The survey aims to conduct high-resolution aquifer mapping

The Union minister for Jal Shakti, Gajendra Singh Shekhawat, along with minister of state (independent charge) science and technology; minister of state (independent charge) earth sciences, MoS PMO, personnel, public grievances, pensions, atomic energy and space, Dr Jitendra Singh, had inaugurated the heli-Borne survey for groundwater management in arid regions on October 5 in Jodhpur.

The survey aims to conduct high-resolution aquifer mapping using heli-borne geophysical studies, including identification of sites for artificial recharge, 3D geophysical model, geophysical thematic maps at horizontal and vertical planes, aquifer geometry of principal aquifer with the demarcation of de-saturated and saturated aquifers, aquifer system with relatively fresh and saline zones. TNN

Media Coverage of ongoing Heliborne Survey in Pokhran, Rajasthan

CGWB, Bhubaneswar conducted Public Interaction Program under the banner @ 75 years of India's Independence at Jawahar Navodaya Vidyalaya, Nabarangpur.

Field Demonstration of blast well constructed under PMKSY - HKKP GW Scheme at Rajgarh, Mirzapur district, Uttar Pradesh to the trainees of Induction Level Training course

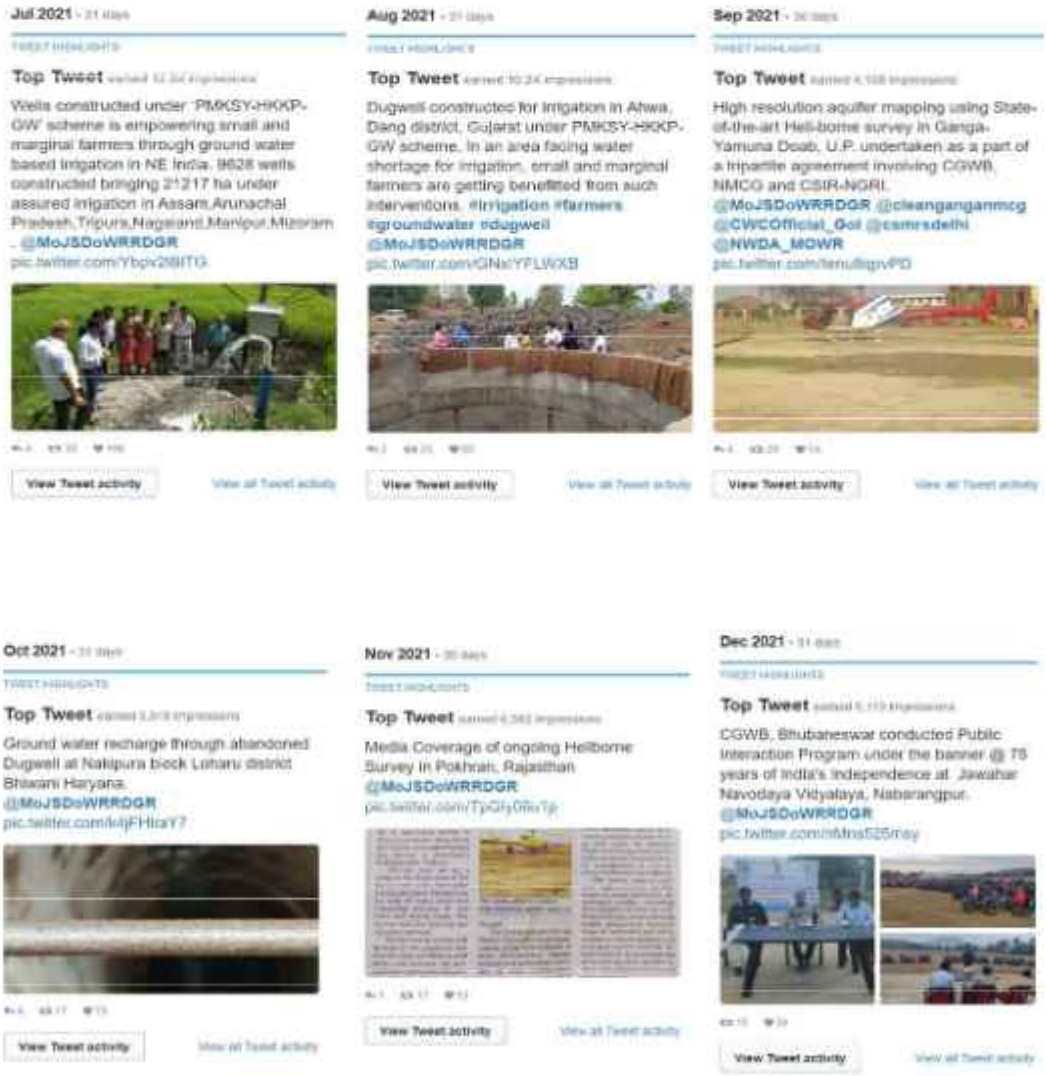
पटना में माननीय संसदीय राजभाषा समिति ने केंद्रीय भूमि जल बोर्ड, मध्य पूर्व क्षेत्र, पटना के साथ निरीक्षण बैठक की। इस दौरान समिति ने मंत्रालय एवं विभाग के वरिष्ठ अधिकारियों की उपस्थिति में हो रहे राजभाषा हिंदी के कार्यों का अवलोकन किया।

Field demonstration on surface geophysical survey carried out by Geophysicists of CGWB, Bhubaneshwar as a part of Tier-II training programme at Baripada, Mayurbhanj, Odisha.

21.3 Followers on Social Media

As on 01.04.2022	Facebook	Twitter	Instagram
Tottal Follower/Likes	5,488 likes / 5,888 follows	10,300 Followers	1,588 Followers
Tottal Posts	-	6,386 Tweets	1,814 Posts
Reach of Post	top post Reach 4k on 07.03.2022	Top Tweet earned 6,319 impressions	top post 542 impressions

21.4 Highlights Tweets



Jan 2022 - 31 days

TWEET HIGHLIGHTS

Top Tweet earned 6,219 impressions

Field Demonstration of blast well constructed under PMKSY – HKKP OW Scheme at Rajgarh, Mirzapur district, Uttar Pradesh to the trainees of Induction Level Training course. @MoJSDoWRRDGR
pic.twitter.com/INH2byrn2bf

View Tweet activity

View all Tweet activity

Feb 2022 - 28 days

TWEET HIGHLIGHTS

Top Tweet earned 2,244 impressions

आज पटना में माननीय संसदीय राजभाषा समिति ने केंद्रीय भूमि जल बोर्ड, मध्य पूर्व क्षेत्र, पटना के साथ त्रिपक्षीय बैठक की। इस दौरान समिति ने मंत्रालय एवं विभाग के वरिष्ठ अधिकारियों की उपस्थिति में हो रहे राजभाषा हिंदी के कार्यों का अवलोकन किया।
@MoJSDoWRRDGR
pic.twitter.com/dEMudkEPoe

View Tweet activity

View all Tweet activity

Mar 2022 - 31 days

TWEET HIGHLIGHTS

Top Tweet earned 2,448 impressions

Field demonstration on surface geophysical survey carried out by Geophysicists of CGWB, Bhubaneshwar as a part of Tier-II training programme at Baripada, Mayurbhanj, Odisha.
@MoJSDoWRRDGR
pic.twitter.com/bECIZWAUN65

View Tweet activity

View all Tweet activity

facebook

	Retweet	Engagement
Cleanliness drive Special campaign 2.0 carried out by CGWB, State Unit Office, Delhi at Ward Block, R.K. Puram, Delhi #SpecialCampaign2 #SwachhBharat #SwachhBharatAzadiKaAmrutMahotsav #SwachhBharatMission #SwachhBharat2022 #MyCleanIndia Ministry of Jal Shakti, Department of Water Resources, RD & GR	7502	256
Sh. Pankaj Kumar, Secretary (DoWR, RD&GR), along with Sh. Subodh Yadav, Joint Secretary, Ministry of Jal Shakti interacted with the Regional Directors and officers of Central Ground Water Board, Ministry of Jal Shakti, Department of Water Resources, RD & GR	7332	254
Sh Subodh Yadav, Joint Secretary (Admin IC & GW) Ministry of Jal Shakti visited Regional office of Central Ground Water Board on 19.08.2022 and reviewed the progress of various activities of the Region Ministry of Jal Shakti, Department of Water Resources, RD & GR	7019	320
Cleanliness drive Special campaign 2.0 carried out by CGWB, State Unit Office, Vishakhapatnam at Tenali Park Beach, Vishakhapatnam, Andhra Pradesh Ministry of Jal Shakti, Department of Water Resources, RD & GR	6062	115
Central Ground Water Board welcomes Sh. Pankaj Kumar, Secretary (DoWR, RD&GR), Ministry of Jal Shakti at CHQ, Bhopal Bhawan, Faridabad. Ministry of Jal Shakti, Department of Water Resources, RD & GR	5862	242
#SpecialCampaign2.0 Central Ground Water Board , Lucknow carried out national cleanliness drive at Kudia Ghat on the bank of river Gomti, Lucknow #SpecialCampaign2 #SwachhBharat #SwachhBharatAzadiKaAmrutMahotsav #CleanIndia #CleanNess #SwachhBharatMission #SwachhBharat2022 #MyCleanIndia Ministry of Jal Shakti, Department of Water Resources, RD & ...	3997	88
#SpecialCampaign2 #SwachhBharat #SwachhBharatAzadiKaAmrutMahotsav #CleanIndia Drive Special Campaign 2.0 by Central Ground Water Board, Jaipur Ministry of Jal Shakti, Department of Water Resources, RD & GR #CleanIndia #CleanNess #SwachhBharatMission #SwachhBharat2022 #MyCleanIndia	2968	54
Scientist from CGWB, NCR, Bhopal shared AHGWM report of Balaghat district, Madhya Pradesh with district administration, Balaghat Ministry of Jal Shakti, Department of Water Resources, RD & GR	2006	140
Monitoring of Piezometric Head at Dhawai district, Tripura Ministry of Jal Shakti, Department of Water Resources, RD & GR	1972	133
शिवनाथपुरम में संसदीय राजभाषा समिति ने केंद्रीय भूमि जल बोर्ड, शिवनाथपुरम कार्यालय के साथ त्रिपक्षीय बैठक की। इस दौरान समिति ने मंत्रालय एवं विभाग के वरिष्ठ अधिकारियों की उपस्थिति में हो रहे राजभाषा हिंदी के कार्यों का अवलोकन किया। #azadikaamrutmahotsav #swachhbharatmission Ministry of Jal Shakti, Department of Water Resources, RD & GR	1650	258

21.5 Bhujal Samvad

Bhujal Samvad is a quarterly magazine of Central ground Water Board started from Jan 2018. Till March 2022, 16 editions have been published and uploaded on official website of CGWB. The magazine comprises of various sections namely Cover Story, In focus, Success Story of NAQUIM studies, Report, Pathshala and Social Media Highlights and Collectables.We try to cover the important & major tasks being carried out in the Board in the 'Cover Story' section. The 'In Focus' section incorporates the major activities and achievements of the board during the quarter so that readers have a glance on the major highlights of CGWB. The main idea behind the Report and Pathshala Section is to impart knowledge to our vivid readers on different hydrogeological aspects. As NAQUIM is the major flagship being taken up by CGWB, its success stories are being shared in anexclusive section of the magazine.

21.6 Azadi Ka Amrit Mahotsva (AKAM)

"Azadi Ka Amrit Mahotsav" mirrors India's socio-cultural, political and economic progress. The official journey of "Azadi ka Amrit Mahotsav" commencedfrom12th March, 2021with a 75 week countdown to our 75th anniversary of Independence, and shall end after a year on 15th August, 2023. The Department has prepared programmes for all of the 75 weeks to celebrate this Mahotsav at large scale by using resources of various wings, organizations, institutions and attached and subordinate offices of this Department spread all over India. Apart from the programmes mentioned in the calendar for 75 weeks, the Department is carrying out some special programmes as per the themes decided by Cabinet Secretariat from time to time, viz special programmes for North East Region, Celebrating Rivers of India, Nadi Utsav (ghat cleaning, afforestation, awareness campaign, quiz, jal yatra, painting completion, debate etc) in different ghats etc.

21.6.1 Public Interaction Programme:

Under Azadi ka Amrit Mahotsva, Public Interaction Programmes were organized to sensitize people in respect of NAQUIM outcomes and water related aspects in all States. Details are as follows:

Sl. No.	Name of the State/UT	Achievement 2021-22 (Since Apr. 2021)*	No of people sensitized 2021-22 (Since April 2021)		Achievement 2021-22 (Since 15 Mar. 2021)*	No of people sensitized 2021-22 (Since 15 Mar. 2021)	
			TOTAL	Female		TOTAL	Female
1	Andhra Pradesh	8	665	159	8	665	159
2	Bihar	18	1687	446	19	1725	454
3	Chandigarh	0	0	0	1	38	9
4	Chhattisgarh	19	1923	879	20	1979	896
5	Goa	2	97	31	2	97	31
6	Gujarat	21	761	136	22	805	146
7	Haryana	16	1281	563	17	1345	593
8	Himachal Pradesh	9	592	253	10	637	279
9	Jammu &Kashmir	11	428	45	11	428	45
10	Ladakh	2	41	1	2	41	1
11	Jharkhand	17	1341	597	17	1341	597
12	Karnataka	19	1551	435	20	1725	474
13	Kerala	11	1066	459	11	1066	459
14	Madhya Pradesh	19	937	108	20	1036	139
15	Maharashtra	24	2410	628	24	2410	628
16	NER States	26	1546	698	26	1546	698
	Assam	14	577	224	14	577	224
	Arunachal Pradesh	0	0	0	0	0	0
	Manipur	1	33	5	1	33	5
	Meghalaya	7	701	359	7	701	359
	Mizoram	0	0	0	0	0	0
	Nagaland	0	0	0	0	0	0
	Tripura	4	235	110	4	235	110
17	Odisha	18	2472	1003	18	2472	1003
18	Punjab	7	333	115	8	504	184
19	Rajasthan	21	898	212	21	898	212
20	SUO- Delhi	9	635	218	10	696	230
21	Tamil Nadu & U.T. P	29	2930	1268	29	2930	1268
22	Telangana	7	755	295	7	755	295
23	Uttar Pradesh	49	2960	1000	56	4946	1553
24	Uttarakhand	8	356	129	9	417	138
25	West Bengal	14	947	285	14	947	285
	Total	**384**	**28612**	**9963**	**402**	**31449**	**10776**

Achievement 2021-22 (Since Apr. 2021)* Achievement 2021-22 (Since 15 Mar. 2021)*

PIP@75 under Azadi ka .. . initiated from 15 Mar. 2021 in these States

21.7 Nadi Utsav

The River Festival- 2021 program was formally launched through the flagging-in of *Ganga Mashal* on 16th December 2021 by the Hon'ble Jal Shakti Minister, Sh. Gajendra Singh Shekhawat to augment, manage and sustain the natural resources i.e. the Surface Water and Ground Water, sustaining the nature and ecology, cleanliness of the environment and also for providing mass awareness among the individuals for maintenance of the ecological stability and sustainability of the environment. Nadi Utsav 2021 is a pan India celebration which is jointly organized by Department of Water Resources, River Development & Ganga Rejuvenation under the Ministry of Jal Shakti in collaboration with Ministry of Culture, Govt. of India and National Mission for Clean Ganga (NMCG) from 17th to 23rd Dec, 2021. The event is being organized to celebrate rivers such as River Brahmaputra, Indus, Narmada, Sabarmati, Mahanadi, Krishna, Godavari, Cauvery, Ganga, Yamuna, amongst several others. The celebrations are being organized, as a part of Azaadi ka Amrut Mahotsav campaign commemorating 75 year of India's independence, under the four chosen themes of Cleanliness, Patriotism, Nature and Ecology, Devotion and Spirituality.

15 Nadi Utsav programs were organized by CGWB in 14 States of the country along the River banks thereby spreading a message of river cleaning, awareness for water conservation and river protection, rain water harvesting, afforestation etc. The event was attended by 1751 peoples including 749 females also. Under the program following activities were organized during 17th – 23rd Dec. 2021:

S.NO.	State: District	River	Date of Prog.	Activities Proposed
1.	**Madhya Pradesh:** Burhanpur	Tapi	17-12-2021	1. Awareness programme on Kundi/ Khooni Bhandara at Burhanpur (a network of well like structures interconnected through underground tunnel) 2. Cleanliness Drive 3.Preparation of Booklet on Kundi/ KhooniBhandara of Burhanpur

S.NO.	State: District	River	Date of Prog.	Activities Proposed
2.	**Maharashtra** :Wardha	Dham (Tributary of Wardha River) Wardha	17.12.2021	1. Awareness 2. Cleanliness Drive 3. Preparation of Document on Water supply security following Tamaswada pattern (TP) Nala treatment by 21/12/2021
3.	**Karnataka:** Bangalore	Vishrabhavathi (Tributary of arkavathi River)	17-12-2021	1. Shram Daan 2. Afforestation
4.	**Odisha:** Khurdha	Kuakhai Khurdha	17-12-2021	Awareness programme on importance of rivers in Sustenance of life
5.	**Uttar Pradesh:** Lucknow	Gomti	17-12-2021	1. Open Quiz 2. Extempore 3. Tree Plantation 4. Nature Biodiversity walk
6.	**Assam:** Kamrup	Brahmaputra	18-12-2021	Awareness Programme and plantation on the river bank
7.	**Chhattisgarh** :Janjgir	Mahanadi	20-12-2021	1. Cleanliness Drive 2. Afforestation 3. Awareness Program
8.	**Haryana:** Yamuna Nagar	Yamuna	21-12-2021	1. Sharm Daan 2. Painting competition 3. Afforestation 4. Jal Yatra 5. Pledge
9.	**Gujarat:** Patan	Saraswati River	21-12-2021	1. Quiz competition 2. Poster competition 3. Lecture on River conservation and rejuvenation.
10.	**Tamil Nadu:** Thiruchirapalli	Cauvery	22-12-2021	1. Shram Daan and Pledge Activity 2. Preparation of Compendium on Rain Water Harvesting structures in the temples along the rivers of Tamil Nadu by 23/12/2021

S.NO.	State: District	River	Date of Prog.	Activities Proposed
11.	**West Bengal:** North 24 Pargana	Hoogly	22-12-2021	1. Quiz Programme on River of India
				2. Debate
				3. Elocution
				4. Story Telling on Local freedom fighters
				5. Pledge
12.	**Jammu:** Udhampur	Devika (Trivutary of Tawi River)	22-12-2021	Awareness Program
13.	**Kerala:** Trivandrum	Neyyar	22-12-2021	1. River Cleaning campaign
				2. Painting Competition
14.	**Kerala:** Trivandrum		23-12-2021	1. River Claning campetition
				2. Painting Competition
15.	**Telangana:** Hyderabad	Musi	23-12-2021	Kala Jatarato Save Musi From Pollution

22.0 VIGILANCE

Under Vigilance, a total of 13 complaints were brought forward wef 1.4.2021 and 2 new complaint cases were received during the period. Out of these15 complaints, 3 were closed. 4 cases of disciplinary proceedings were there during the period and 2 cases were disposed off.

23.0 RTI

The opening balance of RTI applications as on 01.04.2021 was 177 Requests and 16 Appeals. During the year 2020-21, 806 RTI requestsand 70 appeals were received. Number of cases which were transferred to other public authorities is 11. Total RTI requests and RTI appeals disposed of were 730and 67 respectively.. Details are given in table

STATUS OF RTI APPLICATIONSFOR YEAR 2021-22

Opening balance as on 01.04.2021	Received during the year (including cases transferred from other public authorities)	No. of cases transferred to other public authorities	Decisions where requests/ appeals rejected
Requests : 177	Requests : 806	Requests :11	Requests : 7
Appeals: 16	Appeals: 70	Appeals : 0	Appeals: 0

24.0 Budget and Expenditure

Central Ground Water Board,						
Summery of Monthly Expenditure Return for the Month of March, 2022						
Demand No 61						**(Figure in Lakhs)**
Sl. No.	Name of Activity	Budget Estimates 2021-22	Revised Estimates 2021-22	Modified Final RE F.Y 2021-22	Expen. During the Month 03/2022	Cumulative Expend. Up to 03/2022
1	**Establishment Expenditure (REVENUE + CAPITAL)**					
(i)	01 : Investigation (Major Head) HQ, CGWB (Reveune)	23673.95	25767.00	25112.00	229.12	25082.84
(ii)	05 : Investigation (Major Head) HQ, CGWB (Capital)	0.00	1200.00	920.00	0.00	920.00
(iii)	01.96 Swachhta Action Plan (Revenue)	1.06	1.00	1.00	0.01	0.01
(iv)	01.99 Information Technology (Revenue)	5.00	200.00	140.00	74.52	76.84
TOTAL Establishment Expenditure (Revenue+ Capital)		**23680.01**	**27168.00**	**26173.00**	**303.65**	**26079.69**
2	**2702-02 & 4702-07 : Rajiv Gandhi National GWT & R I**					
(i)	03 - Rajiv Gandhi National GWT & R I (Revenue)	230.00	300.00	287.24	51.66	249.79
(ii)	07 - Rajiv Gandhi National GWT & R I (Capital)	120.00	3.00	3.00	0.00	2.91
Total Rajiv Gandhi National GWT&RI (Revenue + Capital)		**350.00**	**303.00**	**290.24**	**51.66**	**252.70**
Total Non Scheme (Revenur + Capital)		**24030.01**	**27471.00**	**26463.24**	**355.31**	**26332.39**
3	**2702-16- PLAN GWM&R (Revenue + Capital)**					
(i)	16-Ground Water Management & Regulation (Revenue)	7747.00	7820.00	7817.00	852.62	7659.58
(ii)	16-Ground Water Management & Regulation (Capital)	20400.00	10050.00	9577.21	4136.07	9696.66
Total GWM&R (Revenue + Capital)		**28147.00**	**17870.00**	**17394.21**	**4988.69**	**17356.24**
4	TOTAL (Deduct Recoveries)	1500.00	1500.00	1200.00	173.53	1106.02
NET GWM&R (PLAN)		**26647.00**	**16370.00**	**16194.21**	**4815.16**	**16250.22**
5	**01-796 - Tribal Sub Plan (TSP) (Revenue + Capital)**					
(i)	01.796 Tribal Sub Plan (Revenue)	453.00	380.00	380.00	38.13	394.78
(ii)	00.796 Tribal Sub Plan (Capital)	400.00	250.00	250.00	4.93	248.62
Total Tribal Sub Plan (Revenue + Capital)		**853.00**	**630.00**	**630.00**	**43.06**	**643.40**
Total GWM&R + TSP (Rev + Capti) Net :		**27500.00**	**17000.00**	**16824.21**	**4858.22**	**16893.62**
Total Gross (GWM&R+TSP) (Revenue + Capital) :		**29000.00**	**18500.00**	**18024.21**	**5031.75**	**17999.64**
6	**2701 NHP (Revenue) & 4701 NHP (Capital)**					
(i)	2701 NHP-(REVENUE) Ext. Supp.	15.25	3.50	0.44	0.29	0.42
(ii)	2701 NHP-(REVENUE) Dom. Supp.	22.25	3.50	2.90	0.66	2.42
Total 2701 NHP (Revenue)		**37.50**	**7.00**	**3.34**	**0.95**	**2.84**
(i)	4701 NHP-(Capital) Ext. Supp.	225.00	502.73	502.73	47.39	272.33
(ii)	4701 NHP-(Capital) Dom. Supp.	725.00	502.74	502.74	65.00	353.14
Total 4701 NHP (Capital)		**950.00**	**1005.47**	**1005.47**	**112.39**	**625.47**
Total NHP (Revenue + Capital)		**987.50**	**1012.47**	**1008.81**	**113.34**	**628.31**

Central Ground Water Board,					
Summary of Monthly Expenditure Return for the Month of March, 2022.					
Demand No. 61					**(Figure in Lakhs)**
SL. No.	Name of Activity	Budget Estimates 2021-22	Revised Estimates 2021-22	**Modified Final RE F.Y 2021-22**	**Cumulative Expenditure Up to 03/2022**
1	**16 - Ground Water Management & Regulation (Revenue + 06 Capital)**				
(i)	TOTAL GWM&R (PLAN) GROSS	28147.00	17870.00	17394.21	17356.24
2	**01.796 Tribal Sub Plan (Reve+ Capital)**				
(i)	TOTAL TSP	853.00	630.00	630.00	643.40
	Gross Total (Plan)	**29000.00**	**18500.00**	**18024.21**	**17999.64**
3	**2701-08.01 and 2701-08.02 : NHP (Revenue)**				
(i)	2701 NHP (REVENUE) Ext. Supp.	15.25	3.50	0.44	0.42
(ii)	2701 NHP-(REVENUE) Dom. Supp.	22.25	3.50	2.90	2.42
	Total NHP (Revenue)	**37.50**	**7.00**	**3.34**	**2.84**
4	**4701.08.01 and 4701.08.02 :NHP (Capital)**				
(i)	4701 NHP (Capital) Ext. Supp.	225.00	502.73	502.73	272.33
(ii)	4701 NHP-(Capital) Dom. Supp.	725.00	502.74	502.74	353.14
	Total NHP (Capital)	**950.00**	**1005.47**	**1005.47**	**625.47**
	Total NHP Ext. Supp. & Dom. Supp. (Revenue + Capital)	**987.50**	**1012.47**	**1008.81**	**628.31**
	Non Scheme (Non Plan)				
5	**Establishment Expenditure (REVENUE + CAPITAL)**				
(i)	2702 : Establishment Expenditure (Revenue)	23673.95	25767.00	25112.00	25082.84
(ii)	4702 : Establishment Expenditure (Capital)	0.00	1200.00	920.00	920.00
6	01.96 : Swachhta Action Plan (Revenue)	1.06	1.00	1.00	0.01
7	01.99 : Information Technology (Revenue)	5.00	200.00	140.00	76.84
Total Estt. Expenditure (Revenue + Capital)		**23680.01**	**27168.00**	**26173.00**	**26079.69**
8	Rajiv Gandhi National Ground Water Trainig and Research Institute (06-Revenue + 07-Capital)	350.00	303.00	290.24	252.70
	Total RGI	**350.00**	**303.00**	**290.24**	**252.70**
	Total (Non Scheme)	**24030.01**	**27471.00**	**26463.24**	**26332.39**
*** Note: RGI Scheme has been shifted to Non Scheme in RE 2021-22**					

Monthly Expenditure Report For the Month of March, 2022							
Demand No. 61							PLAN (GWM &R)
							2702 (REVENUE)
Major Head: 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION							(Figure in lacs)
Sub Head: 16-GROUND WATER MANAGEMENT & REGULATION –01 (Ext.)02(Dom.)							
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
1	2	3	4	5	6	7	8
16.02 Dom.Supp.(GWM&R) PLAN							
16.02.03 - समयोपरिभत्ता -OTA	1.00	0.10	0.10	0.05	**0.04**	0.09	0.01
16.02.06 - चिकित्सा - M/Treat.	350.00	250.00	250.00	212.22	**32.61**	244.83	5.17
16.02.11 - यात्राव्यय - D.T.E.	2190.00	2500.50	2530.31	2334.28	**176.50**	2510.78	19.53
16.02.12 - विदेशयात्राव्यय - F.T.E.	100.00	100.00	0.00	0.00	**0.00**	0.00	0.00
16.02.13 - कार्यालयव्यय - O.E.	1230.00	1200.00	1200.00	1102.66	**84.46**	1187.12	12.88
16.02.14 - किराया, दरएवंकर - R.R.T.	450.00	413.90	493.90	420.12	**50.25**	470.37	23.53
16.02.16 - प्रकाशन - Publications	50.00	0.50	0.50	0.25	**0.25**	0.50	0.00
16.02.20 - O.A.E.	160.00	110.00	90.00	39.80	**31.35**	71.15	18.85
16.02.24 - पेट्रोल / डीजल -P.O.L.	1525.00	1525.00	1632.00	1398.82	**185.15**	1583.97	48.03
16.02.27 - माइनरकार्य - Min/Works	300.00	250.00	250.00	228.94	**0.00**	228.94	21.06
16.02.28 - वृत्तिकसेवाएँ - P.S.	300.00	450.00	355.00	300.06	**38.16**	338.22	16.78
16.02.30 - Other Contractual Services	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.33 - Subsidies	1.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.43 - उचंतस्टाक - S/Stock	1000.00	1000.00	997.00	751.57	**253.85**	1005.42	-8.42
16.02.50 - अन्यप्रभार - O. C.	20.00	20.00	18.19	18.19	**0.00**	18.19	0.00
16.99.13 - OE (Information Technology)	70.00	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Dom Support (PLAN) GWM&R	**7747.00**	**7820.00**	**7817.00**	**6806.96**	**852.62**	**7659.58**	**157.42**

							PLAN (GWM & R)
Demand No. 61							**4702 (CAPITAL)**
Major Head: 4702-MINOR IRRI-02-005-Capital Nature Expenditure (GWM & R)							
Sub Head: 06.01-GROUND WATER MANAGEMENT & REGULATION							
इकाई का कोड/नाम Unit of Appropriation	**Budget Estimate 2021-22**	**Revised Estimate 2021-22**	**Modifide Final RE FY 2021-22**	**Prog. Exp. upto the Pre. Month of 02/2022**	**Expend. during the Month of 03/2022**	**Progr. Exp. upto the Month of 03/2022**	**Variation (+) Exces (-) Saving**
4702.00.800.06 (Capital Nature Expenditure) GWM & R Dom. Supp. PLAN							
1	2	3	4	5	6	7	8
06.01.51 - मोटरवाइकल - M.V.	500.00	50.00	55.00	49.83	**5.74**	55.57	-0.57
06.01.52 - मशीनरीएवंउपस्कर - M&E	4500.00	1200.00	1035.21	996.48	**37.84**	1034.32	0.89
06.01.53 - मुख्यकार्य - M/Works	15400.00	8800.00	8487.00	4514.28	**4092.49**	8606.77	-119.77
Total Capital Nature Expenditure	**20400.00**	**10050.00**	**9577.21**	**5560.59**	**4136.07**	**9696.66**	**-119.45**
Grant Total (Revenue + Capital)	**28147.00**	**17870.00**	**17394.21**	**12367.55**	**4988.69**	**17356.24**	**37.97**

							Tribal Sub Plan
Demand No. 61 **01.796 (REVENUE)**							
Major Head: 01.796 Tribal Sub Plan (Rewvnue)MINOR IRRI-00-445-GROUND WATER INVESTIGATION							
Sub Head: 16-GROUND WATER MANAGEMENT & REGULATION –02.01 (Ext.)02(Dom.)							
Monthly Expenditure Report For the Month of March, 2022							**(Fig.in lacs)**
इकाई का कोड / नाम Unit of Appropriation	**Budget Estimate 2021-22**	**Revised Estimate 2021-22**	**Modifide Final RE FY 2021-22**	**Prog. Exp. upto the Pre. Month of 02/2022**	**Expend. during the Month of 03/2022**	**Progr. Exp. upto the Month of 03/2022**	**Variation (+) Exces (-) Saving**
1	2	3	4	5	6	7	8
02.01 Dom.Supp.(GWM&R) PLAN							
16.02.06 - चिकित्सा - M/Treat.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.11 –यात्रा व्यय - D.T.E.	230.00	230.00	230.00	222.23	**7.31**	229.54	0.46
16.02.12 – विदेशयात्रा व्यय -F.T.E.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.13 –कार्यालय व्यय -O.E.	123.00	70.00	70.00	62.22	**24.25**	86.47	-16.47
16.02.14 -किराया, दरएवंकर - R.R.T.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.16 -प्रकाशन - Pub.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.20 -ओ.ए.ई -O.A.E.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.24 - पेट्रोल / डीजल - P.O.L.	100.00	80.00	80.00	72.20	**6.57**	78.77	1.23
16.02.27 -माइनरकार्य - Min/Works	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.28 -वृत्तिकसेवाएँ - P.S.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.30 -Other Contractual Services	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
16.02.43 -उचंतस्टाक - S/Stock	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Dom Support (PLAN)	**453.00**	**380.00**	**380.00**	**356.65**	**38.13**	**394.78**	**-14.78**
Total Revenue (GWMR + TSP)	**8200.00**	**8200.00**	**8197.00**	**7163.61**	**890.75**	**8054.36**	**142.64**

							00.796- CAPITAL
Monthly Expenditure Report For the Month of March, 2022							
Demand No. 61							**Tribal Sub Plan**
Major Head:00.796-Tribal Sub Plan CAPITAL OUTLAY-00.112 GROUND WATER (M.I.)							
Sub Head: 01.01-Tribal Sub Plan (Capital)							
इकाई का कोड /नाम Unit of Appropriation	**Budget Estimate 2021-22**	**Revised Estimate 2021-22**	**Modifide Final RE FY 2021-22**	**Prog. Exp. upto the Pre. Month of 02/2022**	**Expend. during the Month of 03/2022**	**Progr. Exp. upto the Month of 03/2022**	**Variation (+) Exces (-) Saving**
00.796- CAPITAL							
1	2	3	4	5	6	7	8
01.01.51 –मोटर वाहन -Motor Vehicle	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.52 –मशीनरी एवं उपस्कर -M & E	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.53 -मुख्यकार्य -M/Works	400.00	250.00	250.00	243.69	**4.93**	248.62	1.38
Total: 00.796- CAPITAL	**400.00**	**250.00**	**250.00**	**243.69**	**4.93**	**248.62**	**1.38**
Total CAPITAL (GWMR + TSP)	**20800.00**	**10300.00**	**9827.21**	**5804.28**	**4141.00**	**9945.28**	**-118.07**
Grant Total TSP (Revnue+Capital)	**853.00**	**630.00**	**630.00**	**600.34**	**43.06**	**643.40**	**-13.40**
Grant Total (TSP + GWMR)	**29000.00**	**18500.00**	**18024.21**	**12967.89**	**5031.75**	**17999.64**	**24.57**

Demand No. 61 DEDUCTED RECOVERY	4702 Capital						
केन्द्रीय भूमि जल बोर्ड (जल शक्तिमंत्रालय)					ISSUE TO WORK & OTHER CREDITS		
Monthly Expenditure Report For the Month of March, 2022							
Major Head: 4702-MINOR IRRI-00-800-Capital Nature Expenditure (GWM & R) (PLAN)							
Sub Head: 06 ISSUE TO WORK & OTHER CREDITS							
							(Fig. in lacs)
इकाई का कोड/ नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
1	2	3	4	5	6	7	8
06.01.70 -Issue to Work	1500.00	1500.00	1200.00	932.49	173.53	1106.02	93.98
06.02.70 -Other Suspense Charges	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Recoveries	**1500.00**	**1500.00**	**1200.00**	**932.49**	**173.53**	**1106.02**	**93.98**
NET CGWB (GWM&R)	**26647.00**	**16370.00**	**16194.21**	**11435.06**	**4815.16**	**16250.22**	**-56.01**
NET CGWB (GWM&R + TSP)	**27500.00**	**17000.00**	**16824.21**	**12035.40**	**4858.22**	**16893.62**	**-69.41**

							Non Scheme (Non Plan)
Demand No. 61							Establishment Expenditure 2702 (Revenue)
Major Head: Establishment Expenditure 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION							
Sub Head: 01-CGWB-01.01-H/QRS.							(Fig.in lacs)
Monthly Expenditure Report For the Month of March, 2022							
इकाई का कोड /नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
1	2	3	4	5	6	7	8
16.02 Domestic Support N.PLAN							
01.01.01 - वेतन –Salary	23000.00	24900.00	24250.00	24196.75	**74.73**	24271.48	-21.48
01.01.02 -मजदूरी –Wages	230.00	260.00	255.00	201.26	**48.87**	250.13	4.87
01.01.03 -समयोपरिभत्ता -OTA	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.06 -चिकित्सा -M/Treat.	190.00	160.00	160.00	115.21	**31.99**	147.20	12.80
01.01.11 -यात्राव्यय -D.T.E.	178.71	311.00	311.00	261.73	**41.02**	302.75	8.25
01.01.12 -विदेशयात्राव्यय -F.T.E.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.13 -कार्यालयव्यय -O.E.	46.24	60.00	83.50	45.15	**17.12**	62.27	21.23
01.01.14 -किराया, दरएवंकर R.R.T.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.16 -प्रकाशन -Pub.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.20 -ओ.ए.ई -O.A.E.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.24- पेट्रोल / डीजल -P.O.L.	9.00	20.00	20.00	8.95	**10.75**	19.70	0.30
01.01.26 - ऐड. एवं पब्लिस्ट -Adv. & Publi.	10.00	30.00	14.50	9.64	**2.13**	11.77	2.73
01.01.28 -वृत्तिकसेवाएँ -P.S.	10.00	26.00	18.00	15.03	**2.51**	17.54	0.46
01.01.30 -Other Contractual Services	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
01.01.50 –अन्य प्रभार -O. C	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Dom Support (N.PLAN)	**23673.95**	**25767.00**	**25112.00**	**24853.72**	**229.12**	**25082.84**	**29.16**

Demand No. 61							4702 (CAPITAL)
Major Head: 4702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION (Establishment Expenditure)							
Sub Head: 05. 01-CGWB-01.01-H/QRS.							
इकाई का कोड / नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
4702.00.800.05 (Capital Nature Expenditure) CGWB Head Quarter (NON PLAN)							
1	2	3	4	5	6	7	8
05.01.51 -मोटरवाइकल -M.V.	0.00	0.00	0.00	0.00	0.00	0.00	0.00
05.01.52 -मशीनरीएवंउपस्कर -M & E	0.00	0.00	0.00	0.00	0.00	0.00	0.00
05.01.53 -मुख्यकार्य -M/ Works	0.00	1200.00	920.00	920.00	0.00	920.00	0.00
Total Capital Nature Expenditure	**0.00**	**1200.00**	**920.00**	**920.00**	**0.00**	**920.00**	**0.00**
Grant Total (Non Scheme)	**23673.95**	**26967.00**	**26032.00**	**25773.72**	**229.12**	**26002.84**	**29.16**

Swachhta Action Plan Scheme (Non Plan)							**Non**
Demand No. 61							**Establishment Expenditure 2702 (Revenue)**
Major Head: Establishment Expenditure 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION							
Sub Head: 01-CGWB-01.01-H/QRS.							
Monthly Expenditure Report For the Month of March, 2022							
इकाई का कोड /नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
01.96 Swachhta Action Plan							
01.96.13 – कार्यालय व्यय -O.E.	1.06	1.00	1.00	0.00	0.01	0.01	0.99
Total	**1.06**	**1.00**	**1.00**	**0.00**	**0.01**	**0.01**	0.99
Total (Estt. Exp. + SAP)	**23675.01**	**26968.00**	**26033.00**	**25773.72**	**229.13**	**26002.85**	**30.15**

Information Technology Non Scheme (Non Plan)							
Demand No. 61					Establishment Expenditure 2702 (Revenue)		
Major Head: Establishment Expenditure 2702-MINOR IRRI-02-005-GROUND WATER INVESTIGATION							
Sub Head: 01-CGWB-01.01-H/QRS.							
इकाईकाकोड/नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
01.99 Information Technology							
01.99.13 –कार्यालय व्यय -O.E.	5.00	200.00	140.00	2.32	74.52	76.84	63.16
Total (Information Technology)	**5.00**	**200.00**	**140.00**	**2.32**	**74.52**	**76.84**	**63.16**
Total (Estt. Exp. + SAP +IT)	**23680.01**	**27168.00**	**26173.00**	**25776.04**	**303.65**	**26079.69**	**93.31**

R.G.N.G.W.T.R.I (Training)							
Demand No. 61							2702 (REVENUE)
Major Head: 2702 : Minor Irrigation, 02 : Ground Water, 800 : Other Expenditure (Non Scheme)							
Sub Head: 03 : Rajiv Gandhi National Ground Water Training and Research Institure. Detailed Head : 00 (Default)							
Monthly Expenditure Report For the Month of March, 2022							(Figure in lacs)
इकाई का कोड / नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modified Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
1	2	3	4	5	6	7	8
03.00 : Domestic Support (RGNGWTRI)							
03.00.06 - चिकित्सा -M/Treat.	2.00	11.00	8.00	2.00	**3.00**	5.00	3.00
03.00.11 – यात्रा व्यय -D.T.E.	20.00	13.00	13.00	13.68	**0.00**	13.68	-0.68
03.00.12 –विदेश यात्रा व्यय -F.T.E.	30.00	0.00	0.00	0.00	**0.00**	0.00	0.00
03.00.13 – कार्यालय व्यय -O.E.	63.00	160.00	154.50	92.84	**32.27**	125.11	29.39
03.00.14 -किराया, दर एवं कर R.R.T.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
03.00.16 - प्रकाशन - Pub.	5.00	3.00	0.24	0.24	**0.00**	0.24	0.00
03.00.24 - पेट्रोल/ डीजल -P.O.L.	10.00	8.00	6.50	4.60	**1.08**	5.68	0.82
03.00.30 - Other Contractual Services	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
03.00.28 – वृत्तिक सेवाएँ -P.S.	100.00	105.00	105.00	84.77	**15.31**	100.08	4.92
Total Dom Support (RGI)	**230.00**	**300.00**	**287.24**	**198.13**	**51.66**	**249.79**	**37.45**

R.G.N.G.W.T.R.I (Training)							
Demand No. 61							4702 (CAPITAL)
Major Head: 4702-06.01 HRD/Capacity Building (PLAN)							
Sub Head: 07.01-RGN. G/W TRG AND RESURCH INSTITUTE.							
इकाई का कोड / नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
4702 Capital Nature Expenditure RGNT &RI (PLAN)							
1	2	3	4	5	6	7	8
07.01.51 –मोटर वाइकल - Motor Vehicle.	20.00	0.00	0.00	0.00	**0.00**	0.00	0.00
07.01.52 –मशीनरी एवं उपस्कर -M&E	100.00	3.00	3.00	2.91	**0.00**	2.91	0.09
07.01.53 –मुख्य कार्य - M/Works	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
Total Capital Nature Expenditure	**120.00**	**3.00**	**3.00**	**2.91**	**0.00**	**2.91**	**0.09**
Grant Total (RGI) PLAN	**350.00**	**303.00**	**290.24**	**201.04**	**51.66**	**252.70**	**37.54**

Hydrology Project-Phase-III							
Demand No. 61				**2701**		REVENUE	
Major Head: 2701-.80.004.08.01(Ext.Supp.) & 08.02 (Dom. Supp) National Hydrology Project-Phase-III **(PLAN)**							
Monthly Expenditure Report For the Month of March, 2022							
इकाई का कोड /नाम Unit of Appropriation	**Budget Estimate 2021-22**	**Revised Estimate 2021-22**	**Modifide Final RE FY 2021-22**	**Prog. Exp. upto the Pre. Month of 02/2022**	**Expend. during the Month of 03/2022**	**Progr. Exp. upto the Month of 03/2022**	**Variation (+) Exces (-) Saving**
08.01-Ext. Support NHP-III	2	3	4	5	6	7	8
08.01.11 - D.T.E.	5.00	0.50	0.00	0.00	0.00	0.00	0.00
08.02.12 – विदेशयात्रा व्यय -F.T.E.	0.00	0.00	0.00	0.00	0.00	0.00	0.00
08.01.13 – कार्यालय व्यय - O.E.	2.75	0.50	0.30	0.00	0.29	0.29	0.01
08.01.20 - Other Adminstative Expe.	5.00	0.10	0.14	0.13	0.00	0.13	0.01
08.01.27 - Minor/Works	0.00	0.00	0.00	0.00	0.00	0.00	0.00
08.01.28 - Professional Services	2.50	0.90	0.00	0.00	0.00	0.00	0.00
08.01.30 - Other Contratual Services	0.00	1.50	0.00	0.00	0.00	0.00	0.00
TOTAL (Ext. Support) NHP	**15.25**	**3.50**	**0.44**	**0.13**	**0.29**	**0.42**	**0.02**
08.02 Dom. Support (NHP)-III							
08.02.11 - D.T.E.	5.00	0.50	0.00	0.00	**0.00**	0.00	0.00
08.02.12 – विदेशयात्रा व्यय - F.T.E.	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
08.02.13 – कार्यालय व्यय - O.E.	2.75	0.50	0.15	0.00	**0.14**	0.14	0.01
08.02.20 - Other Adminstative Expe.	5.00	0.10	0.00	0.00	**0.00**	0.00	0.00
08.02.27 -Minor/Works	0.00	0.00	0.00	0.00	**0.00**	0.00	0.00
08.02.28 - Professional Services	2.50	0.90	0.00	0.00	**0.00**	0.00	0.00
08.02.30 - Other Contratual Services	7.00	1.50	2.75	1.76	**0.52**	2.28	0.47
TOTAL (Dom. Support) NHP	**22.25**	**3.50**	**2.90**	**1.76**	**0.66**	**2.42**	**0.48**
TOTAL (Ext. Sup. + Dom. Support) NHP	**37.50**	**7.00**	**3.34**	**1.89**	**0.95**	**2.84**	**0.50**

Demand No. 61 4701 CAPITAL							Hydrology Project-Phase-III
Major Head: 4701-.80.004.08.01 (Ext. Supp) & 08.02 (Dom. Supp)Hydrology Project-Phase-III (PLAN)							
Sub Head: 06 : National Hyydrology Project – 01 (Ext. Support) & 02 (Dom. Support)							
4701. 80.004.04.02 (Capital Nature Expenditure) National Hydrology Project (PLAN)							
इकाई का कोड / नाम Unit of Appropriation	**Budget Estimate 2021-22**	**Revised Estimate 2021-22**	**Modifide Final RE FY 2021-22**	**Prog. Exp. upto the Pre. Month of 02/2022**	**Expend. during the Month of 03/2022**	**Progr. Exp. upto the Month of 03/2022**	**Variation (+) Exces (-) Saving**
4701. 08.01 Ext. Supp. NHP (Capital)							
08.01.51 - M V	0.00	0.00	0.00	0.00	0.00	0.00	0.00
08.01.52 - M & E	0.00	230.40	230.41	0.00	0.00	0.00	230.41
08.01.53 - Major/Works	225.00	272.33	272.32	224.94	47.39	272.33	-0.01
08.01.60 - Other Capital Expenditure	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Capital Nature (Ext. Supp.)	**225.00**	**502.73**	**502.73**	**224.94**	**47.39**	**272.33**	**230.40**
4701. 08.02 Dom. Supp. NHP (Capital)							
08.02.51 - M V	0.00	0.00	0.00	0.00	0.00	0.00	0
08.02.52 - M & E	500.00	230.41	230.41	63.37	17.44	80.81	149.6
08.02.53 - Major/Works	225.00	272.33	272.33	224.77	47.56	272.33	0
08.02.60 - Other Capital Expenditure	0.00	0.00	0.00	0.00	0.00	0.00	0
Total Capital Nature (Dom. Supp.)	**725.00**	**502.74**	**502.74**	**288.14**	**65.00**	**353.14**	**149.6**
Total Capital (Ext. Sup. + Dom.Sup.)	**950.00**	**1005.47**	**1005.47**	**513.08**	**112.39**	**625.47**	**380**
G.Total (Revenue + Capital) (NHP-III)	**987.50**	**1012.47**	**1008.81**	**514.97**	**113.34**	**628.31**	**380.5**

Demand No. 61
IEC (SCHEME)

केन्द्रीय भूमि जल बोर्ड (जल संसाधन मंत्रालय) **(IEC) ACTIVITIES (PLAN)**

Monthly Expenditure Report For the Month of March, 2022

Major Head: 2701-MINOR IRRI-02-005-GROUND WATER INVESTIGATION (PLAN)

Sub Head : 20 MASS AWARENESS ACTIVITIES (IEC) (PLAN)

(Figure in lacs)

इकाई का कोड /नाम Unit of Appropriation	Budget Estimate 2021-22	Revised Estimate 2021-22	Modifide Final RE FY 2021-22	Prog. Exp. upto the Pre. Month of 02/2022	Expend. during the Month of 03/2022	Progr. Exp. upto the Month of 03/2022	Variation (+) Exces (-) Saving
1	2	3	4	5	6	7	8
20.01.13 - Office Expenses	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20.01.20 - Other Administrative Expenses	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20.01.26 - Advertising & Publicity	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20.01.28 - Professional Services	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20.01.31 - Grant-in-aid-General	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total (IEC) PLAN	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

25.0 RAJBHASHA – राजभाषा हिन्दी का प्रचार-प्रसार एवं कार्यान्वयन

- केंद्रीय भूमिजल बोर्ड और इसके सभी अधीनस्थ क्षेत्रीय, प्रभागीय और राज्य एकक कार्यालयों में वर्ष 2021-2022 के दौरान राजभाषा हिन्दी के कार्यान्वयन और प्रचार–प्रसार के सक्रिय प्रयास जारी रखे गए। केंद्रीय भूमिजल बोर्ड राजभाषा हिंदी के प्रगामी प्रयोग व कार्यान्वयन के लिए प्रतिबद्ध है। राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम के अनुसार निर्धारित लक्ष्य को प्राप्त करने के लिए बोर्ड सतत प्रयत्नशील है। राजभाषा विभाग, गृह मंत्रालय, जल शक्ति मंत्रालय और नगर राजभाषा कार्यान्वयन समिति से समय–समय पर प्राप्त अनुदेशों और दिशानिर्देशों के पूर्ण अनुपालन सुनिश्चित करने तथा संघ की राजभाषा नीति के प्रावधानों के कार्यान्वयन और वार्षिक कार्यक्रम में निर्धारित लक्ष्यों को प्राप्त करने के उद्देश्य से सभी अधीनस्थ कार्यालयों और मुख्यालय के सभी अनुभागों को इससे अवगत कराया गया।
- सरकारी काम काज मेंहिंदी के प्रयोग को बढ़ाने के लिए नियमित रूप से विभिन्न परिपत्र,आदेश और अपील जारी किए गए । कार्यालय में हिन्दी में टिप्पण और आलेखन को बढ़ावा देने के उद्देश्य से राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम 2021-2022 की प्रति,द्विभाषी मानक मसौदे, प्रशासनिक शब्दावली आदि सभी अधिकारियों और कर्मचारियों को उपलब्ध कराये गए। सभी कम्प्यूटरों पर हिन्दी में काम करने की सुविधा प्रदान करने हेतु यूनिकोड सुविधा उपलब्ध कराई गयी ।
- राजभाषा हिन्दी के प्रभावी प्रचार-प्रसार और कुशल कार्यान्वयन के लिए राजभाषा विभाग, गृह मंत्रालय द्वारा जारी समस्त प्रोत्साहन योजनाओं यथा मूल रूप से हिन्दी में टिप्पण –आलेखन, श्रुतलेखन, हिन्दी में टंकण आदि योजनाओं को कार्यालय में लागू किया गया है तथा अधिक से अधिक अधिकारी और कर्मचारी उत्साहपूर्वक इन योजनाओं में भाग ले रहे हैं।
- केंद्रीय भूमिजल बोर्ड की वेबसाइट को द्विभाषी रूप में तैयार किया गया है। राजभाषा से संबन्धित विभिन्न गतिविधियों और महत्वपूर्ण उपलब्धियों को इसमे विशेष स्थान दिया जाता है।
- राजभाषा हिन्दी के प्रभावी कार्यान्वयन और इसकी प्रगति की मॉनिटरिंग के लिए बोर्ड और इसके समस्त अधीनस्थ कार्यालयों में राजभाषा कार्यान्वयन समिति का गठन किया गया है। इस समिति की बैठकें नियमित रूप से प्रत्येक तिमाही में आयोजित की जाती हैं। इन बैठकों में राजभाषा विभाग, गृह मंत्रालय द्वारा जारी वार्षिक कार्यक्रम में निर्धारित लक्ष्यों की प्राप्ति हेतु विभिन्न मुद्दों पर विचार-विमर्श द्वारा कार्यनीति तैयार की जाती है। इन बैठकों में लिए गए निर्णयों पर प्राथमिकता के आधार पर अनुवर्ती कार्रवाई की जाती है।

- गृह मंत्रालय द्वरा जारी दिशा निर्देश के अनुसार सितंबर माह में दिनांक 14.9.2021 से 28.9.2021 के दौरान बोर्ड और इसके समस्त अधीनस्थ कार्यालयों में हिन्दी दिवस, हिन्दी सप्ताह, हिन्दी पखवाड़ा, हिन्दी माह का आयोजन किया गया। इस दौरान अधिकारियों और कर्मचारियों के लिए कई प्रतियोगिताओं जैसे हिन्दी निबंध प्रतियोगिता, टिप्पण-आलेखन, टंकण, वाद-विवाद , प्रश्न मंच आदि का आयोजन किया गया। सभी अधिकारियों और कर्मचारियों ने इन प्रतियोगिताओं में बढ़-चढ़कर भाग लिया।
- भारत सरकार की राजभाषा नीति के अनुसरण में राजभाषा अधिनियम, 1963 की धारा 3(3) के तहत सभी दस्तावेज हिंदी और अंग्रेजी दोनों में तैयार किए जाते हैं। राजभाषा नियम, 1976 केनियम 5 के तहत हिन्दी में प्राप्त सभी पत्रों के उत्तर अनिवार्य रूप से हिन्दी में ही दिये जा रहे हैं। वार्षिक कार्यक्रम में निर्धारित लक्ष्यों को ध्यान में रखते हुए कार्यालय में जांच बिन्दु स्थापित किए गए हैं, ताकि इस दिशा में किए जा रहे कार्यों की मॉनिटरिंग की जा सके।
- राजभाषा विभाग, गृह मंत्रालय द्वारा समय-समय पर जारी दिशानिर्देशों के अनुपालन में केंद्रीय भूमिजल बोर्ड, मुख्यालय और इसके अधीनस्थ कार्यालयों में नियमित रूप से प्रत्येक तिमाही मे हिन्दी कार्यशालाओं का आयोजन किया जा रहा है।
- केंद्रीय भूमिजल बोर्ड, मुख्यालय फ़रीदाबाद में समृद्ध पुस्तकालय है। इस पुस्तकालय में वैज्ञानिक एवं तकनीकी मानक ग्रन्थों, जर्नल आदि के अतिरिक्त प्रशासनिक और अन्य विषयों पर प्रचुर पठनीय साहित्य उपलब्ध है। पुस्तकालय में हिन्दी पुस्तकों की खरीद वार्षिक कार्यक्रम में निर्धारित लक्ष्यों के अनुसार की जा रही है।
- माननीय संसदीय राजभाषा समिति की दूसरी उपसमिति द्वारा दिनांक 15.07.2021 को केंद्रीय भूमि जल बोर्ड,उत्तर पश्चिमी हिमालयी क्षेत्र,जम्मू कार्यालय, दिनांक 27.09.2021 को केंद्रीय भूमि जल बोर्ड, दक्षिणी क्षेत्र , हैदराबाद,दिनांक 20.11.2021 को केंद्रीय भूमि जल बोर्ड , प्रभाग VII, गुवाहाटी , दिनांक 23.11.2021 को केंद्रीय भूमि जल बोर्ड, प्रभाग-XV, कोलकाता, दिनांक 27.12.2021 को केंद्रीय भूमि जल बोर्ड, प्रभाग VI, नागपुर,दिनांक 12.02.2022 को केंद्रीय भूमि जल बोर्ड,मध्य पूर्वी क्षेत्र, पटना और दिनांक 02.03.2022 को केंद्रीय भूमि जल बोर्ड, प्रभाग XII, भोपालकार्यालय का राजभाषायी निरीक्षण किया गया । समिति द्वारा कार्यालयोंमें राजभाषा के प्रचार-प्रसार और इसके कार्यान्वयन की दिशा में किए जा रहे कार्यों पर संतोष प्रकट किया गया।
- राजीव गांधी राष्ट्रीय भूमि जल प्रशिक्षण और अनुसंधान संस्थान में एक प्रकाशन प्रकोष्ठ की स्थापना की गयी ताकि प्रशिक्षण संबंधी समस्त सामग्री को द्विभाषी रूप में तैयार किया जा सके ।

- केंद्रीय भूमि जल बोर्ड के सभी अधीनस्थ कार्यालयों में राजभाषा कार्यान्वयन की समुचित मॉनिटरिंग के लिए सभी कार्यालयों के राजभाषा प्रभारियों की एक सूची तैयार की गयी है और इसे केंद्रीय भूमि जल बोर्ड की वेबसाइट पर उपलब्ध कराया गया है ।
- केंद्रीय भूमि जल बोर्ड हिंदी के प्रगामी प्रयोग व कार्यान्वयन के लिए प्रतिबद्ध है । राजभाषा विभाग द्वारा जारी वार्षिक कार्यक्रम के अनुसार निर्धारित लक्ष्य को प्राप्त करने के लिए बोर्ड सतत प्रयत्नशील है

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26. GALLERY

Sharing of NAQUIM report with D.C., Baska District, Assam

Sharing of NAQUIM report with D.C., Darrang District, Assam

Sharing of NAQUIM report with D.C., Dhubri District, Assam

Sharing of NAQUIM report with D.C., Udalguri District, Assam

Sharing of NAQUIM report with D.C., Hailakandi District, Assam

Discussion with villagers on NAQUIM output in Barpeta District, Assam

Celebration of World Water Day (22nd March 2022) at Shillong, Meghlaya

NadiUtsav was celebrated on 18.12.2021 at Dharapur, Rani Chapori -MoujuliKamrup (M), district, Assam under the banner of "Azadi ki Amrit Mahotsava-Celebrating Rivers of India". Total number of participants=50, Female=27.

Public Interaction Program (PIP) on "Aquifer Mapping and Management in Warangal District with special emphasis on Artificial Recharge at Kakatiya University, Warangal - Azadi Ka Amrut Mahotsav"(Virtual) by Central Ground Water Board in association with Geology Department of Kakatiya University, Telangana

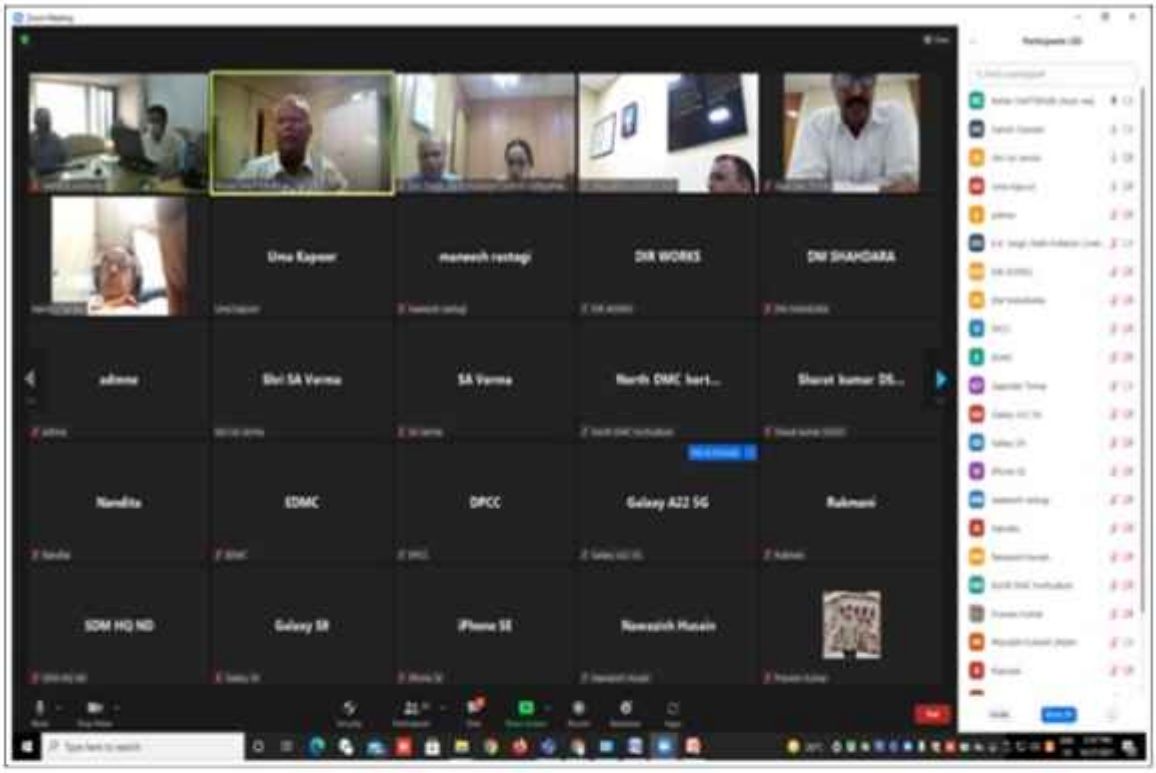
25.10.2021

Total Participants: 73

Male:42
Female:31

CGWB, SR, Hyderabad

MINISTRY OF JAL SHAKTI
एक्विफर मेपिंग, भूजल संसाधन आकलन,
प्रबंधन और विनियमन पर कार्यशाला
WORKSHOP ON AQUIFER MAPPING, GROUND WATER
RESOURCE ASSESSMENT, MANAGEMENT AND REGULATION
आयोजक :
केंद्रीय भूमिजल बोर्ड, राज्य एकक कार्यालय- दिल्ली
CENTRAL GROUND WATER BOARD, STATE UNIT OFFICE DELHI
जल संसाधन, नदी विकास और गंगा संरक्षण विभाग
DEPARTMENT OF WATER RESOURCES, RD & GR
भारत सरकार, जल शक्ति मंत्रालय
GOVERNMENT OF INDIA, MINISTRY OF JAL SHAKTI


Uma Kapoor
manesh rastogi
DIR WORKS
DM SHAHDARA
admin
Shri SA Verma
SA Verma
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Sharat kumar DS...
Nandita
EDMC
DPCC
Galaxy A22 5G
Rukmani
SDM HQ ND
Galaxy S9
iPhone SE
Nawazish Husain

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